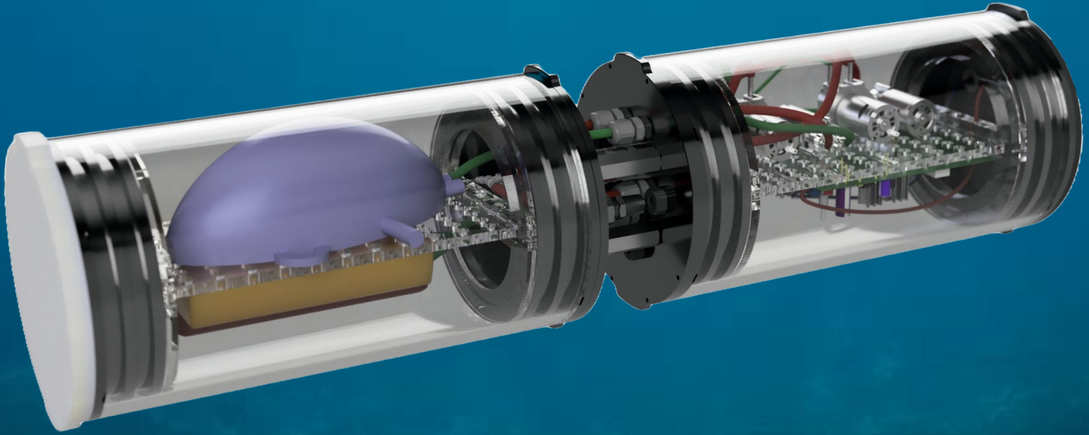
CALIPSO

Chemical **Al**uminum **I**nflation **P**roducing **S**ubmersible **O**scillator



2.014/2.734 Final Presentation

May 9th, 2025

2.014 Team

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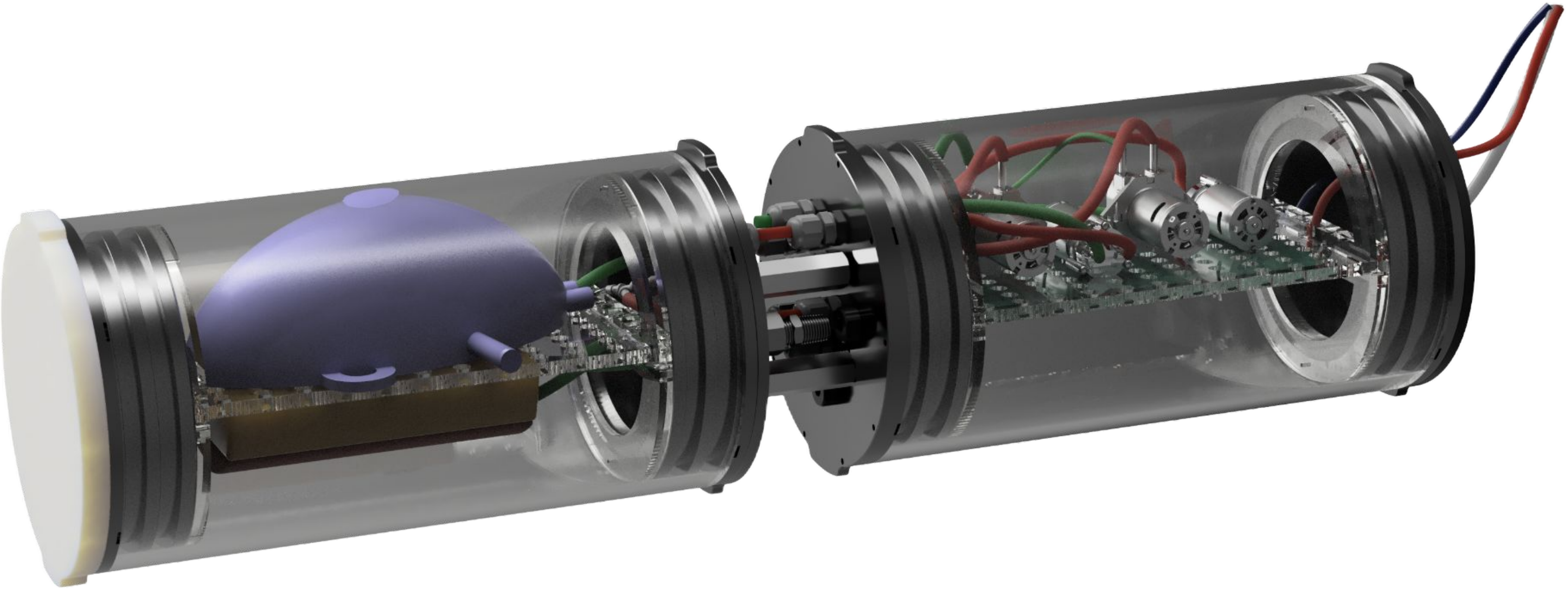
Electronics/Controls

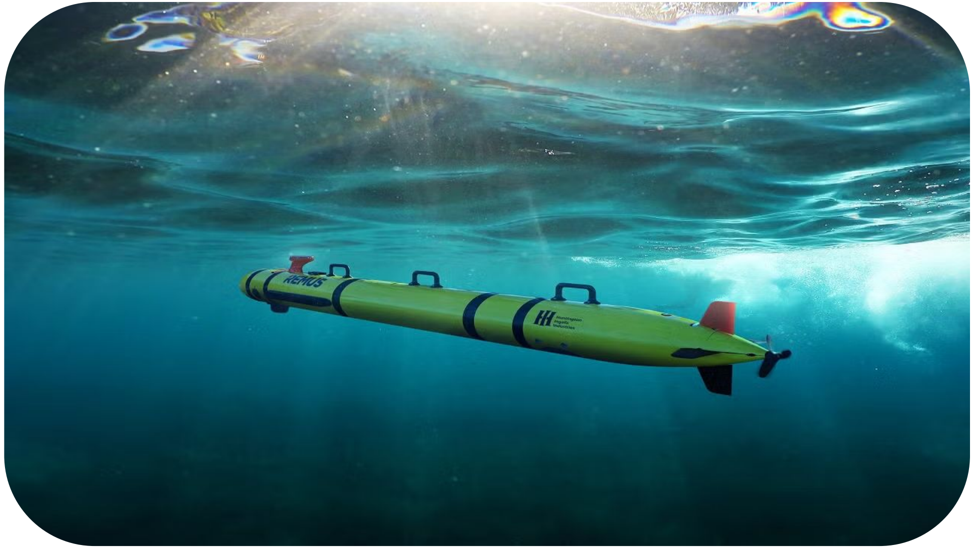
System Snapshot



Unmanned Underwater Vehicles (UUV's)

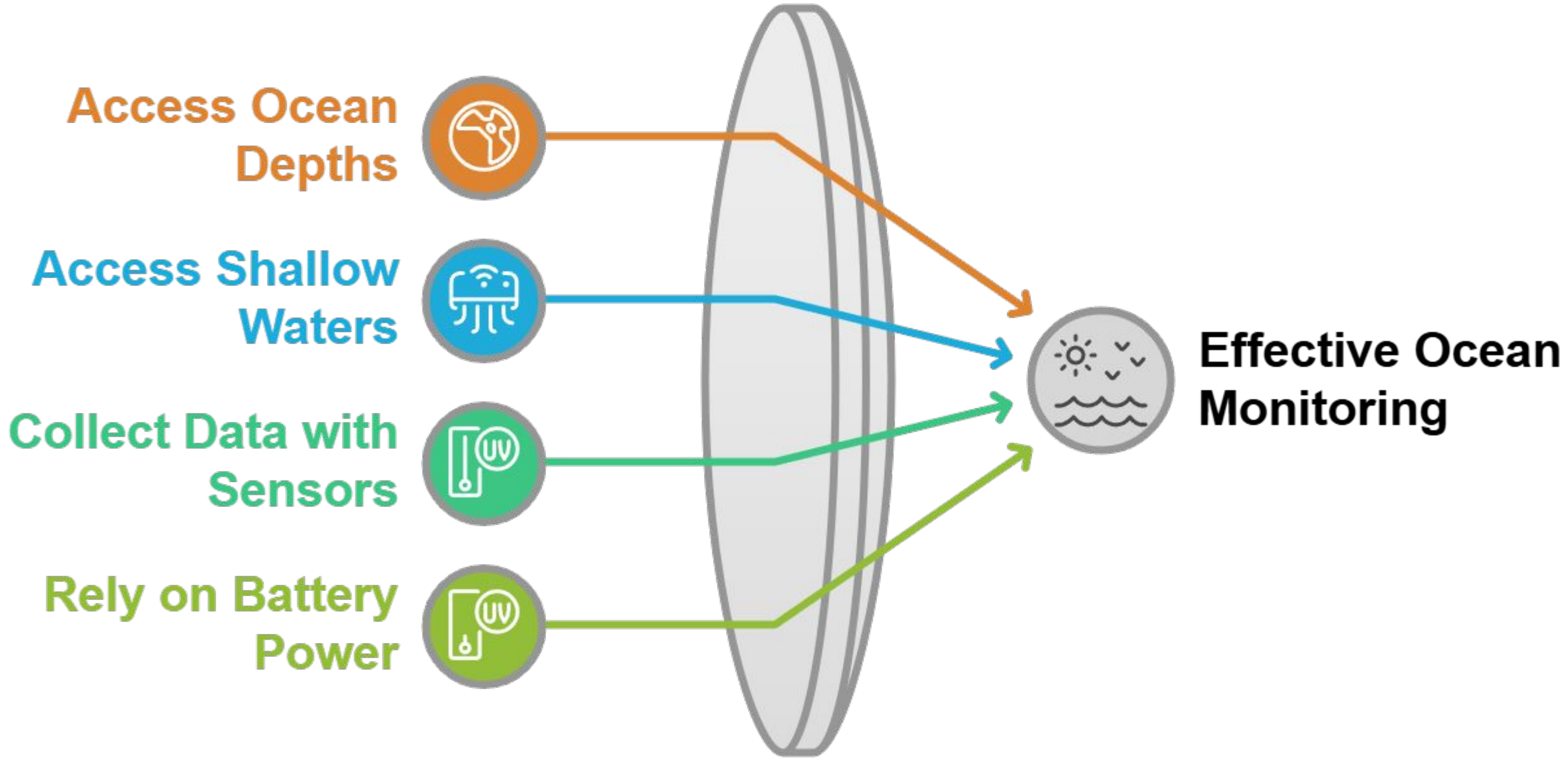
"Only the ocean remains as the **last great unexplored portion of our globe**; so it is to the sea that man must turn to meet the last great challenge of exploration this side of outer space." In Deep Challenge (1966) by H. B. Stewart.

Glider

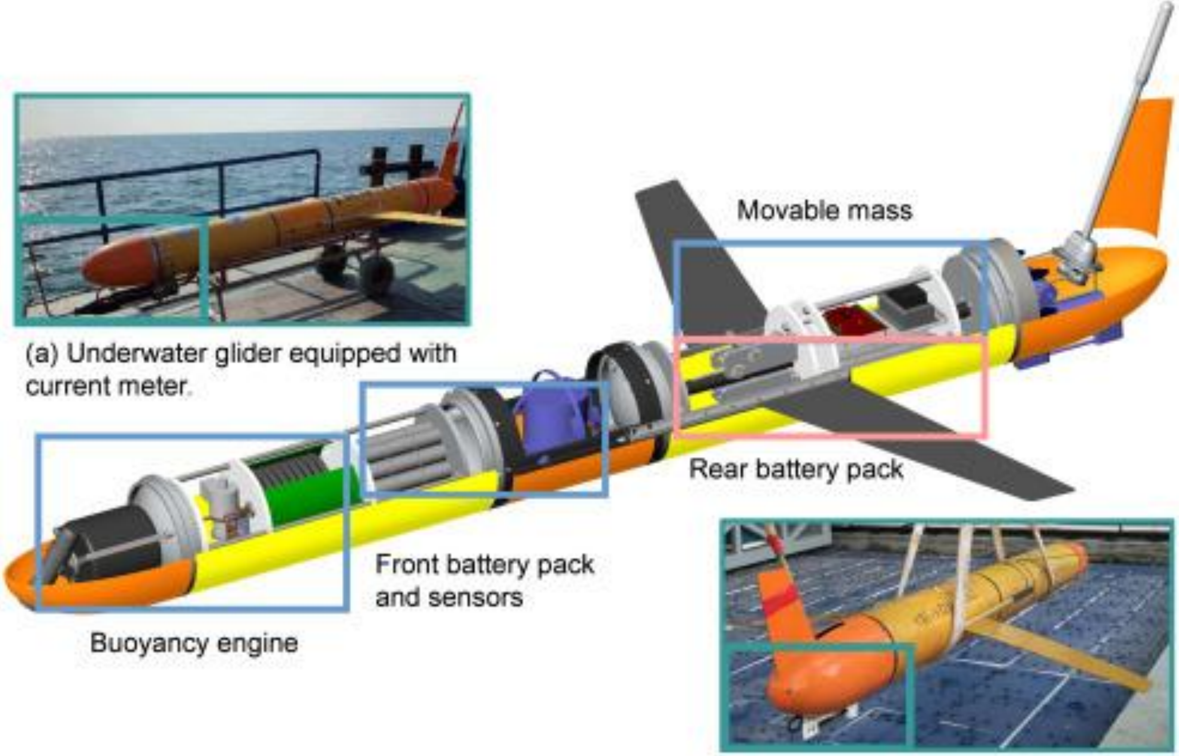


Propeller Driven

Unmanned Underwater Vehicles (UUV's) Use Case



Buoyancy Engines Reduce Power Usage



(a) Underwater glider equipped with current meter.

(b) Sampling sensor mounted on underwater glider.

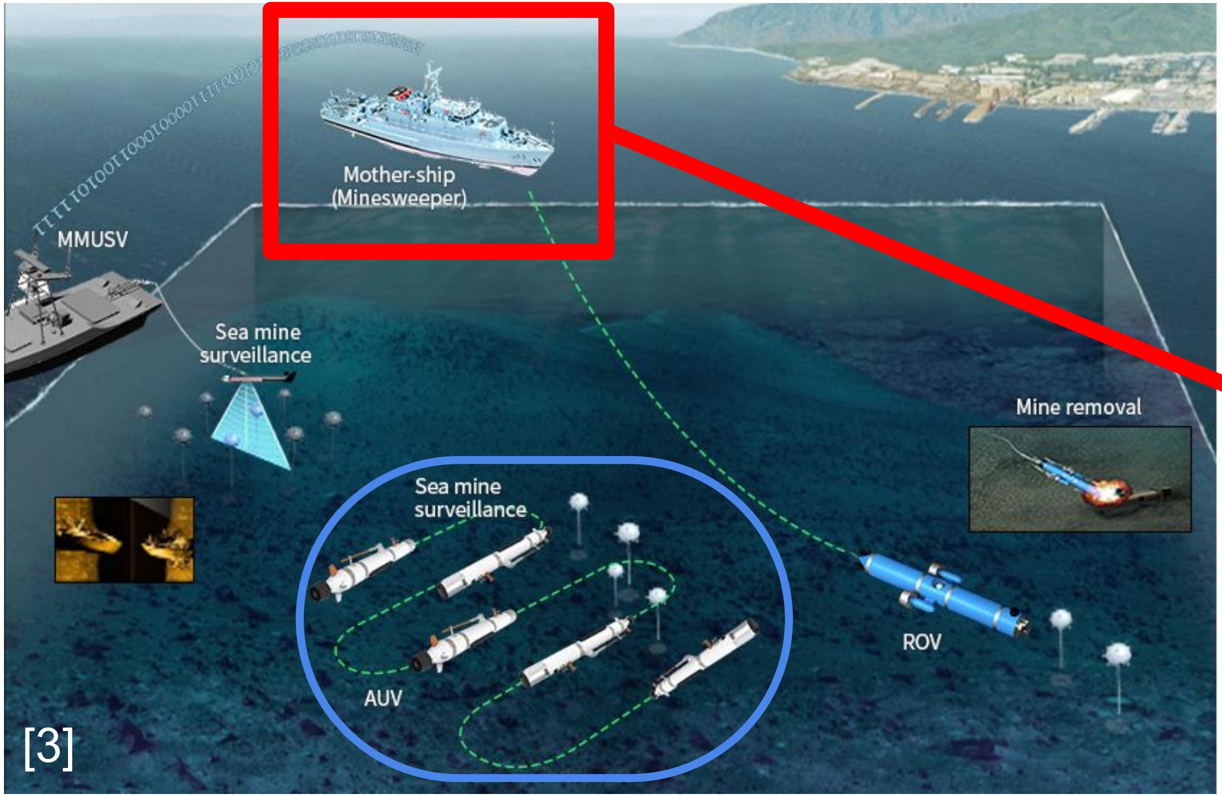
Li-on Battery Range Concerns

~ 3 days of operation for mid-sized AUV's running on Li-Ion batteries [1]

4 - 8 hours to recharge battery [2]

> 10 days of operation for typical underwater missions [1]

Current Range Shortcomings

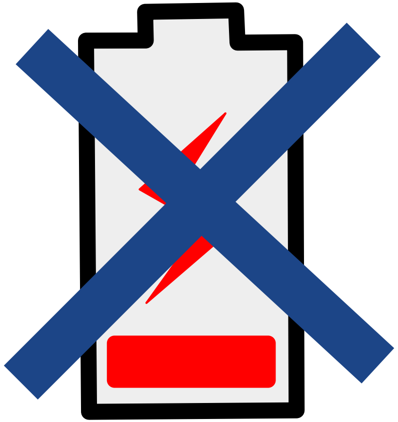


Bluebird

Hanwha Systems

Surface Ships Cost
~$70,000 Per Day

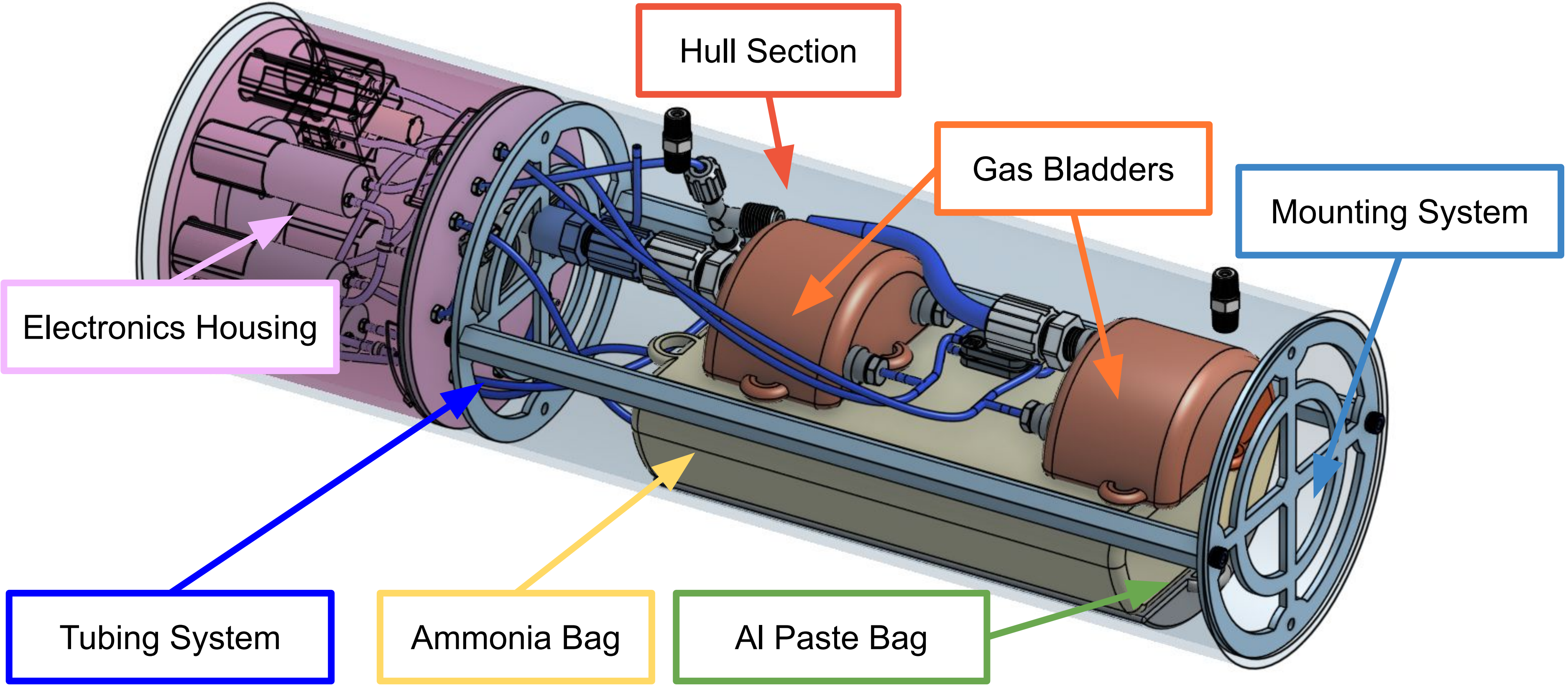
Enhancing UUV Range and Value



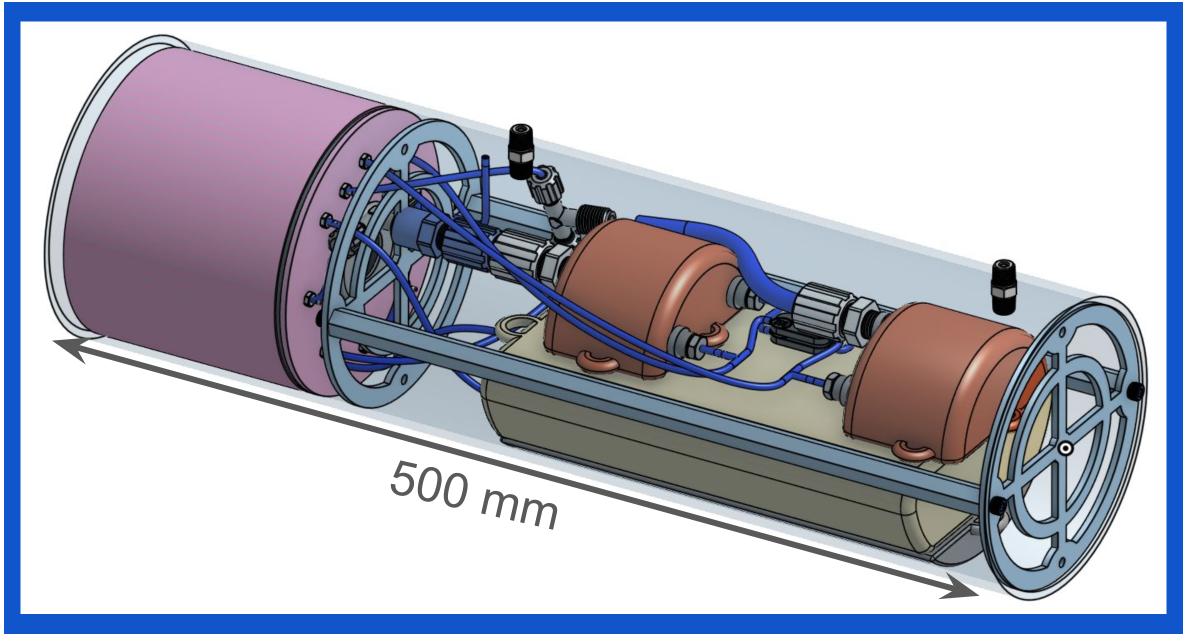
Expensive Surface Ships

Limited-Energy Batteries

Full Scale Design



Last cycle: the concept of a buoyancy engine that runs on Aluminum fuel



Aluminum boasts **42x more energy density** than Lithium-Ion Batteries

Aluminum Fuel Buoyancy Engine

LiOn Batteries

- Long charging times
- Low energy density
- Gradual degradation

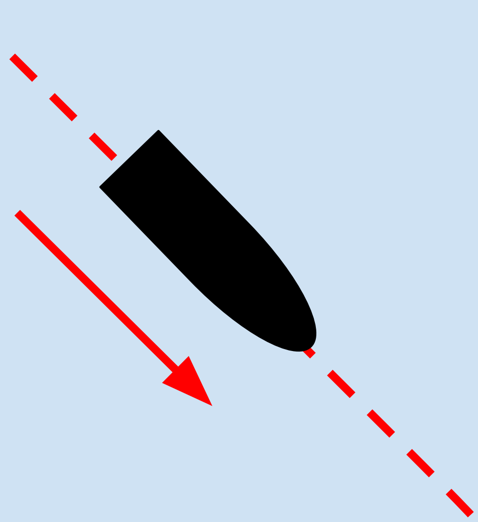
Aluminum Fuel

- Readily available
- High energy density
- Cost effective manufacturing

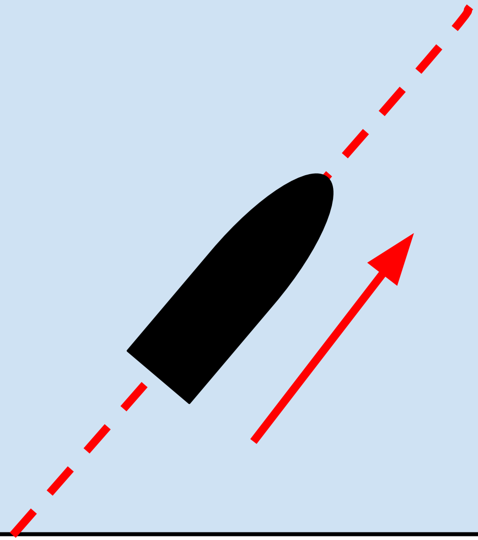
How Buoyancy Engines Typically Work

Displace mass or volume to change the ***<u>net buoyancy</u>*** of a vehicle

$F_b < F_g \rightarrow$ UUV Moves DOWN

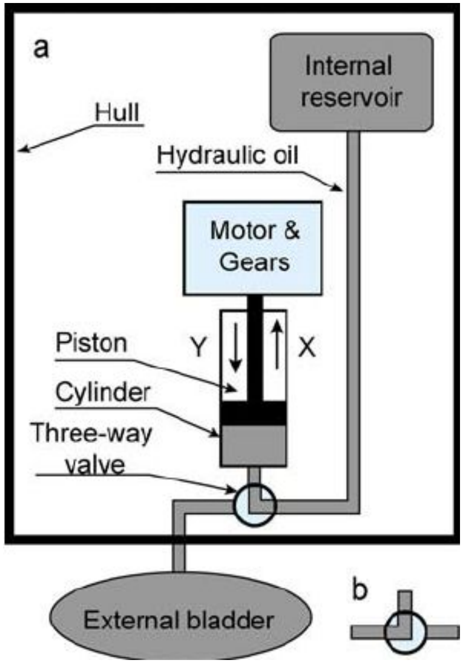


$F_b > F_g \rightarrow$ UUV Moves UP

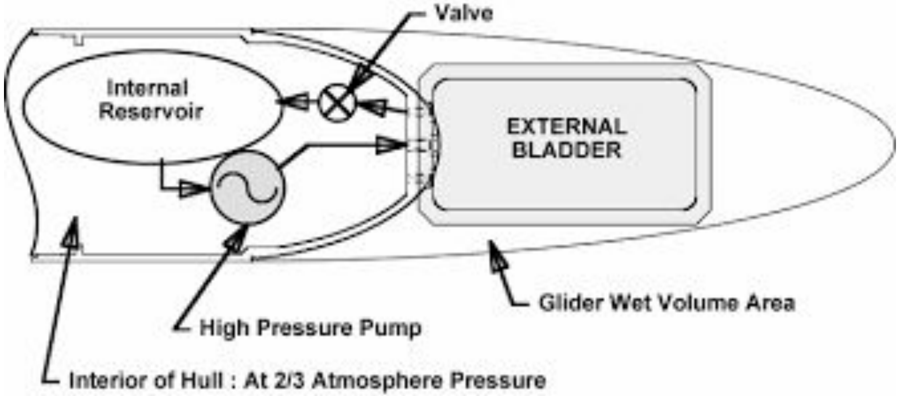


Workings of current buoyancy engines

Motors & Gears



Pumps



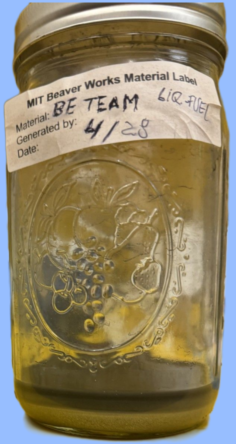
Al Reaction with Ammonia

$$2Al + 6H_2O \rightarrow 3H_2 + 2Al(OH)_3 + Q$$

$$NH_3 + H_2O \leftrightarrow NH_4^+ + OH^-$$

Reactants

Aluminum Paste



+

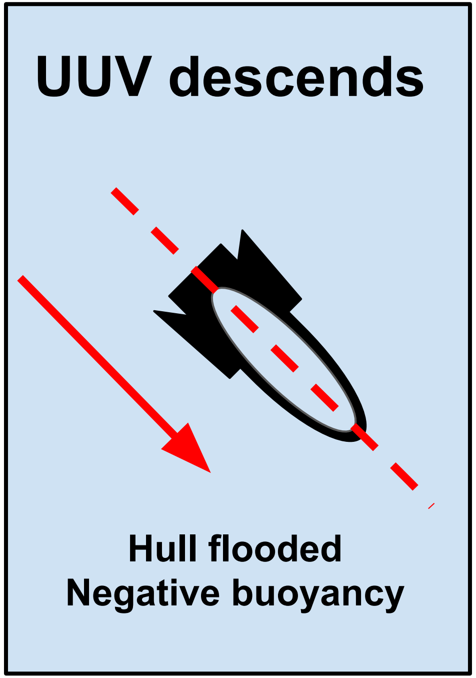
20% Aqueous Ammonia

Products

- Hydrogen Gas
- Ammonia Gas
- Steam
- Solid Precipitates

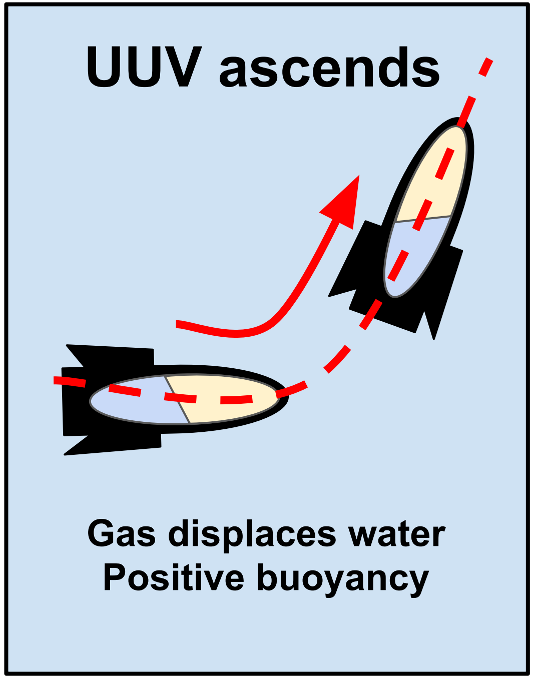
How we change buoyancy

Displace water volume to change the **net buoyancy** of a vehicle

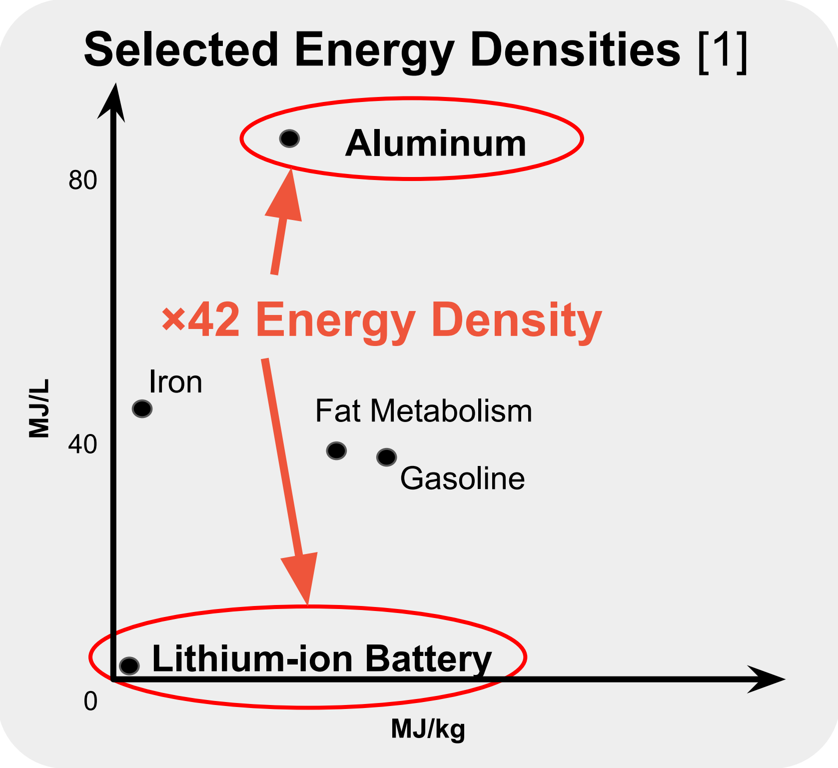


Start Chemical Reaction

Reactants: Al (Aluminum) + H_2O (Aqueous Ammonia) → Gas bubble: H_2 (Hydrogen Gas) + Q (Water & Ammonia Vapor)



Energy Density



Energy Losses

-50% due to not burning hydrogen

-25% due to compressibility

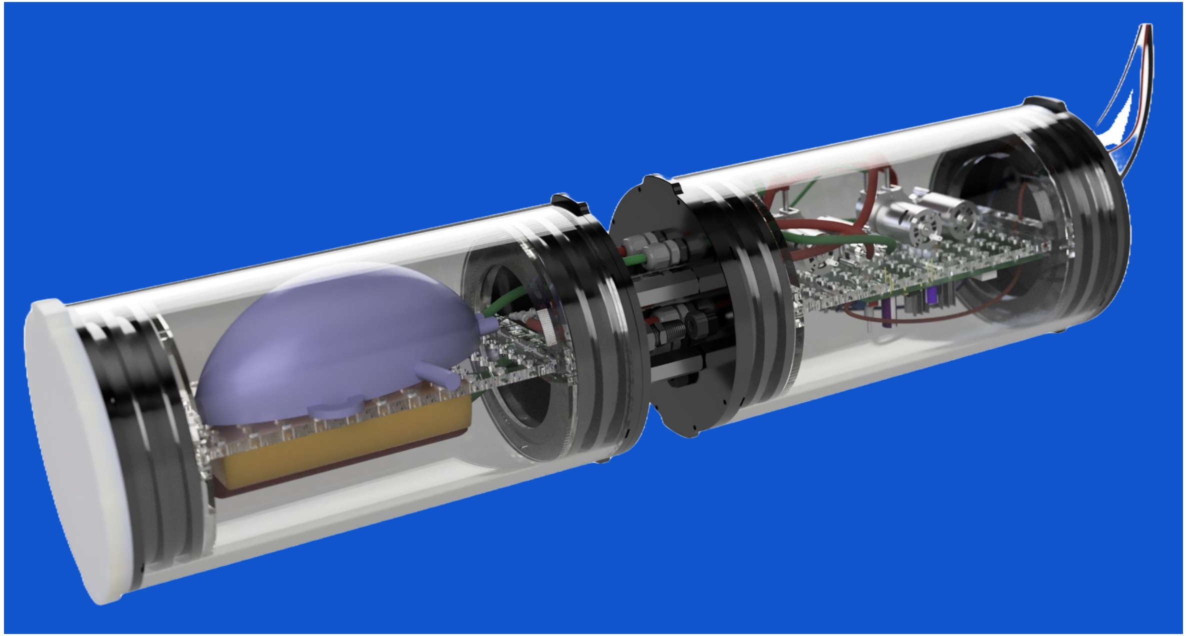
25% Energy Efficient
10x better than batteries

Challenges of Fully Scope Engine

- Dangerous
- High Cost
- Complicated to implement
- Hard to test
- Difficult to rapid prototype

	Requirement	Value
1	Operational Depth	500 m
2	Operational Pressure Range	1 - 50 atm
3	Operational Temperature Range	0 - 49° C
4	Vehicle Compatibility	< 0.15 m
6	Buoyancy Displacement	≥ 350 mL
7	Fine Buoyancy and Moment Control	Yes

Our target this semester: A Simplified Proof of Concept Setup



De-risking the technology through functional requirement simplification

Functional Requirement Evolution

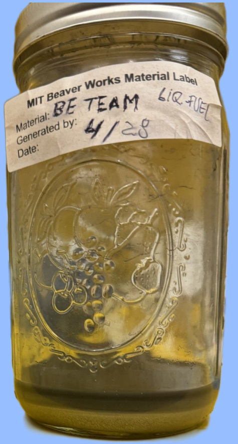
	Requirement	Description	Old Value	New value
1	**Operational Depth**	Depth to target for dive & return	**500 m**	**5m**
2	**Operational Pressure Range**	Pressure range of operation	**1 - 50 atm**	**1 - 1.5 atm**
3	**Operational Temperature Range**	Temperature range of operation	**0 - 49º C**	**15 - 49º C**
4	**Fine Buoyancy Control**	Precise control over depth shifts	**Yes**	**No**
5	**Moment Control**	Limit shifts in CG relative to CB	**Yes**	**No**
6	**Buoyancy Displacement**	Min. volume of water that engine can displace	**≥ 350 mL**	**≥ 350 mL**

Al Reaction

$$2Al + 6H_2O \rightarrow 3H_2 + 2Al(OH)_3 + Q$$

Reactants

Aluminum Paste



+

DI Water

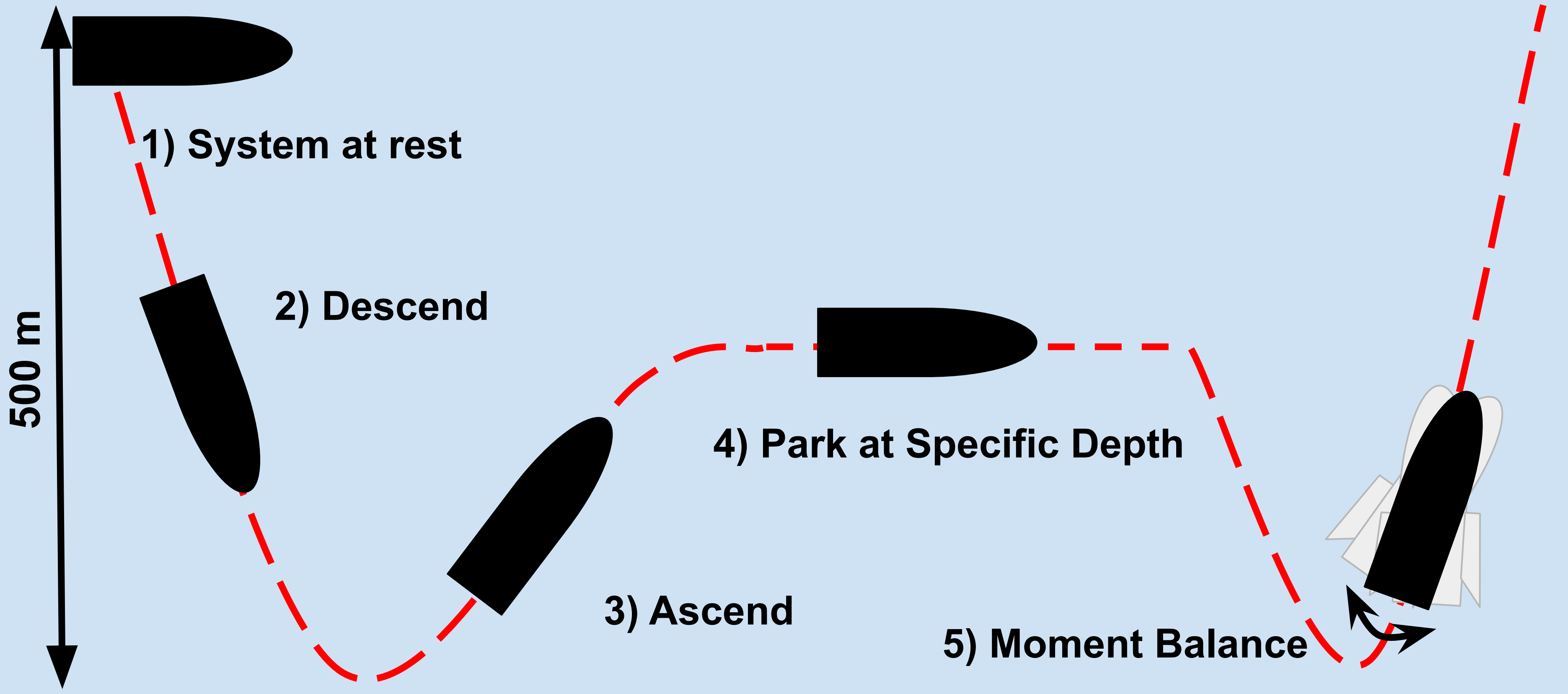
Products

Hydrogen Gas

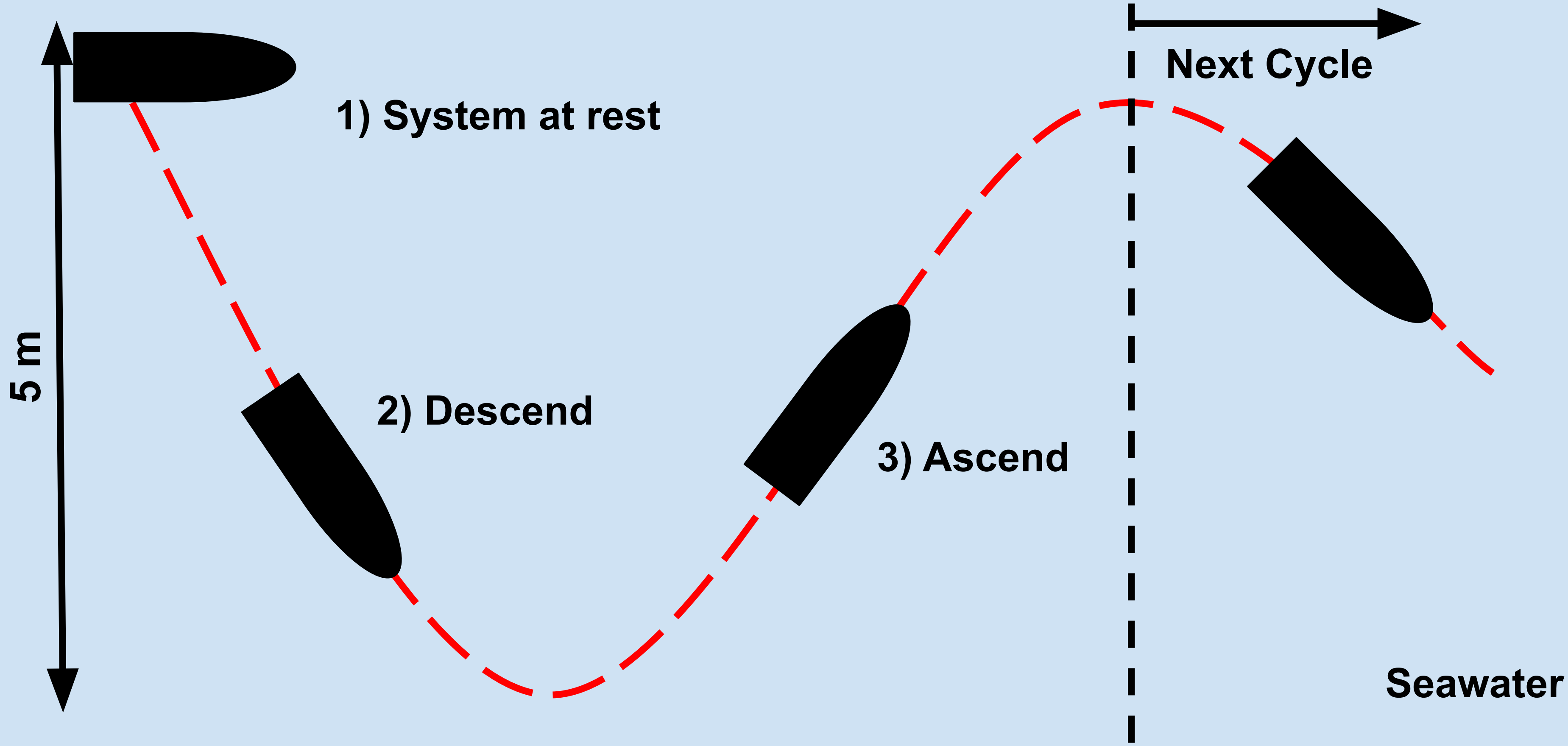
Steam

Solid Precipitates

Full Scope Dive Cycle

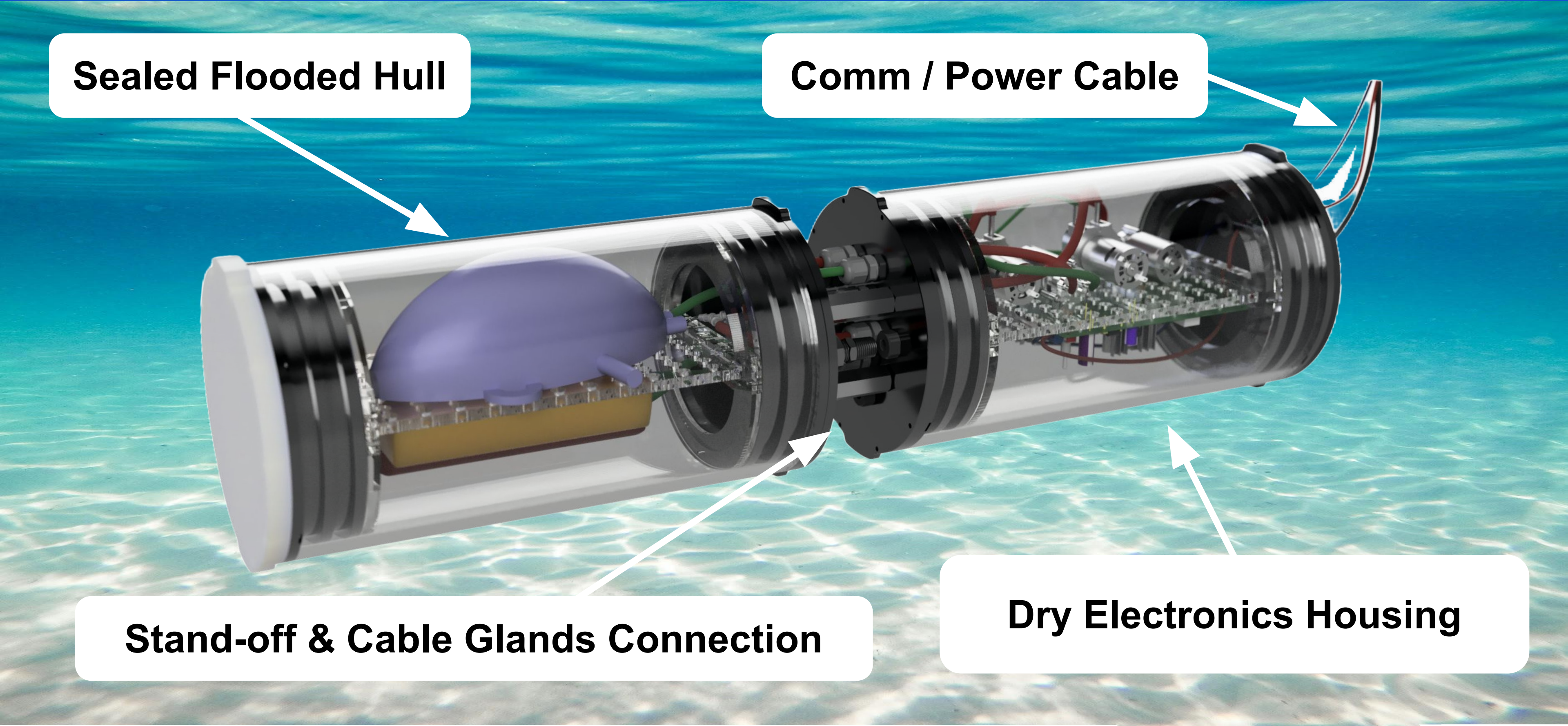


Prototype Dive Cycle

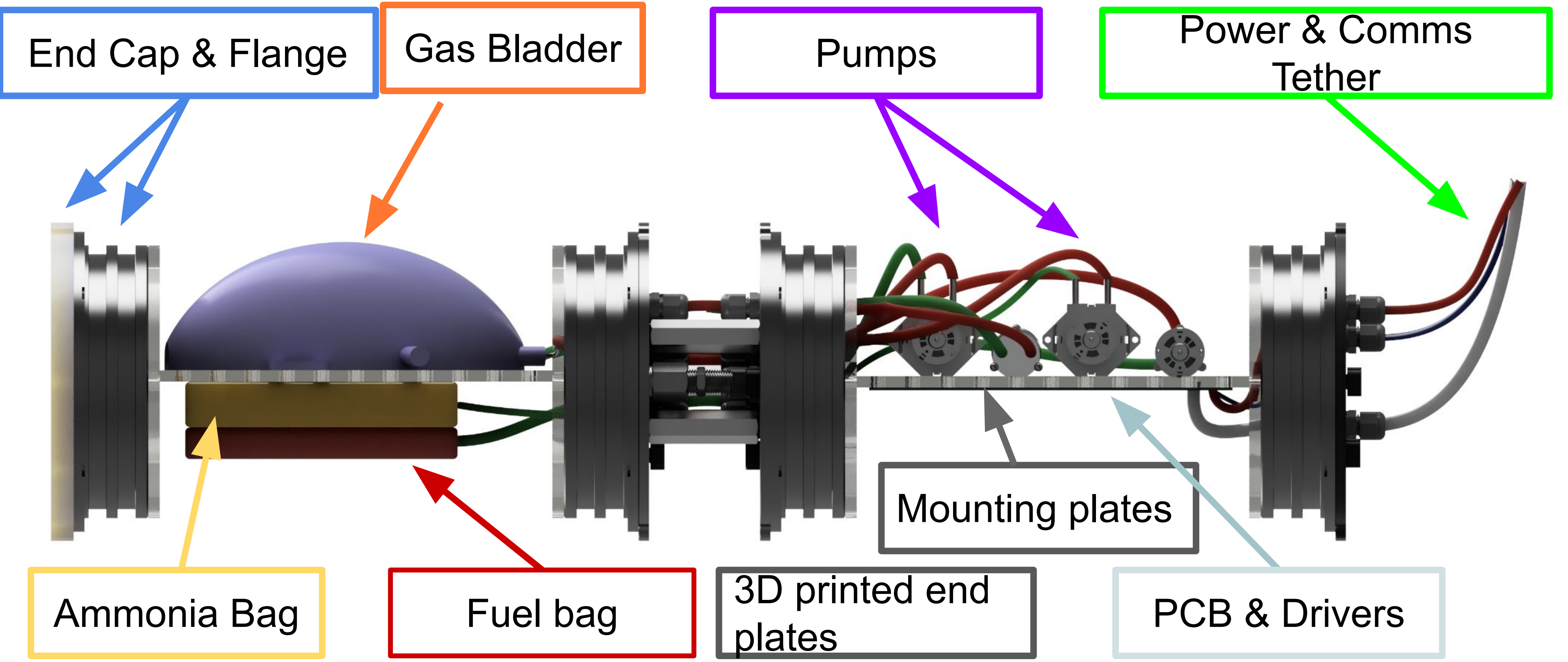


Our Proof of Concept:
CALIPSO

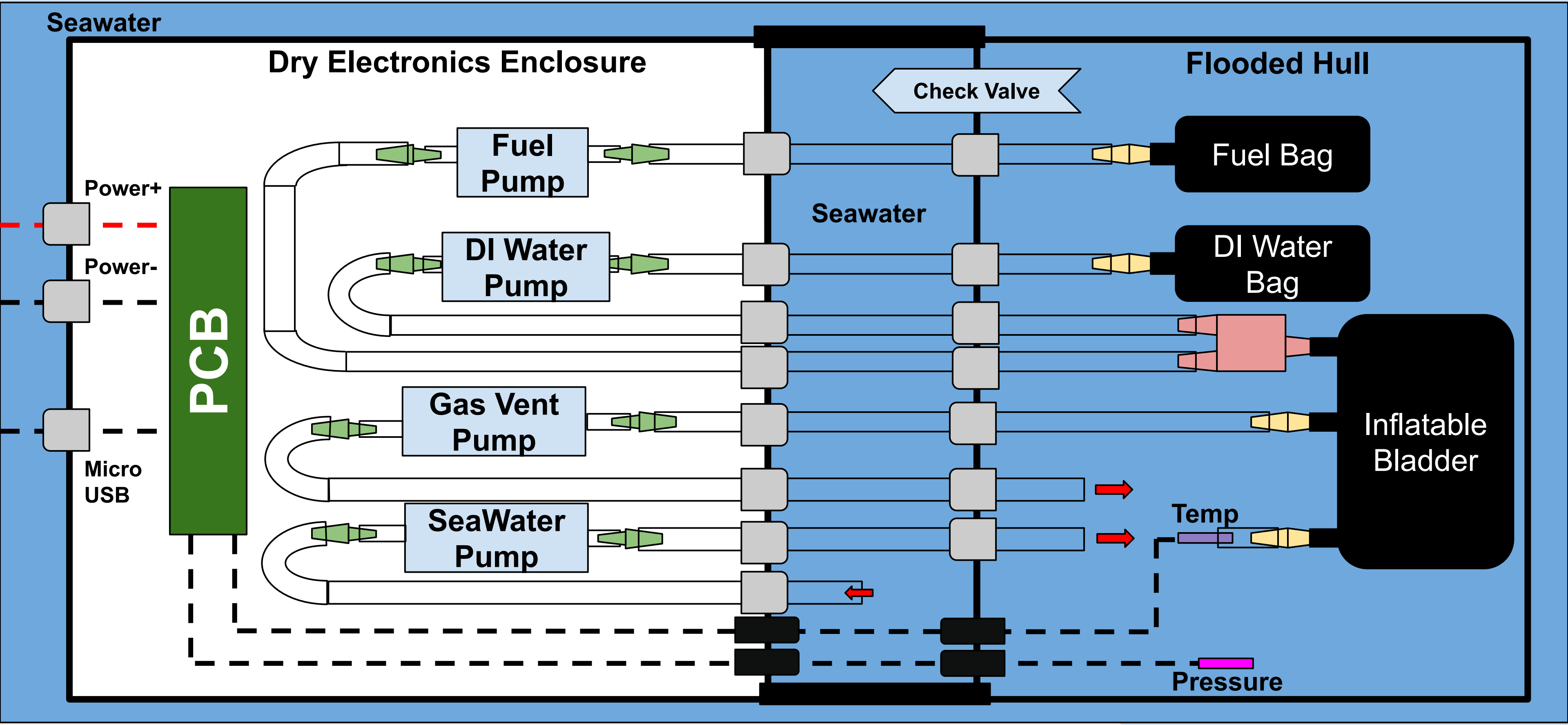
System Overview



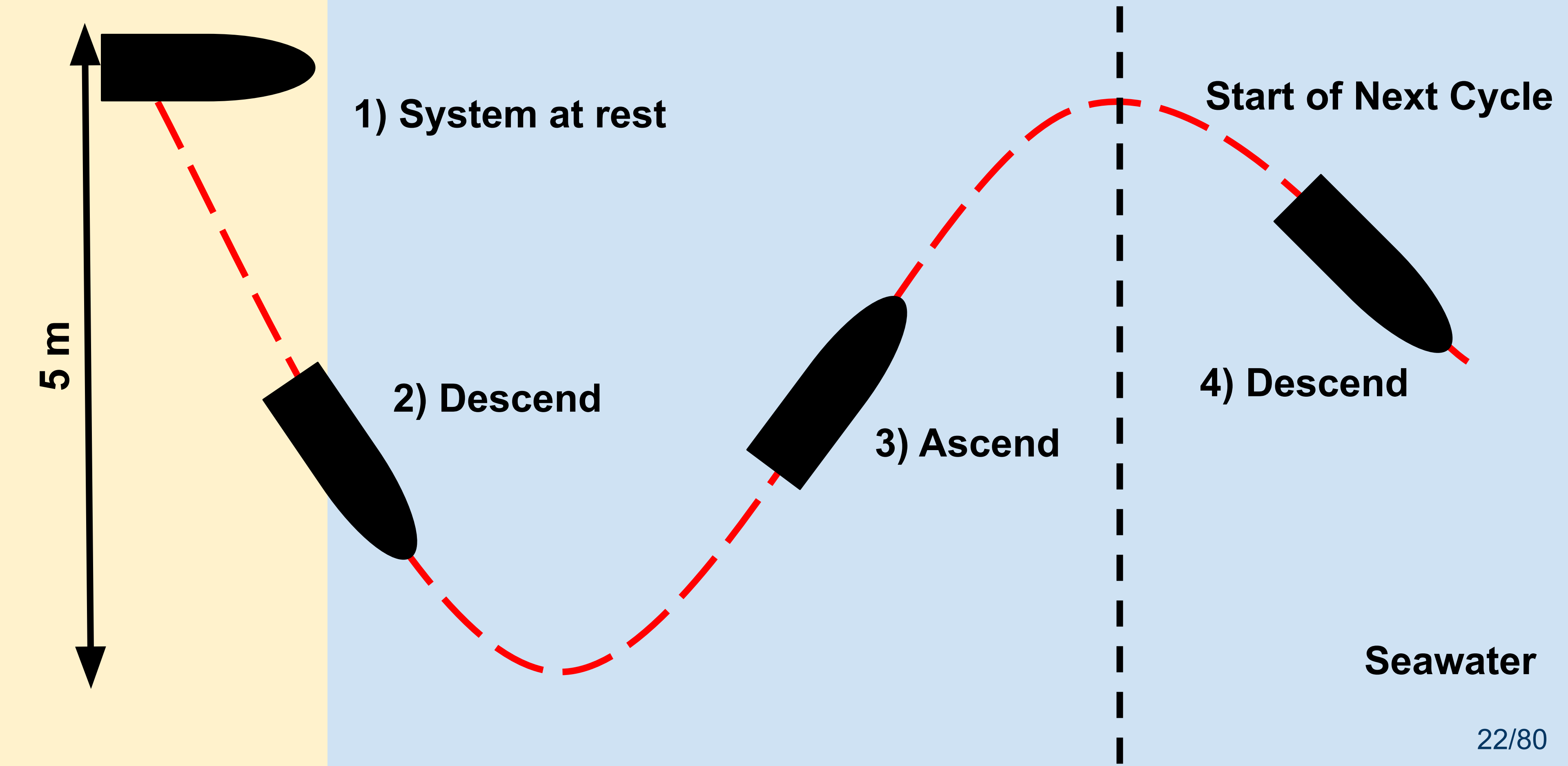
Current Housing Layout



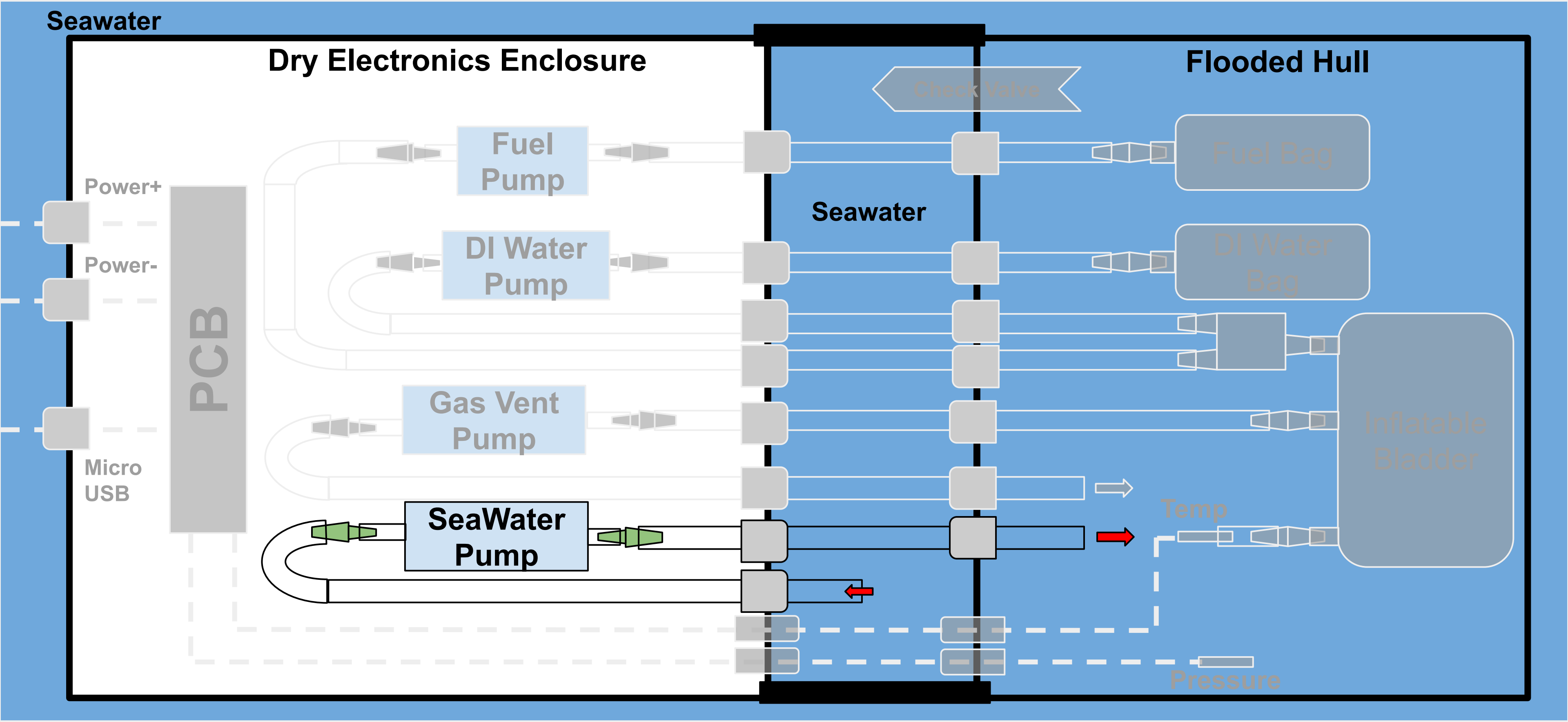
Complete Interface Diagram



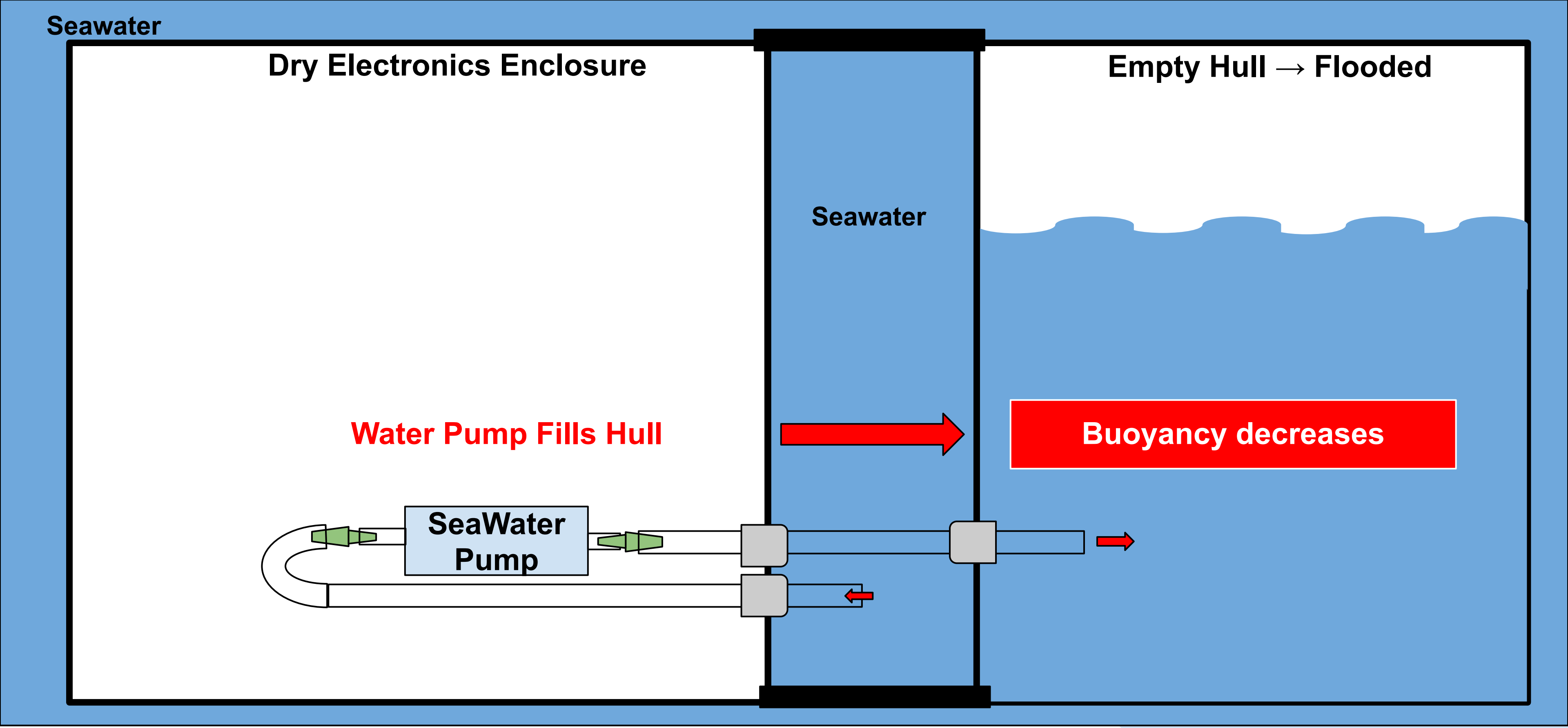
Prototype Dive Cycle



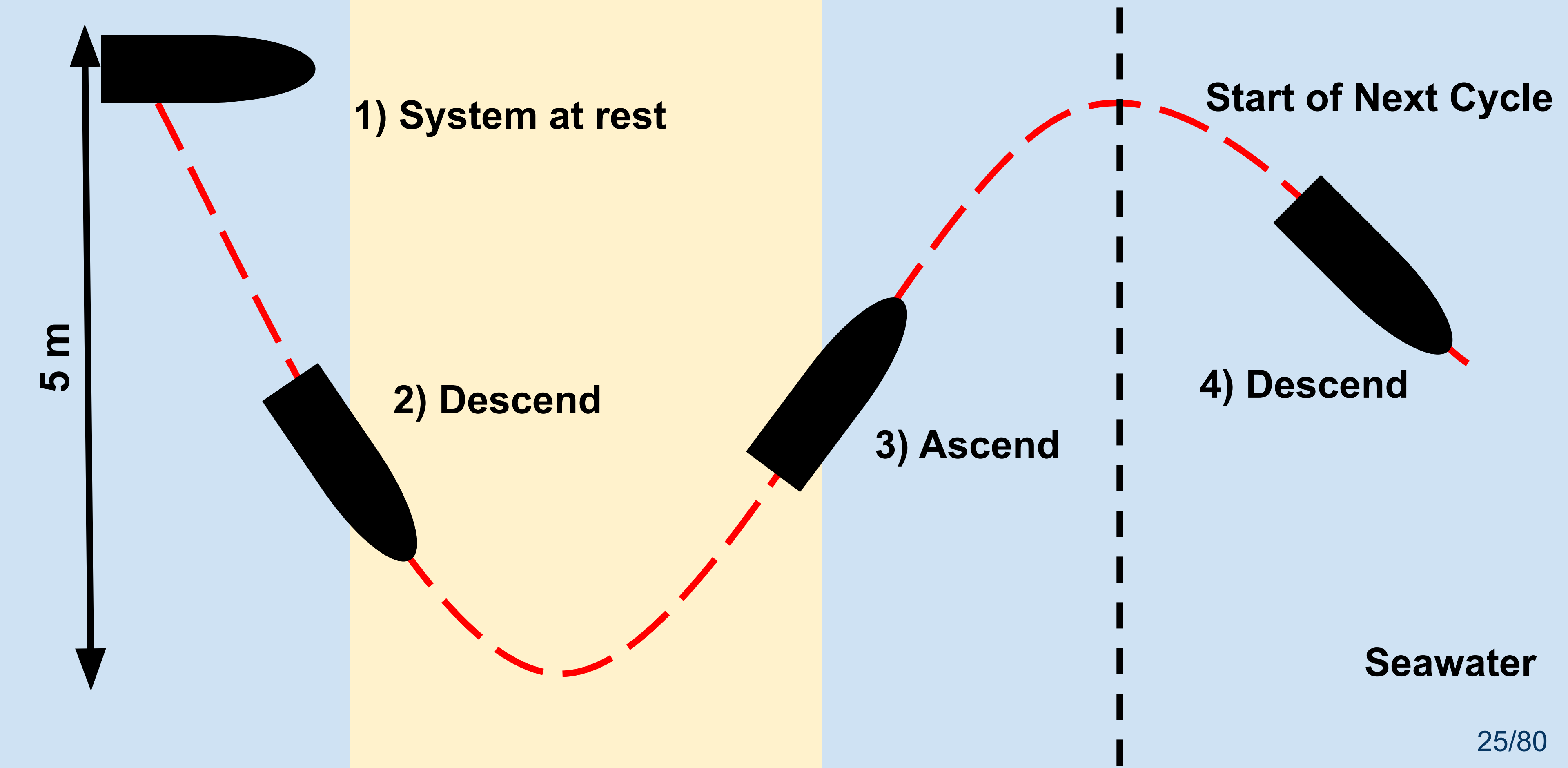
System at rest → Descend



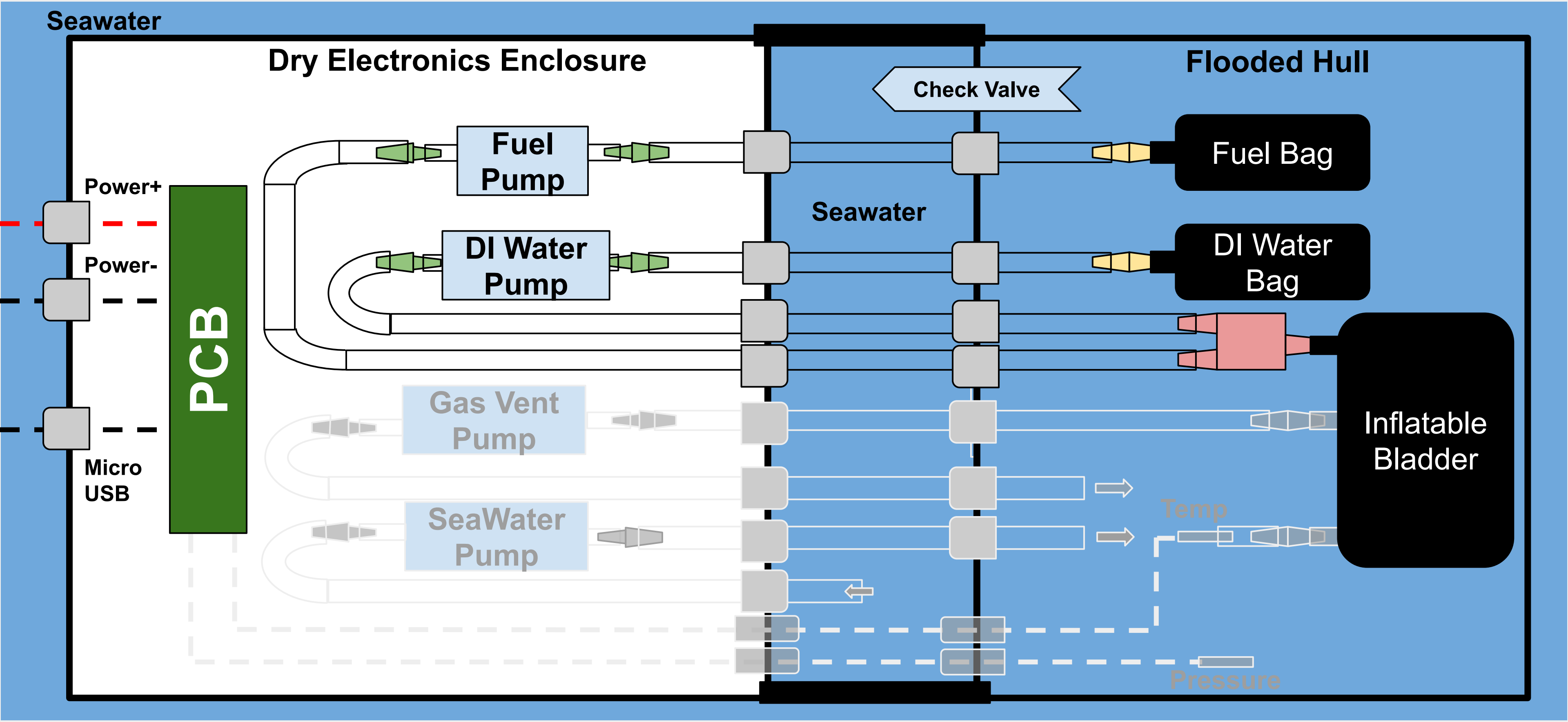
System at rest → Descend



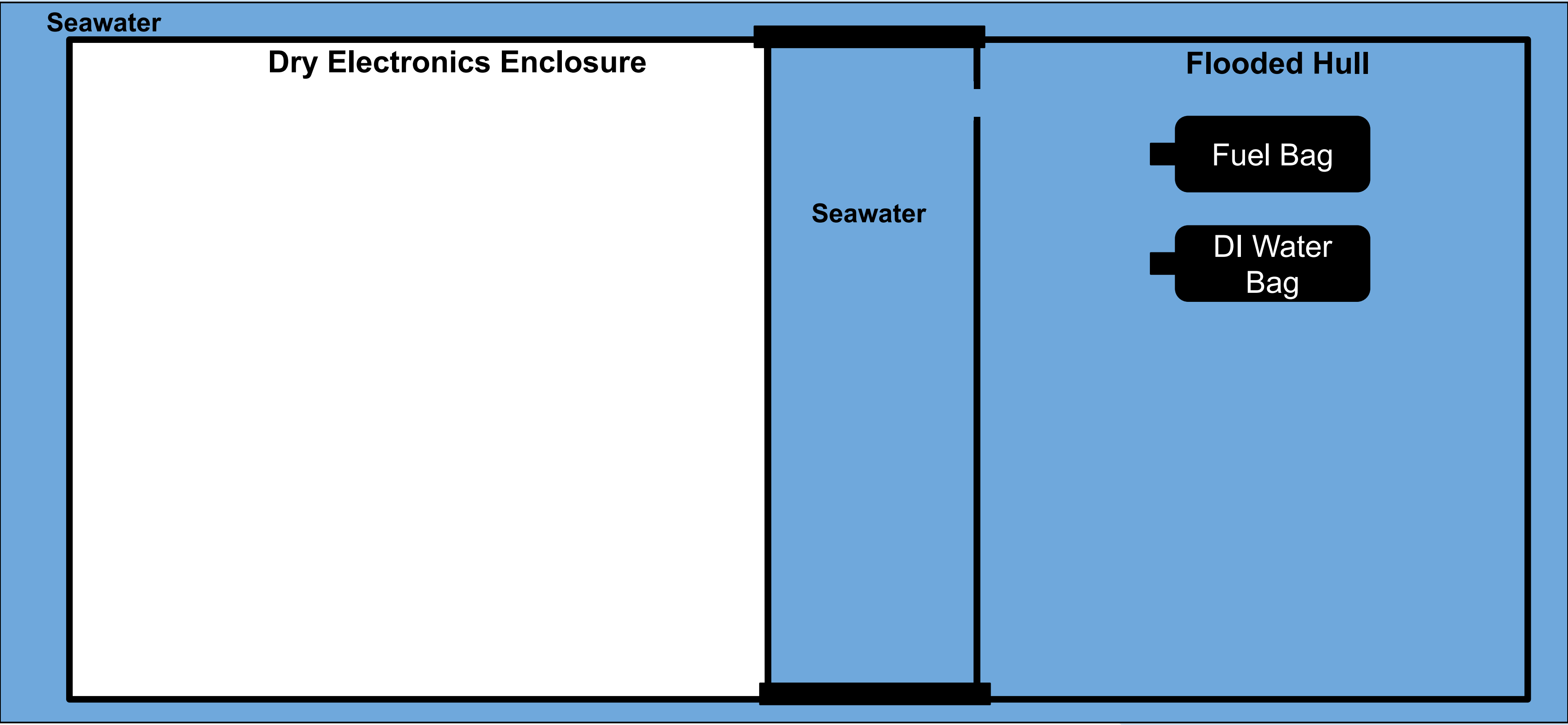
Prototype Dive Cycle



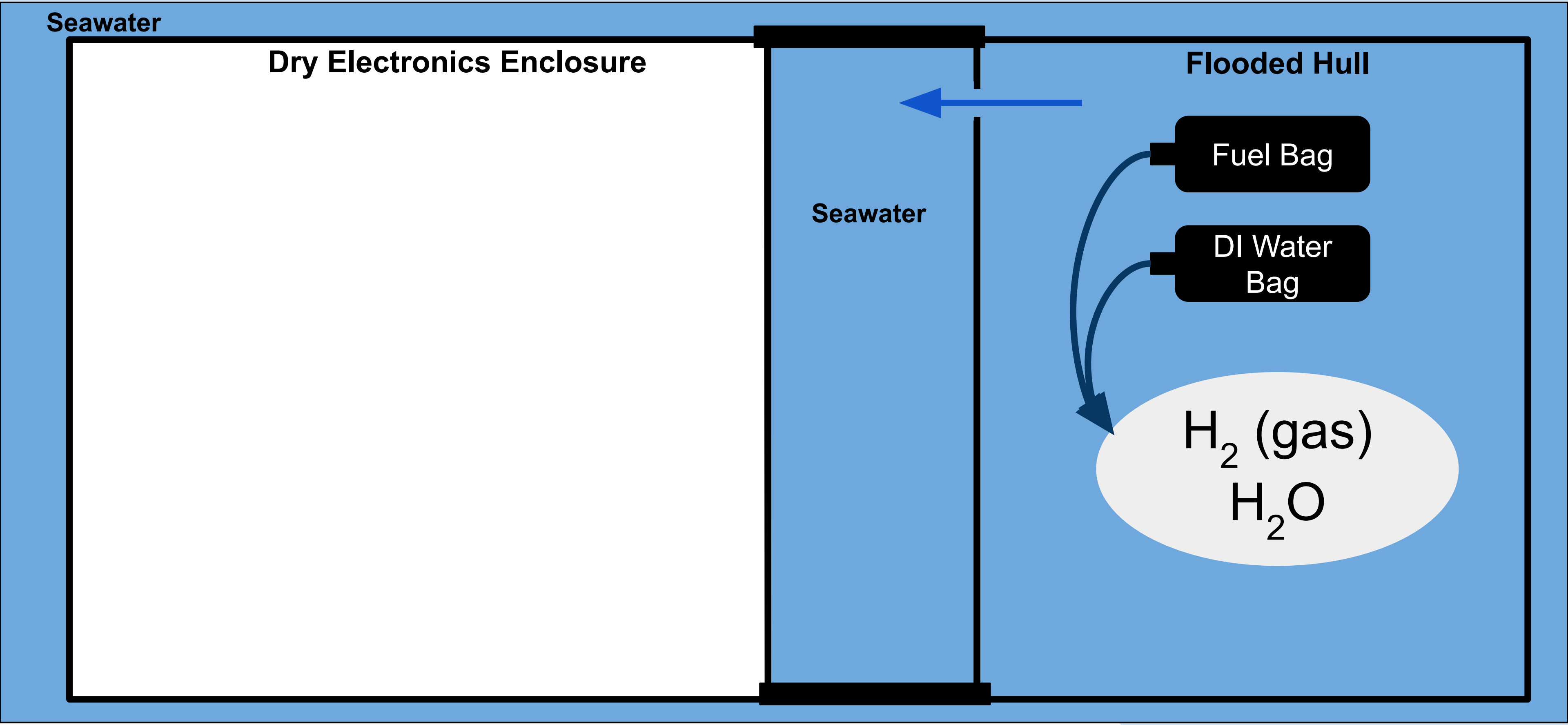
Ascend


Seawater
Dry Electronics Enclosure
Flooded Hull
Check Valve
Fuel Pump
Fuel Bag
DI Water Pump
DI Water Bag
Seawater
Power+
Power-
Micro USB
PCB
Gas Vent Pump
SeaWater Pump
Inflatable Bladder
Temp
Pressure

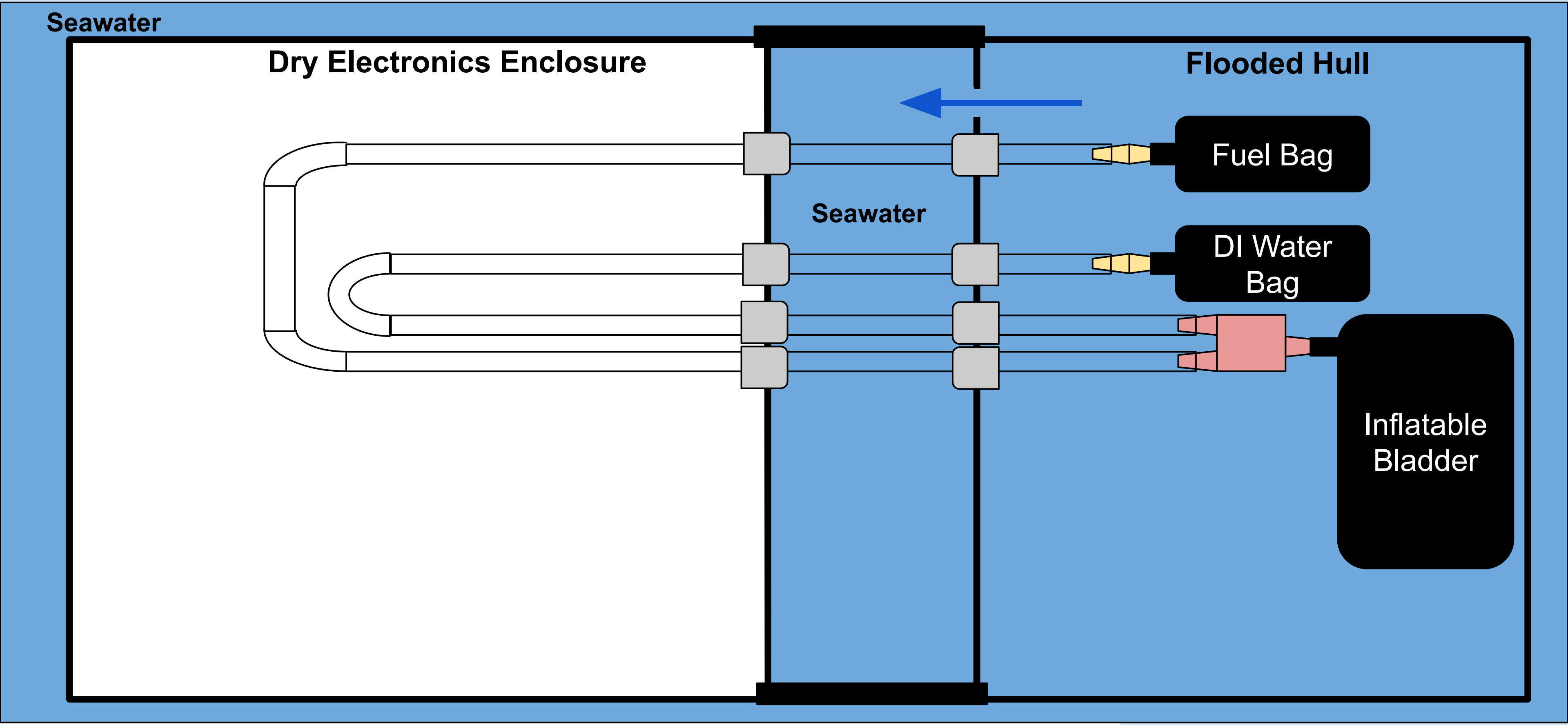
Ascend: Removing Water From Hull



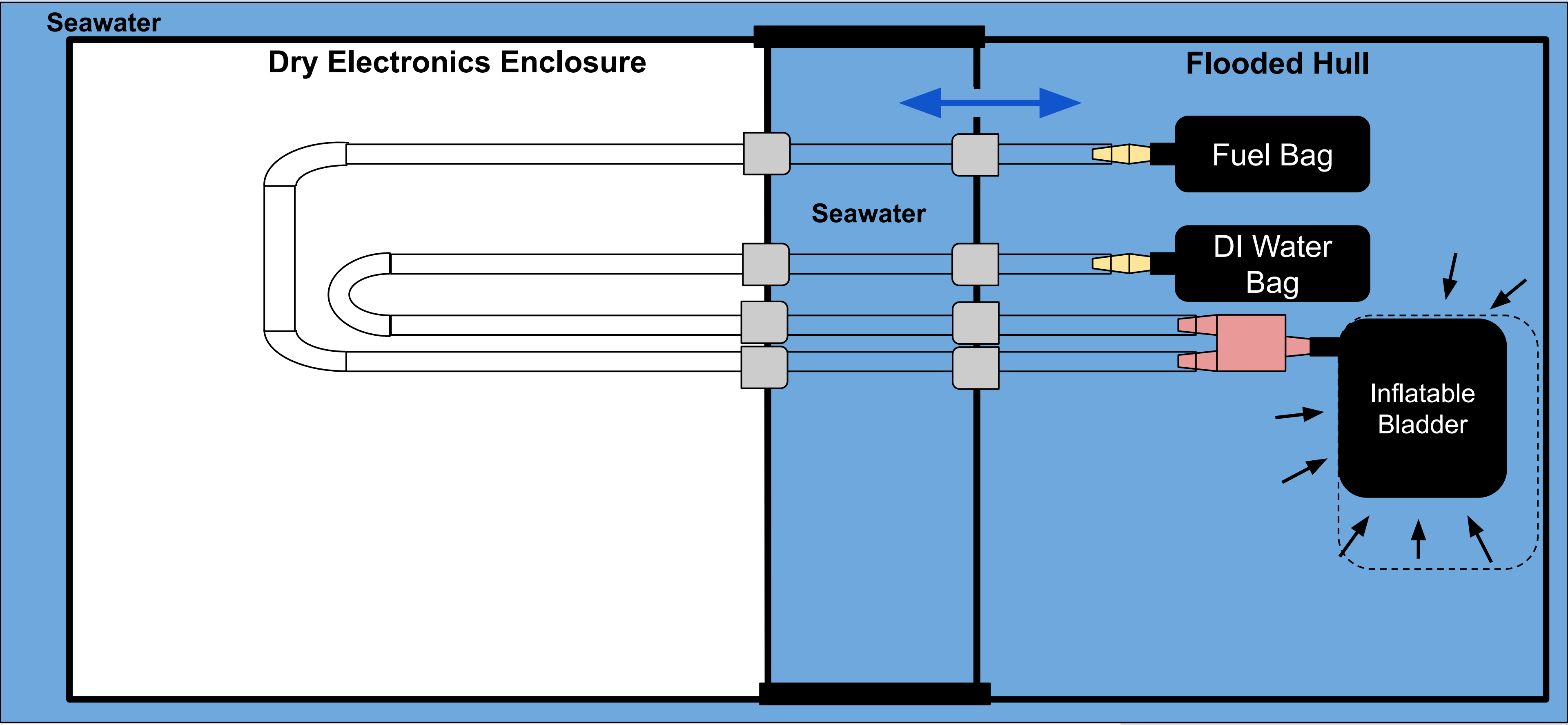
Ascend: Removing Water From Hull



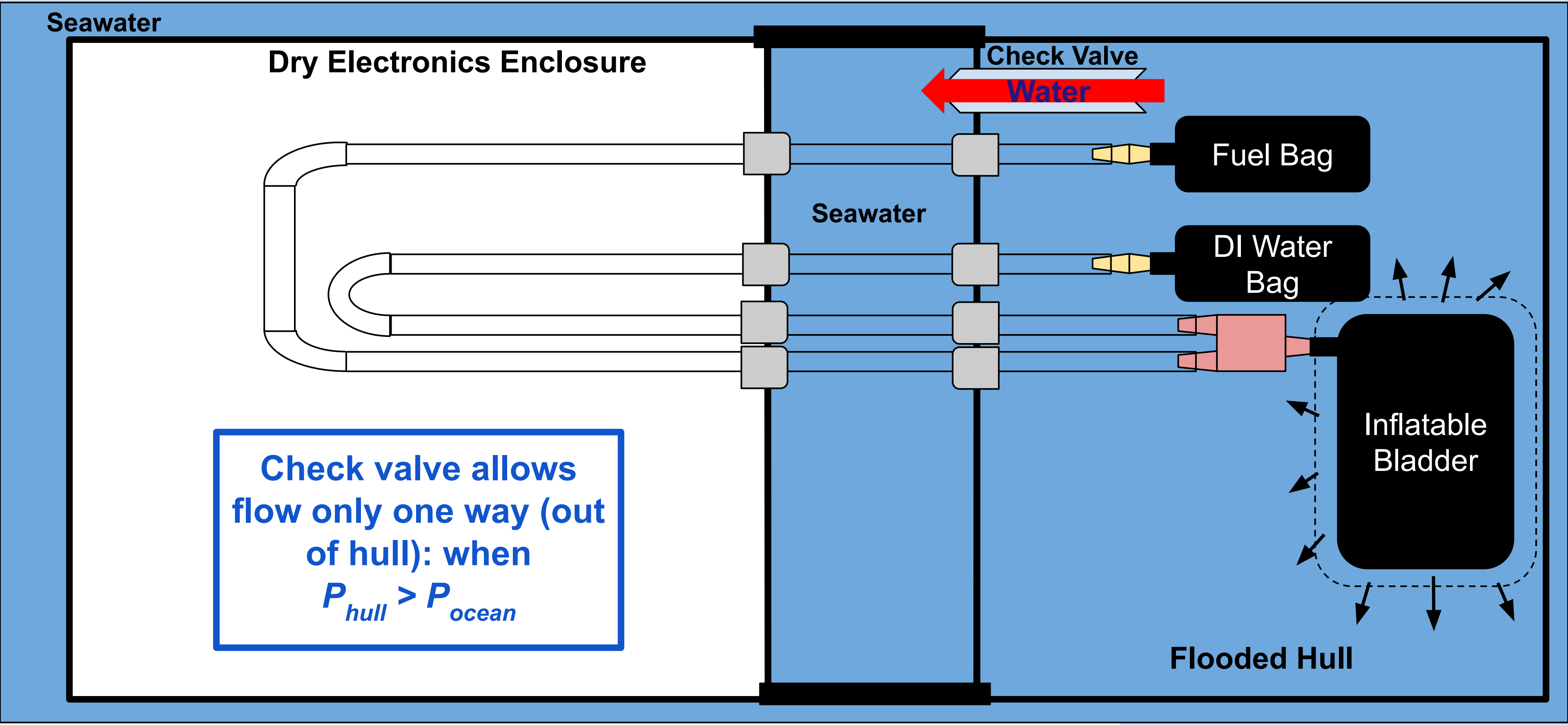
Ascend: Containing Gas



Ascend: Prevent volume loss from cooling

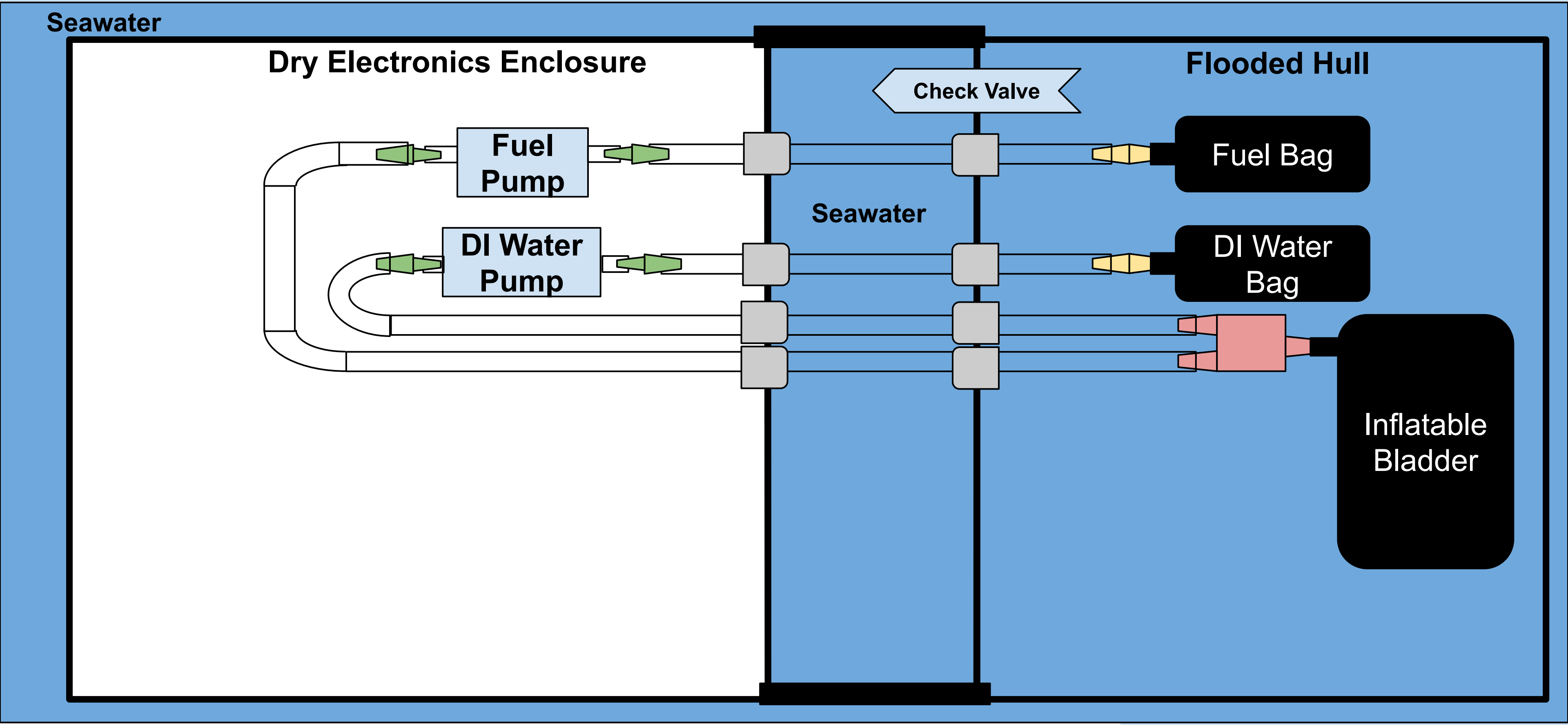


Ascend: Prevent volume loss from cooling

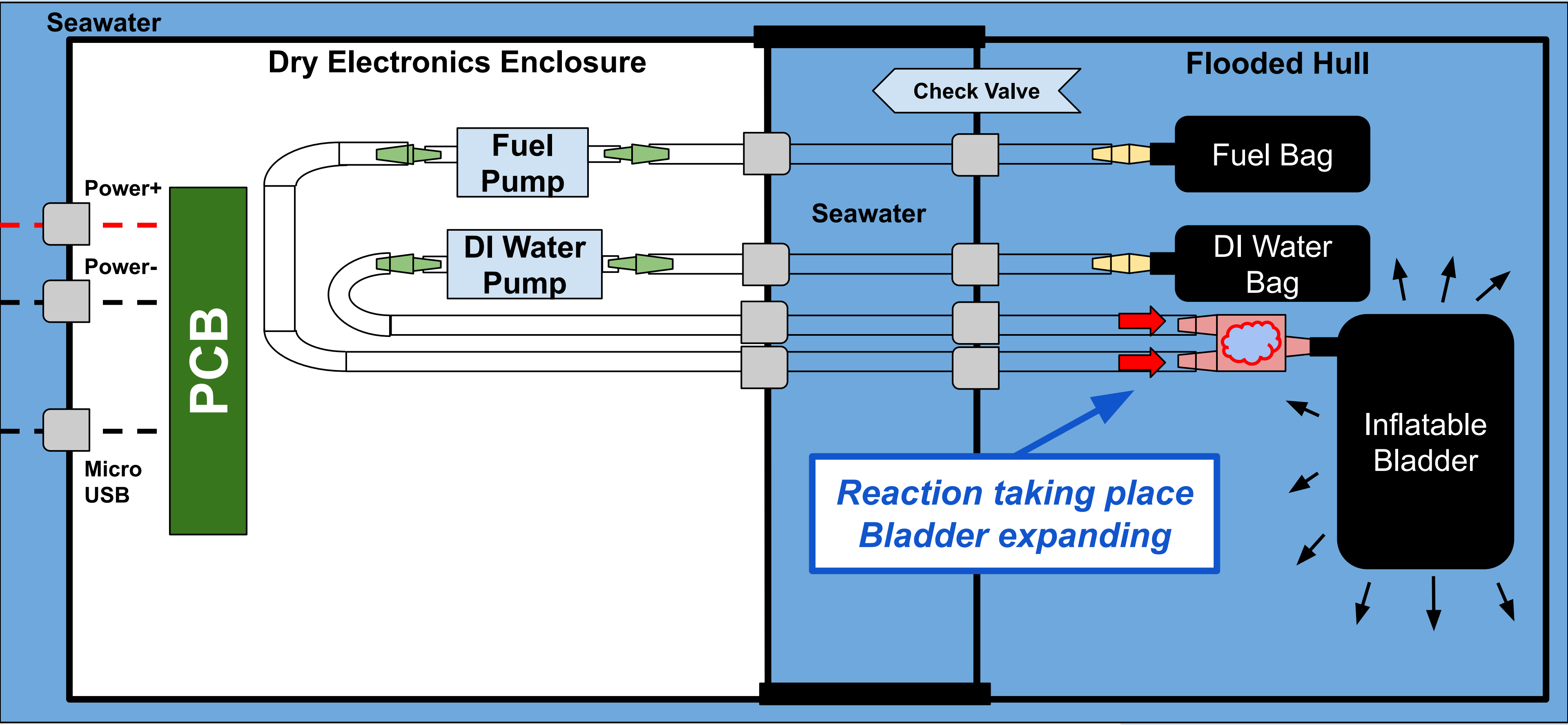


Check valve allows flow only one way (out of hull): when $P_{hull} > P_{ocean}$

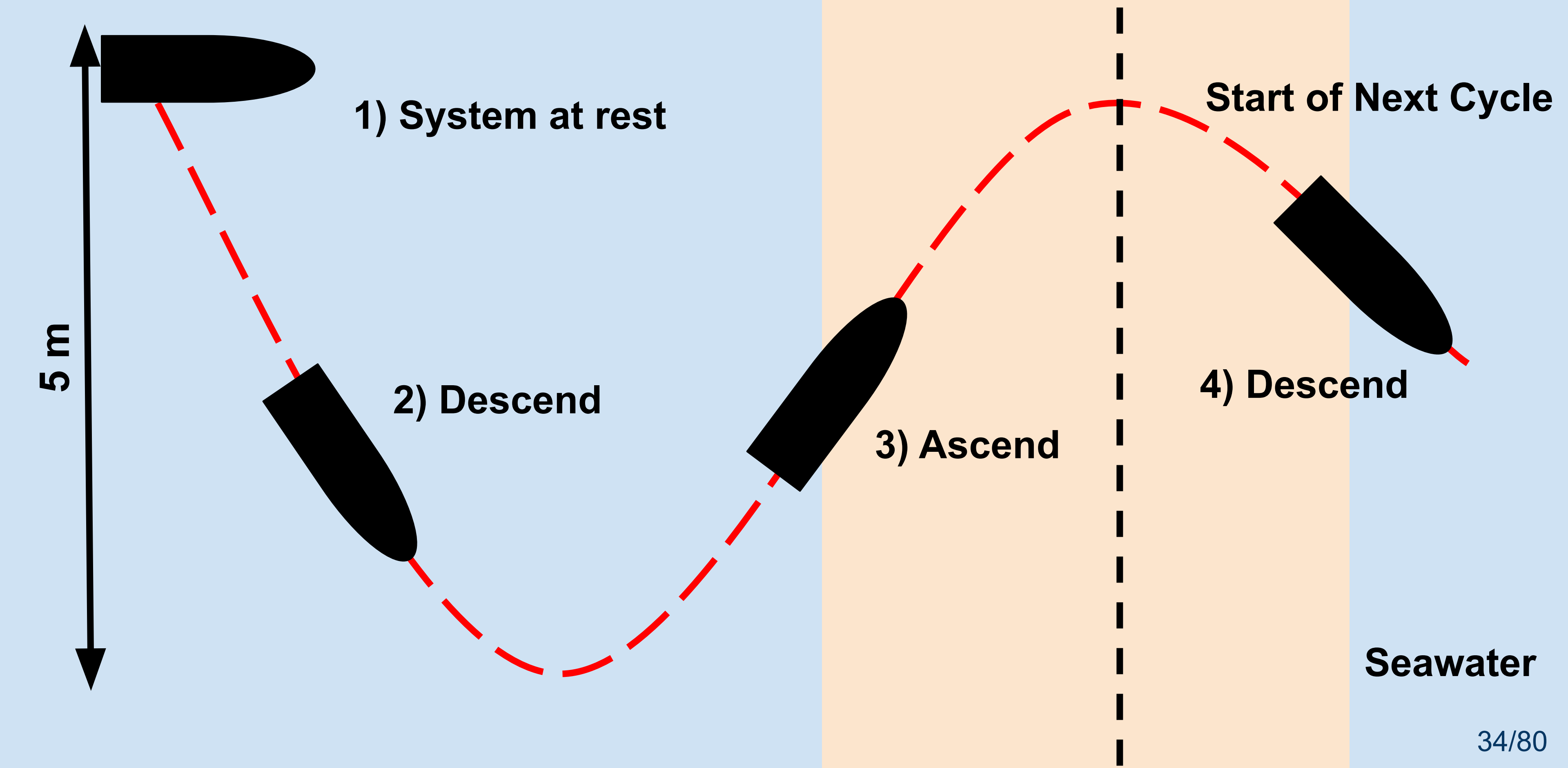
Ascend: Reaction Control



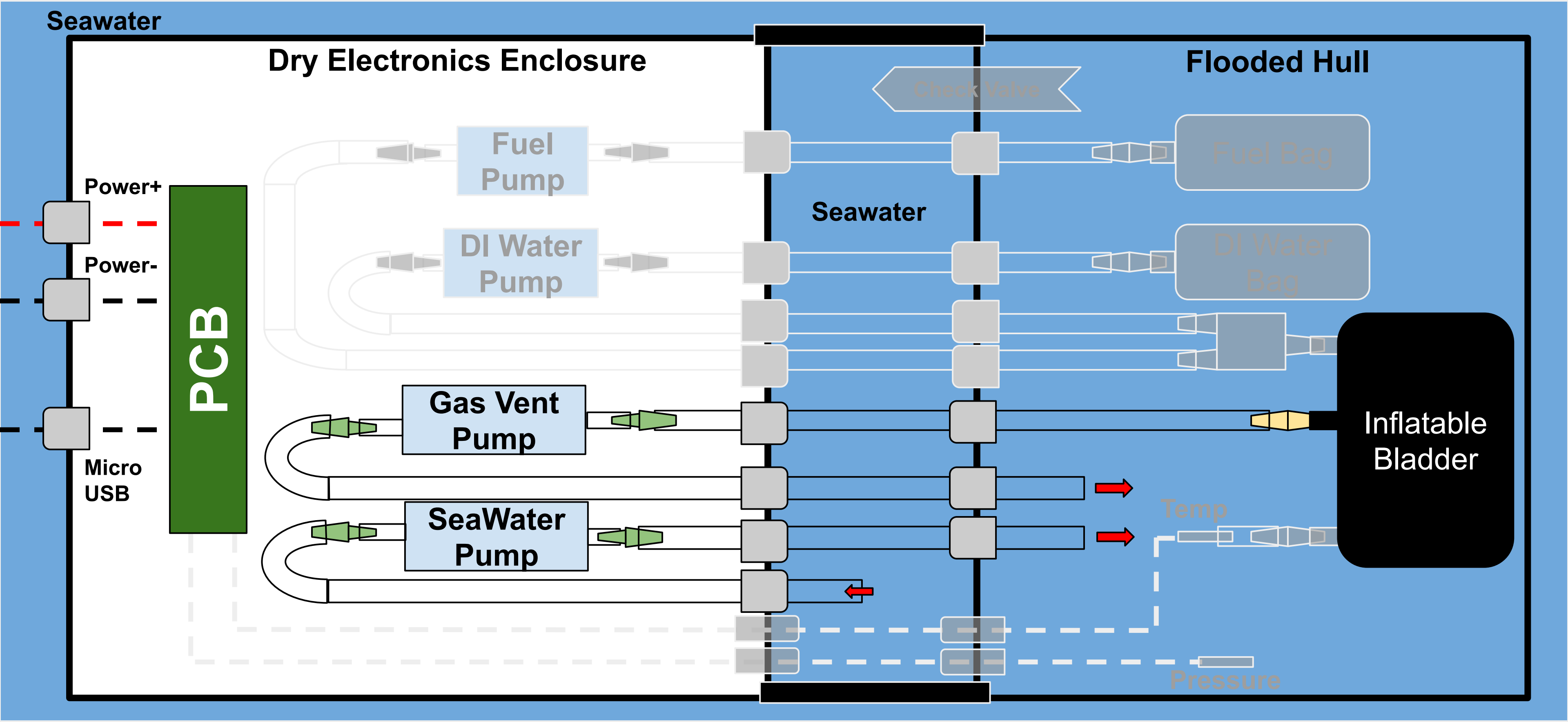
Ascend: Reaction Control



Prototype Dive Cycle

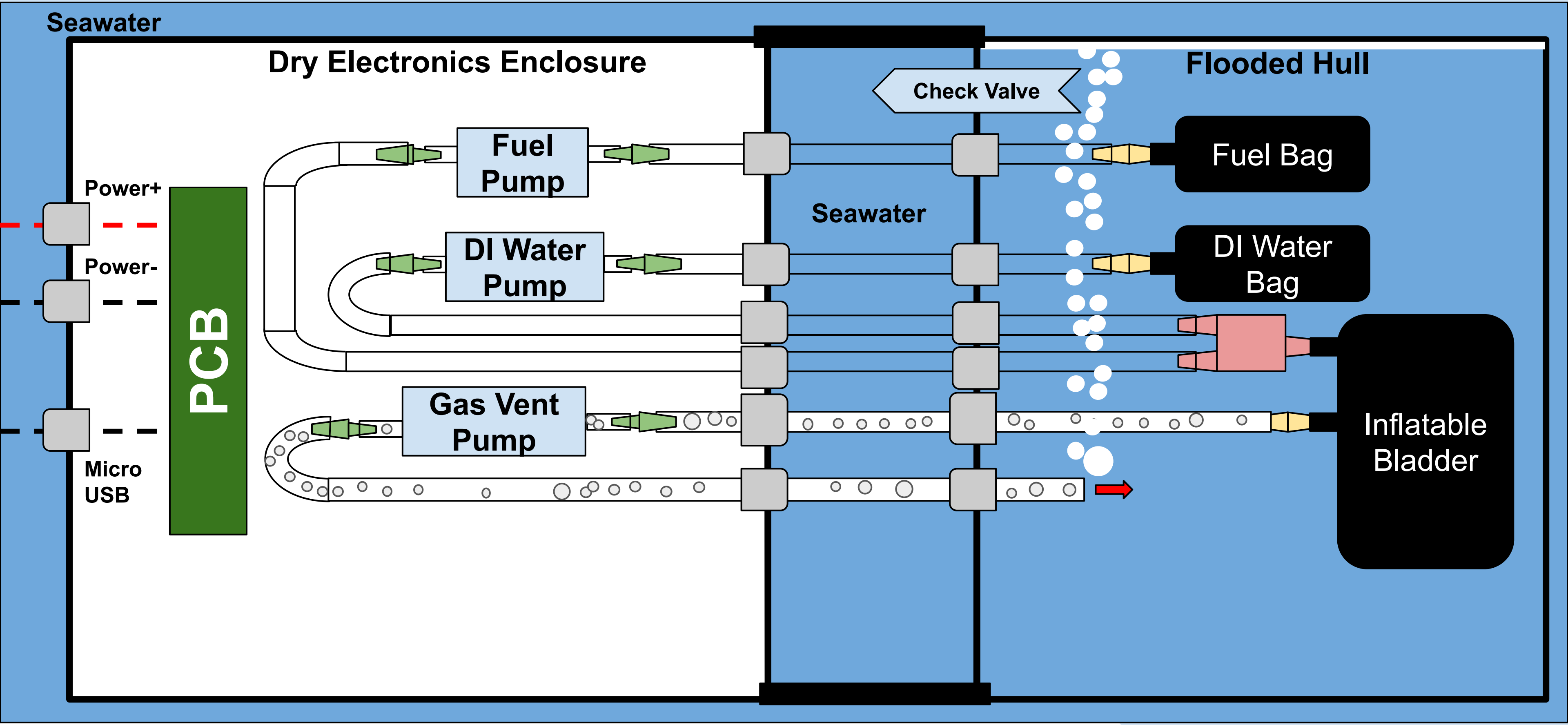


Descend

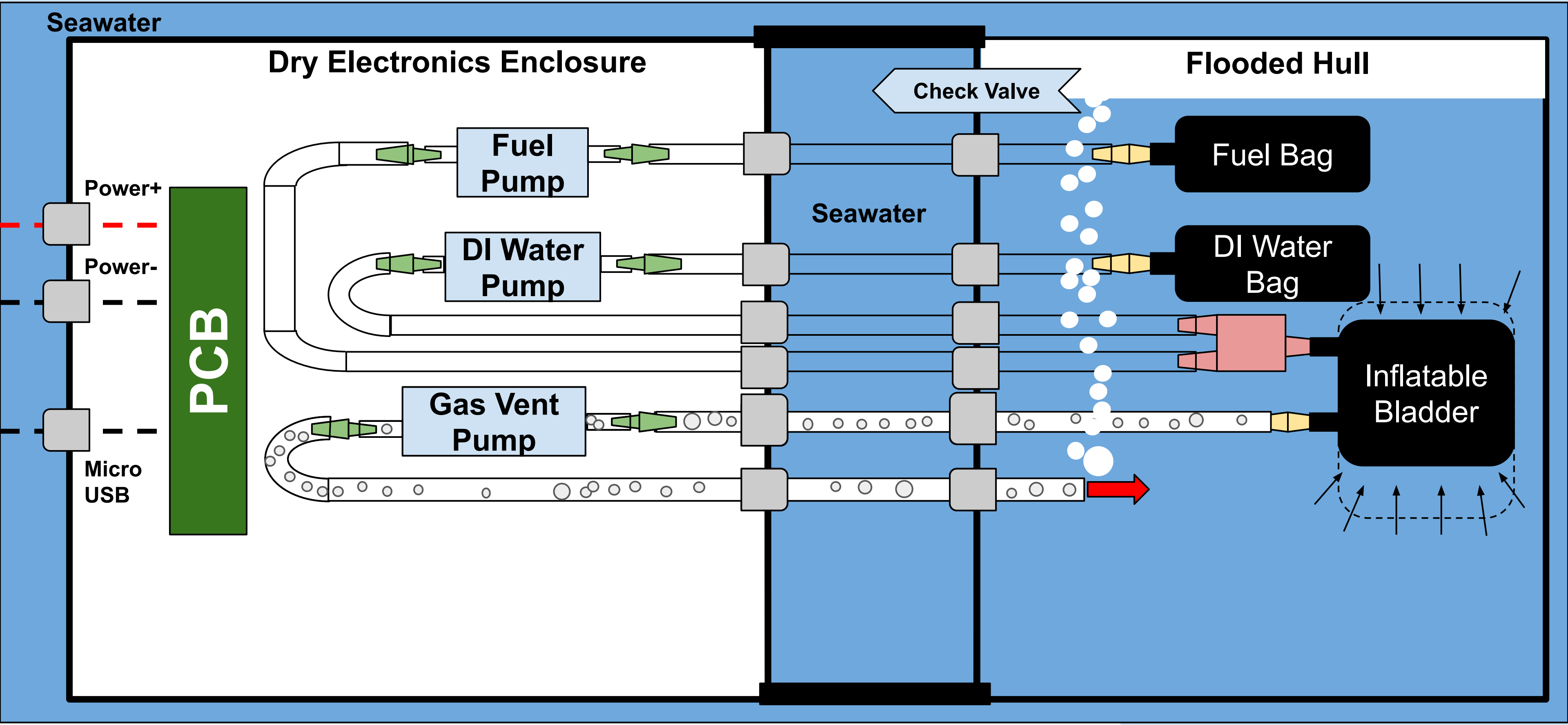


Seawater
Dry Electronics Enclosure
Flooded Hull
Check Valve
Fuel Pump
Fuel Bag
Seawater
DI Water Pump
DI Water Bag
Power+
Power-
Micro USB
PCB
Gas Vent Pump
SeaWater Pump
Inflatable Bladder
Temp
Pressure

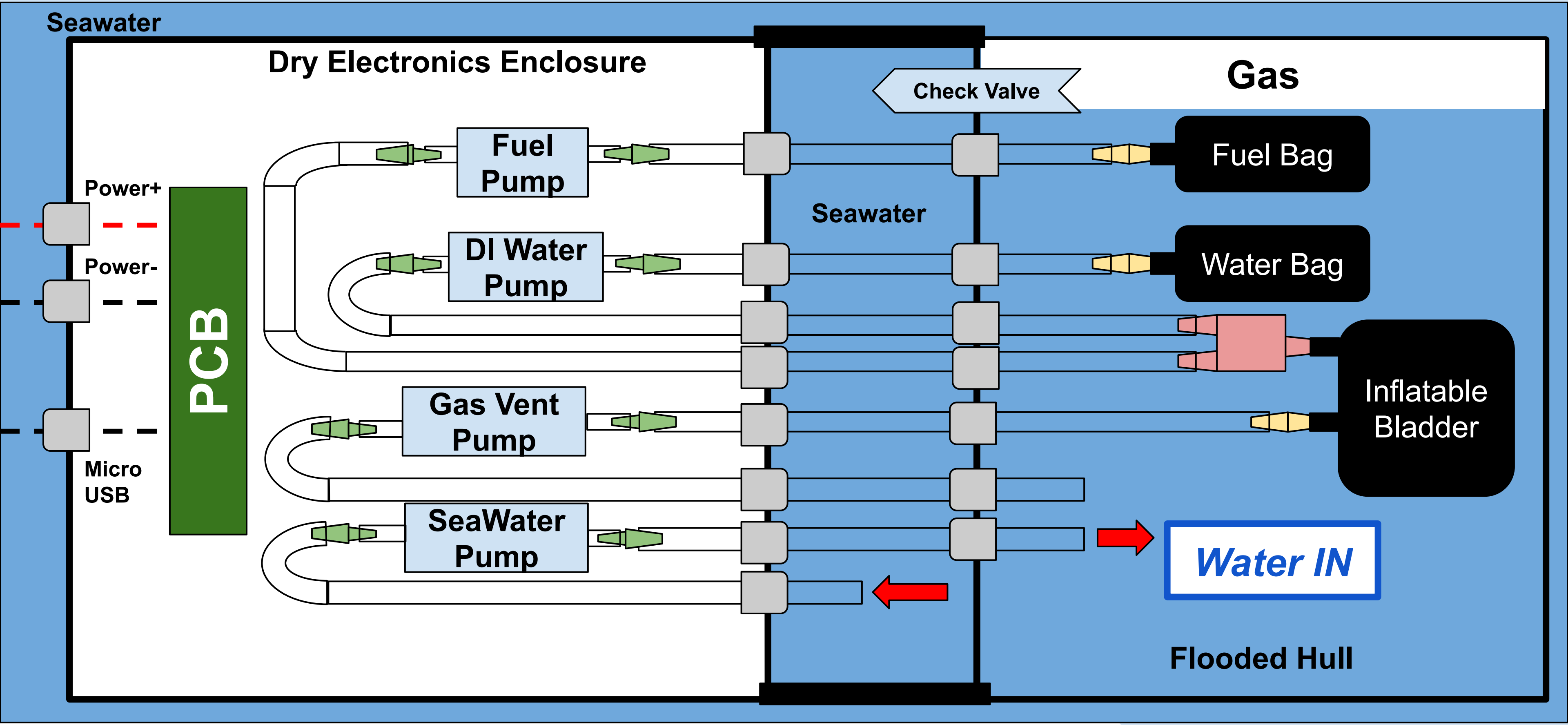
Descend: Removing Gas From Hull



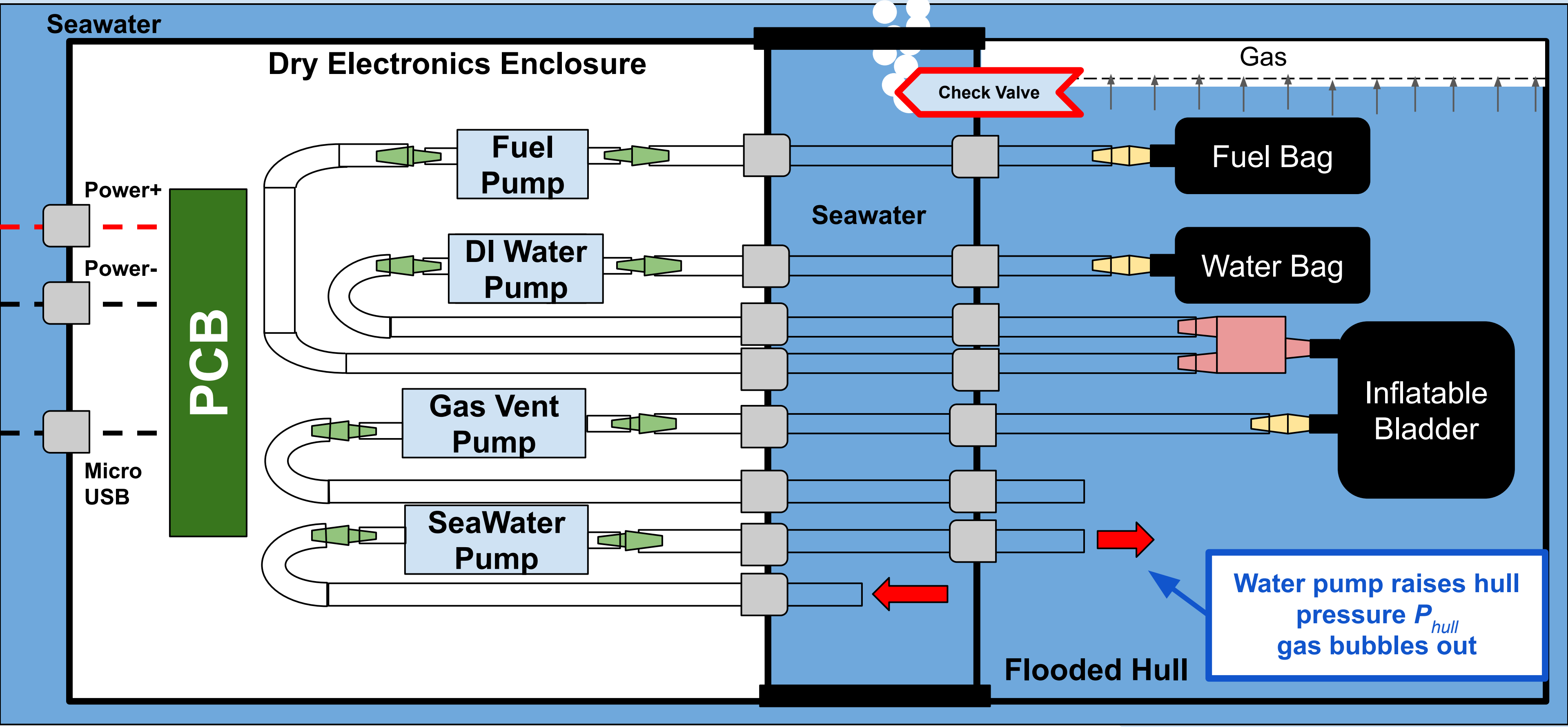
Descend: Removing Gas From Hull



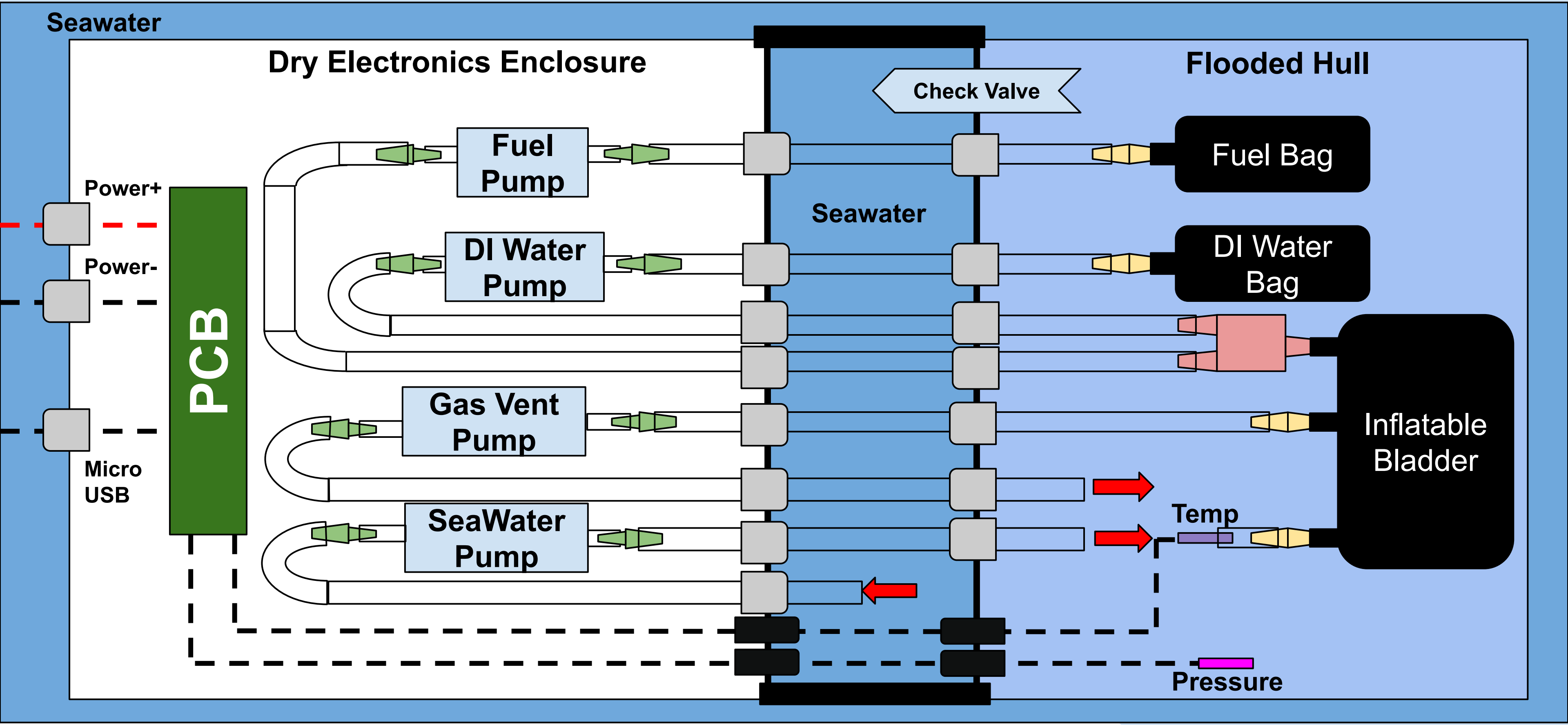
Descend: Removing Gas From Hull



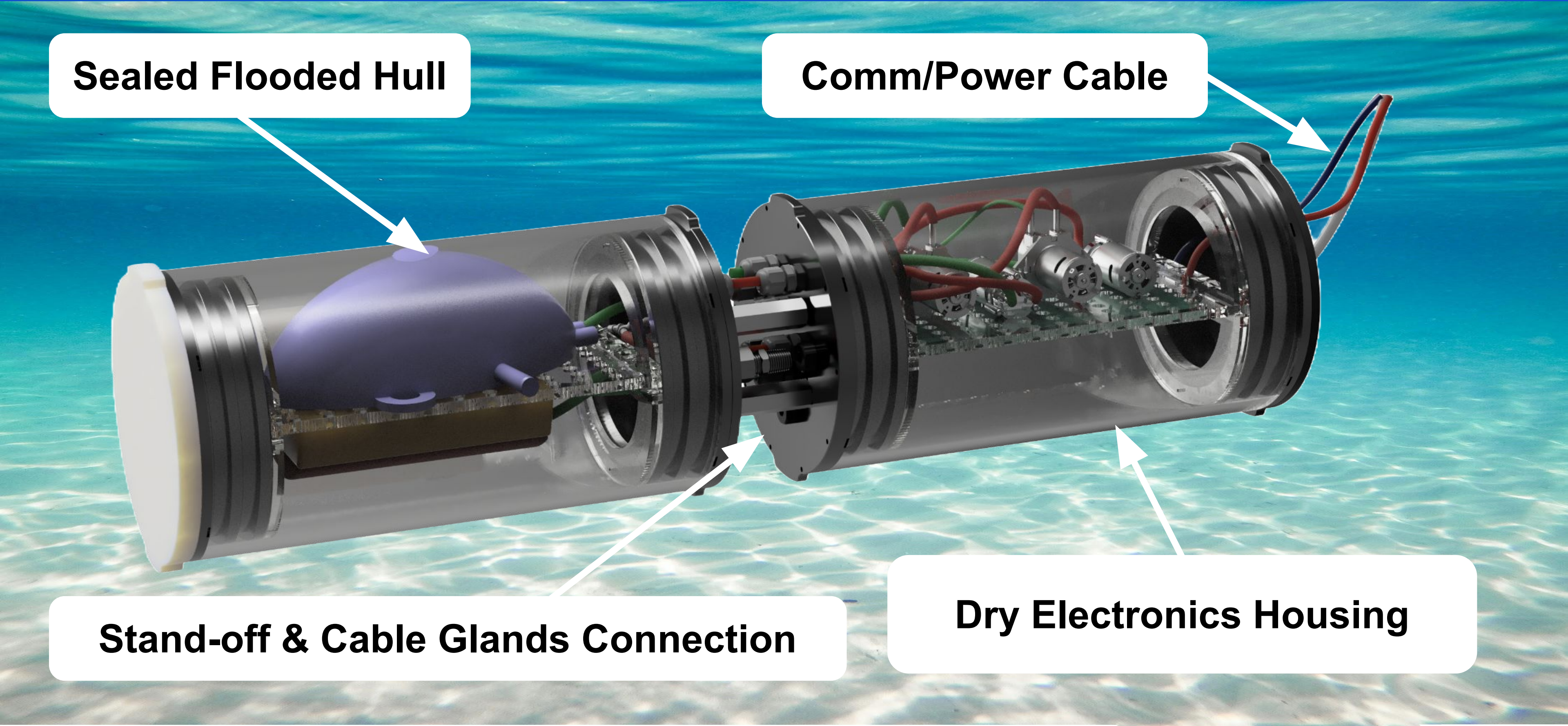
Descend: Removing Gas From Hull



Complete Interface Diagram



Design Overview



Challenges to Validating Concept

1. **Delivering reactants**
2. **Proper reactant mixing and storage of gas**
3. **Waterproofing test configuration**
4. **Sensing and control during experimental testing**

Challenge 1: Delivering Reactants

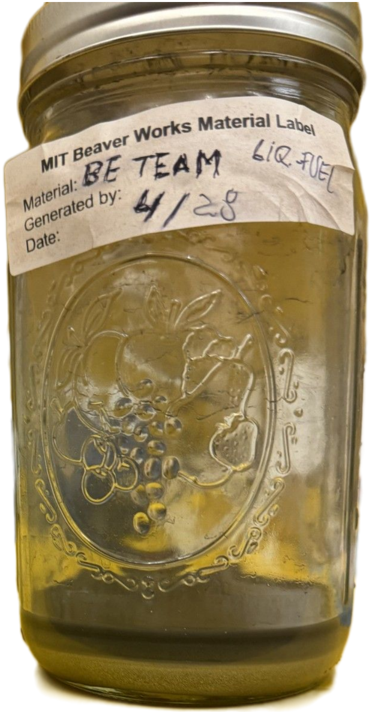
Reaction Control

- ❑ On demand delivery
- ❑ Amount and rate control (gas volume control)

Testing Solutions

- ❑ Used fuel paste of aluminum powder and oil
- ❑ Validate the flow characteristics of fuel
- ❑ Pumps allow for stable reactant delivery

Aluminum Fuel



Composition Percentages by Weight

- 60% Aluminum Powder (25 - 50 µm APS)
- 37% Canola Oil
- 3% Fumed Silica (Thickening/Anti-Caking Agent)

Aluminum particles settling in mixture, will require additional fumed silica thickener

Viscosity

Using Poiseuille's Law:

$$\frac{\eta}{\rho} \sim k\,t$$

k = Mathematical constant

Therefore:

$$\frac{\eta_w \rho_f t_f}{\rho_w t_w} = \eta_f$$

Kinematic Viscosity:

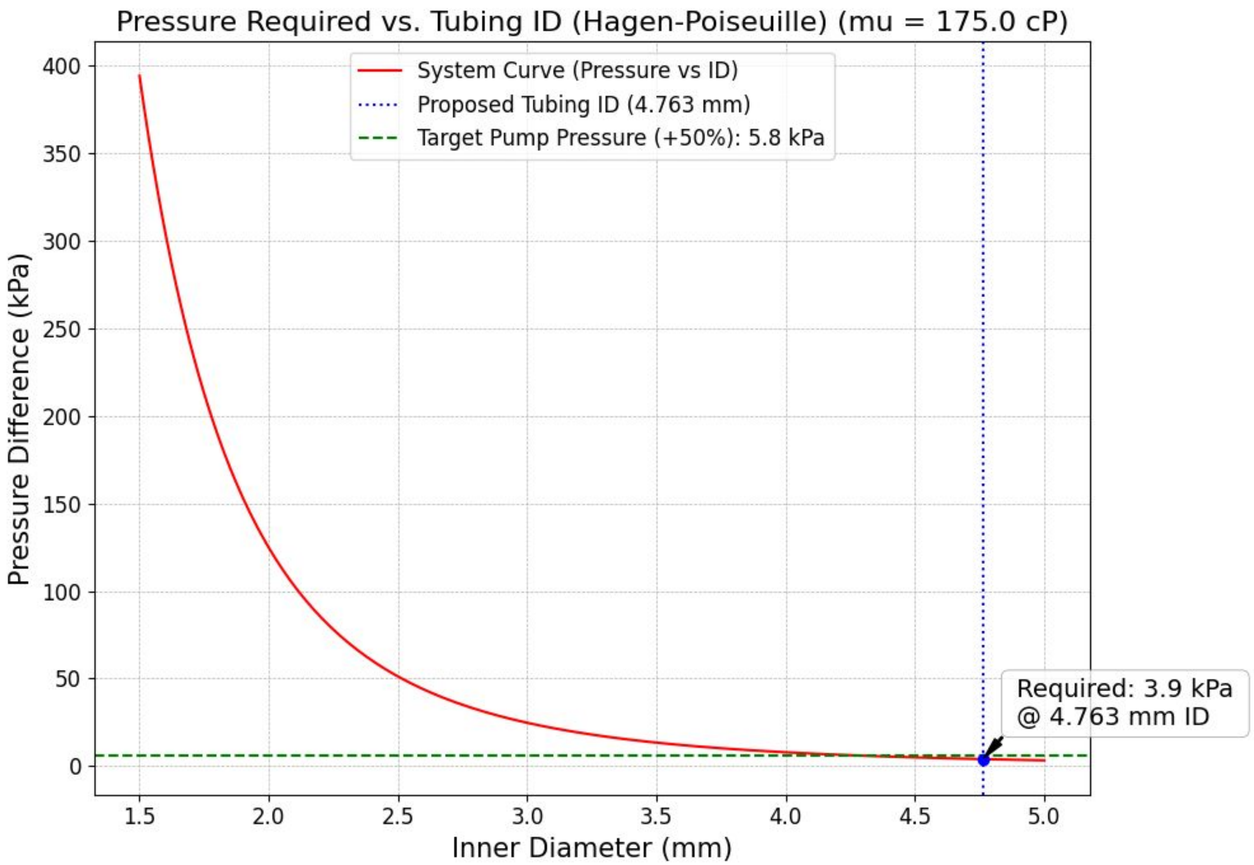
$$\eta = 30.9 \ to\ 48.7\, cSt$$

Dynamic Viscosity:

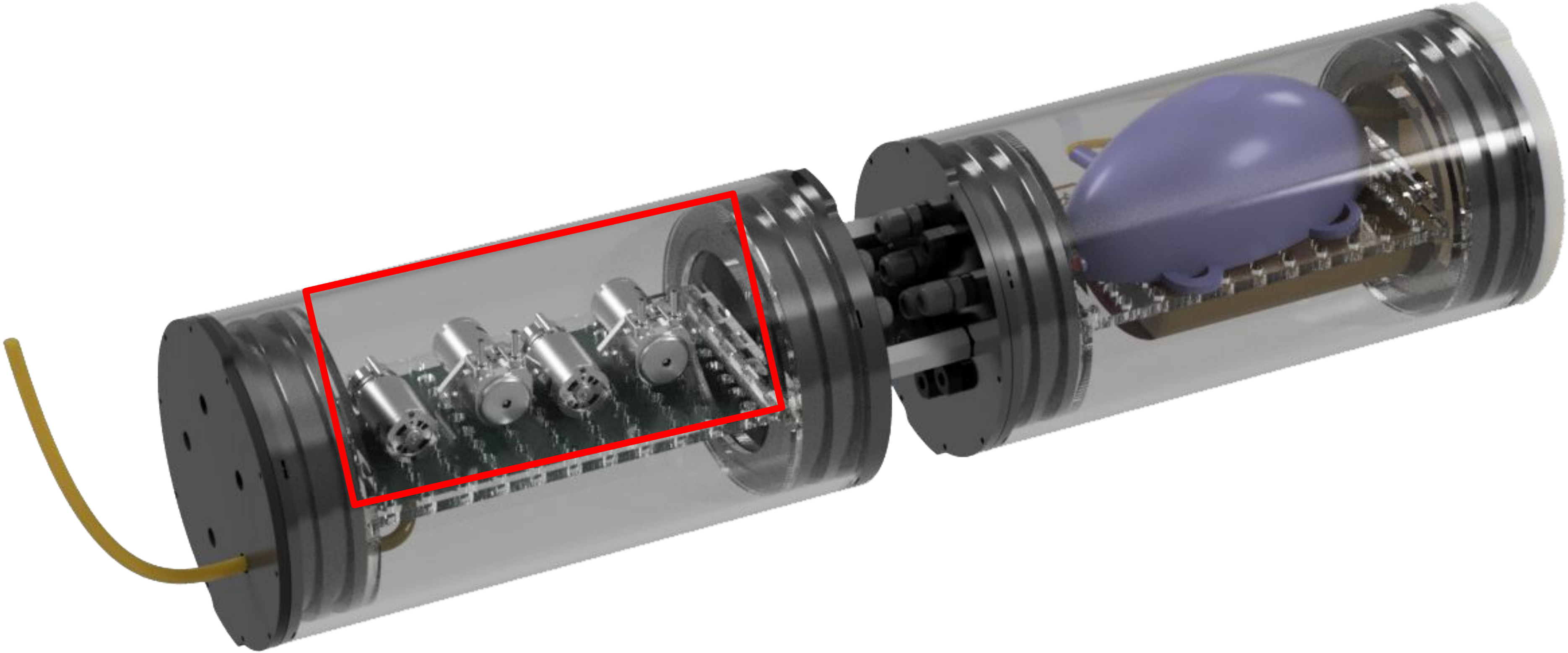
$$\mu = 100.0 \ to\ 157.5\, cP$$

* Silicon oil kinematic viscosity is 50 cSt, viscosity is 3x higher due to colloidal solution and addition of silica.

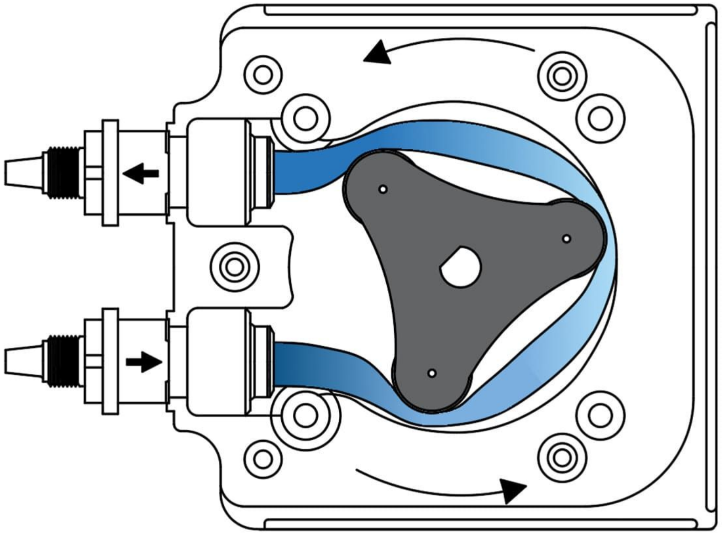
Viscosity Testing



Pumps



Reactant Pumps

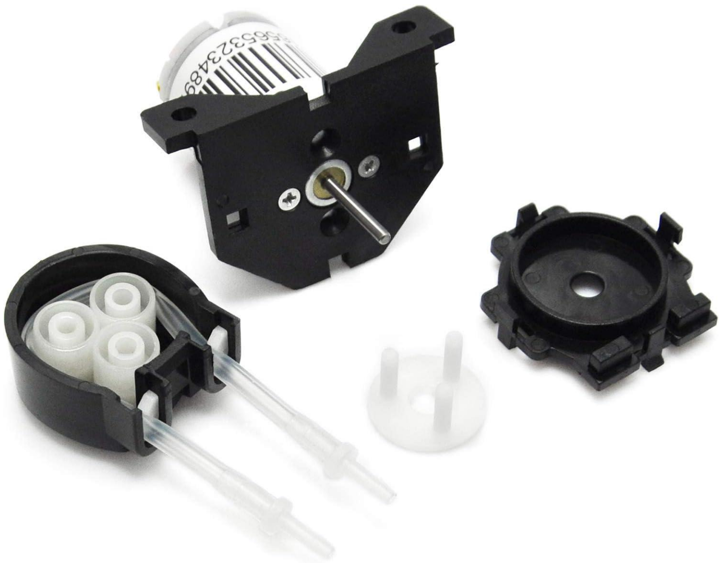


Peristaltic pumps are ideal for:

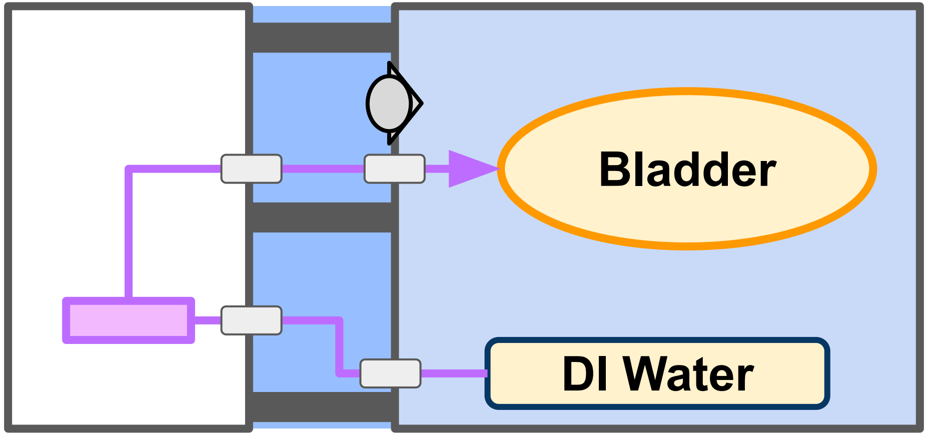
- Small dosing amounts
- Lack of contamination
- Chemical compatibility

DI Water Pump

DI Water Pump
Gikfun Dosing Pump

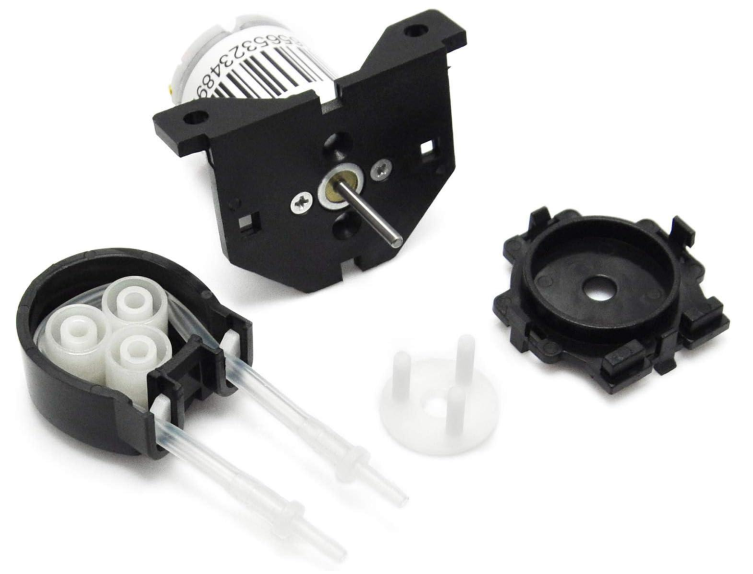


Pumps DI water from reactant bag to bladder against ***no*** pressure

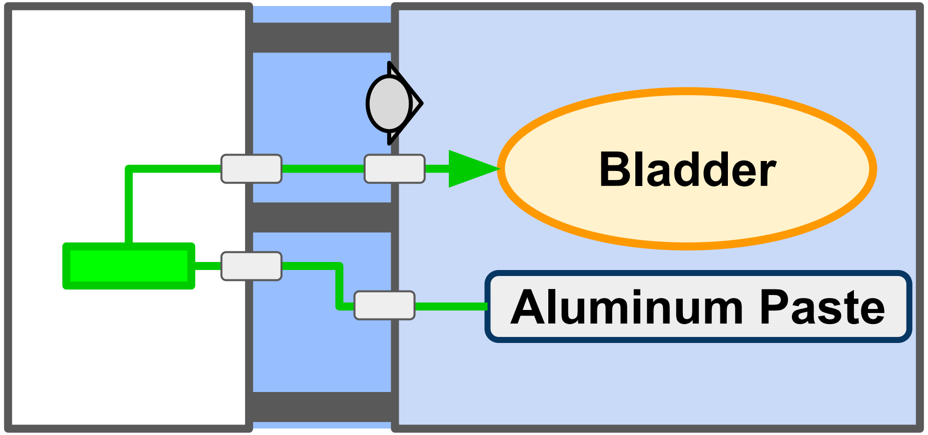


Aluminum Pump

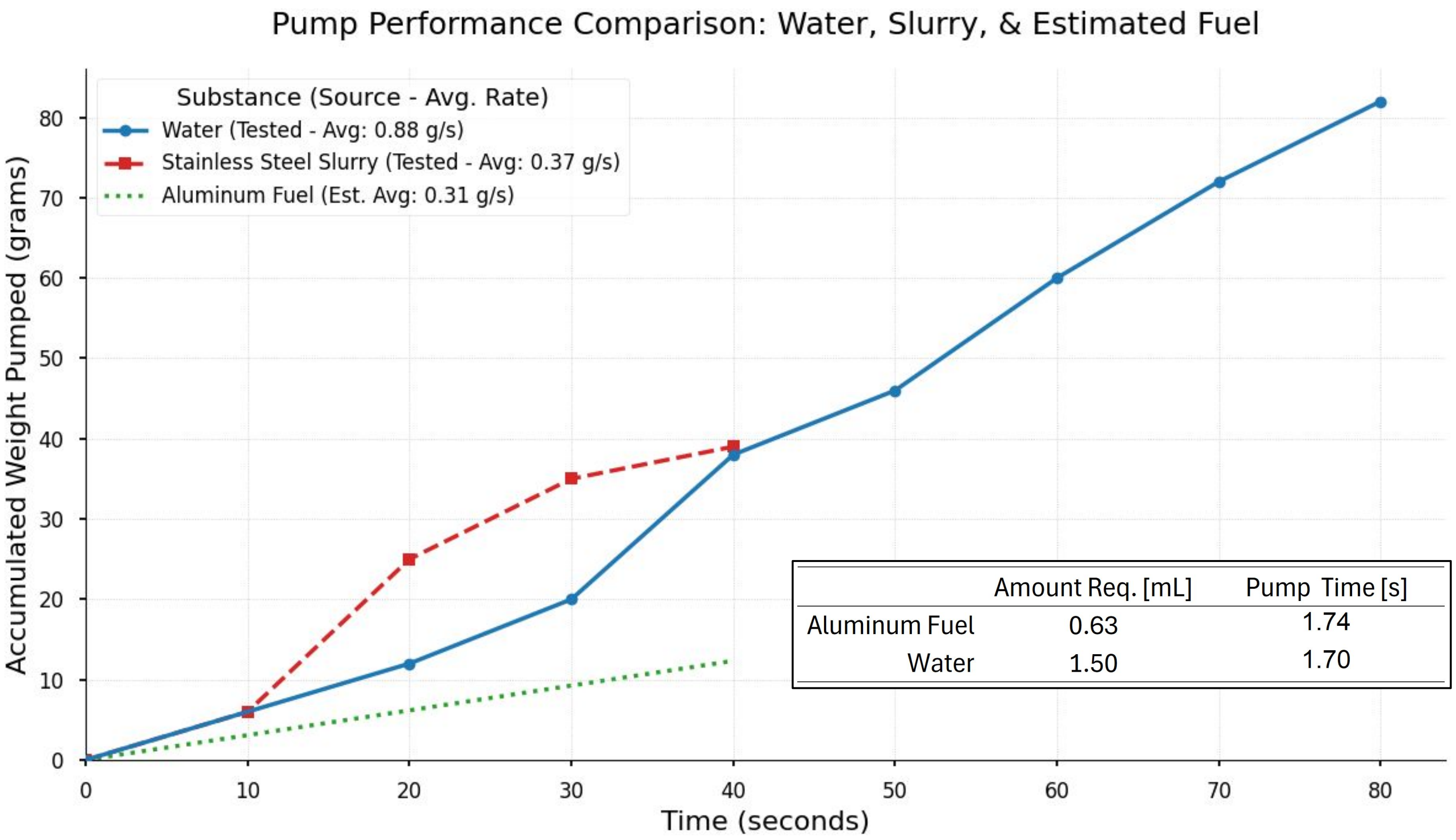
Aluminum Paste Pump
Gikfun Dosing Pump



Generates required pressure head base on Poiseuille's flow equations



Pump Rates



	Amount Req. [mL]	Pump Time [s]
Aluminum Fuel	0.63	1.74
Water	1.50	1.70

Challenge 2: Fuel Reaction and Venting

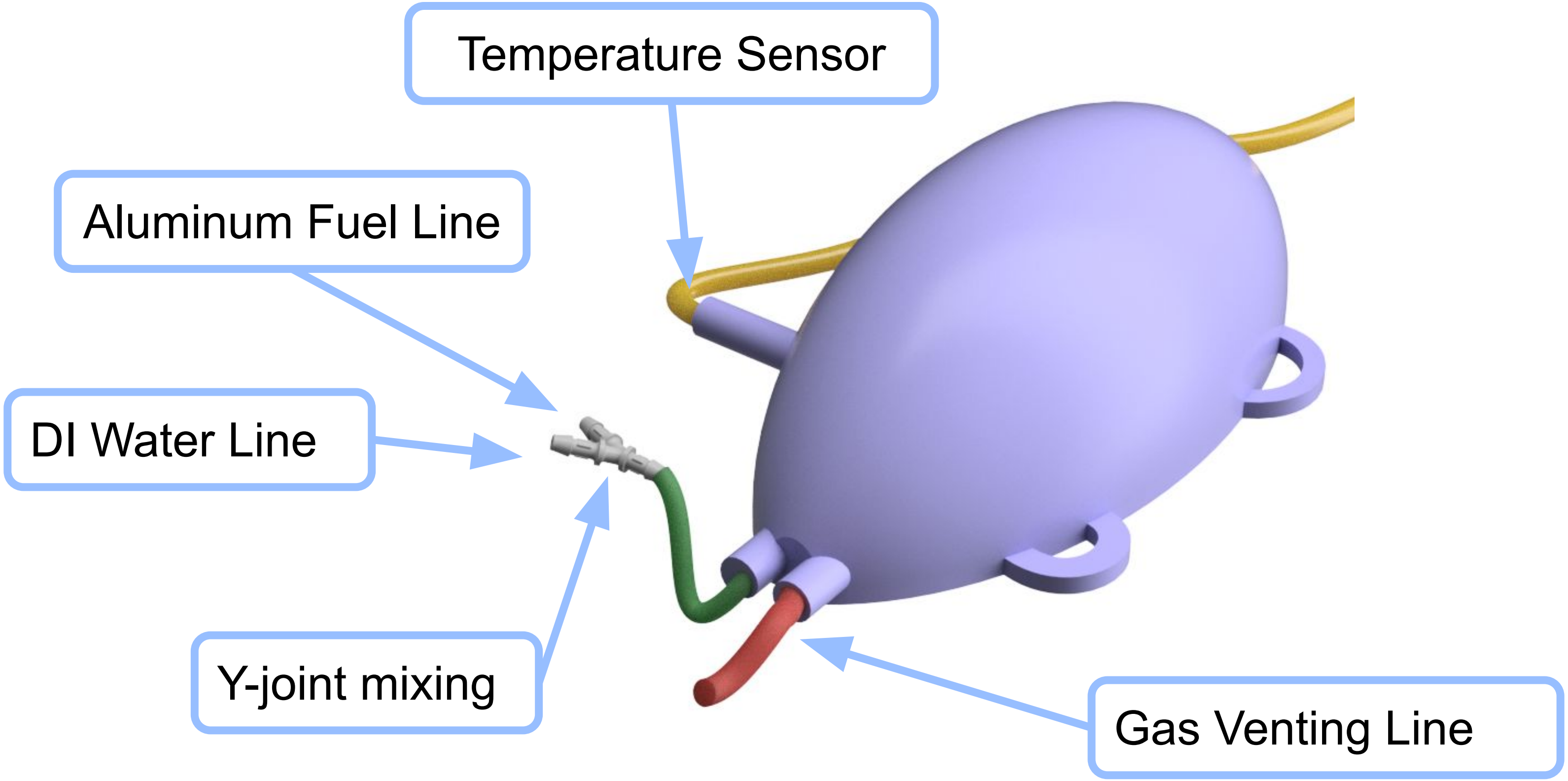
Mixing and Venting

- ❑ Mix reactants thoroughly ensuring reaction
- ❑ Utilize produced gas to displace water
- ❑ Vent gas to regain negative buoyancy

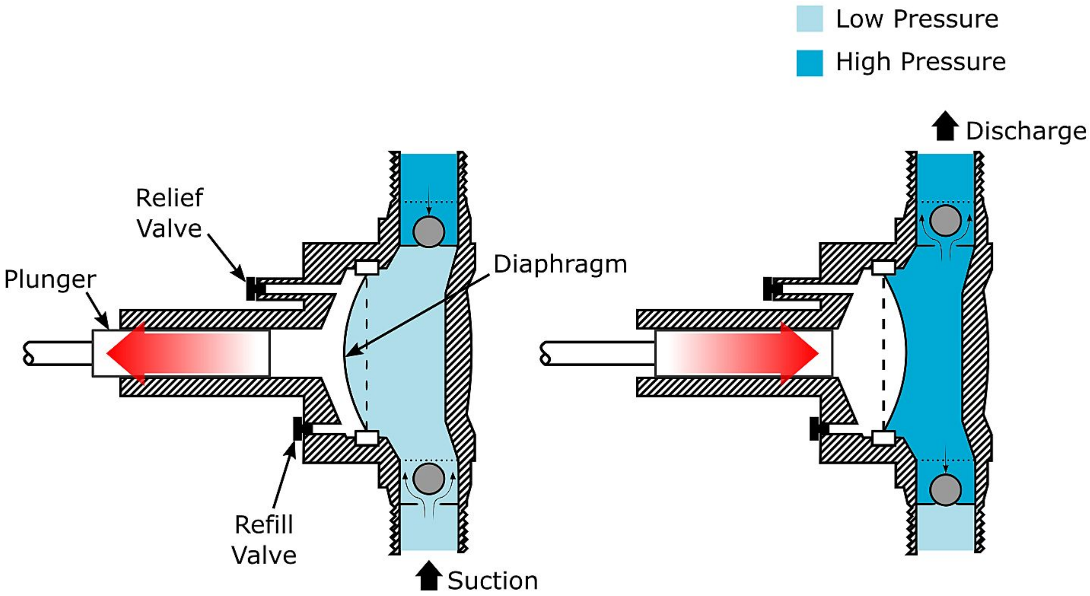
Testing Solutions

- ❑ Y-joint facilitates laminar reactant mixing
- ❑ Flexible gas bladder forces water out of check valve
- ❑ Two part venting avoids large pressure differential

Challenge 2: Fuel Reaction



Venting and Sea Water Pumps



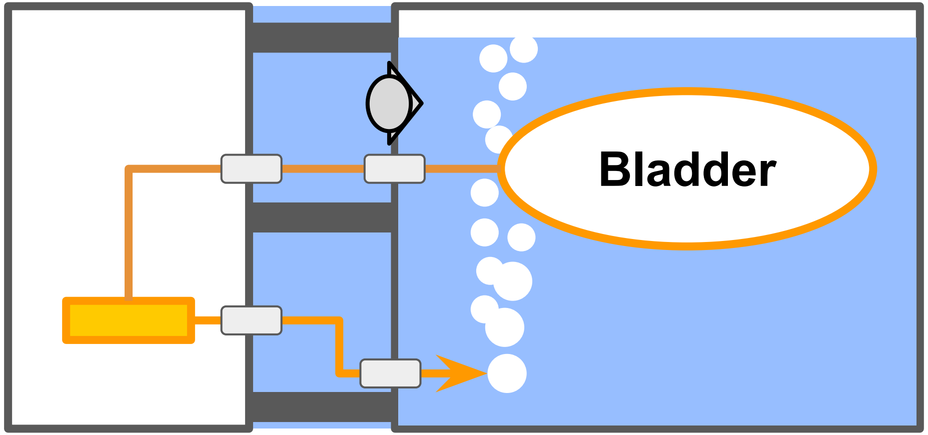
Diaphragm pumps are ideal for:

- Higher flow rates
- Higher pressures
- Excellent self-priming

Gas Venting Pump

Gas Venting Pump
WP-370B Diaphragm Pump

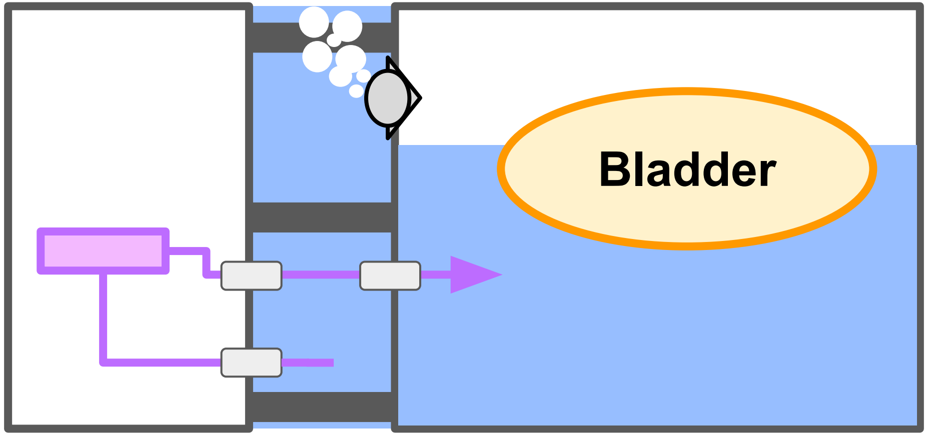
Pumps gas from the bladder into the flooded hull against no pressure differential



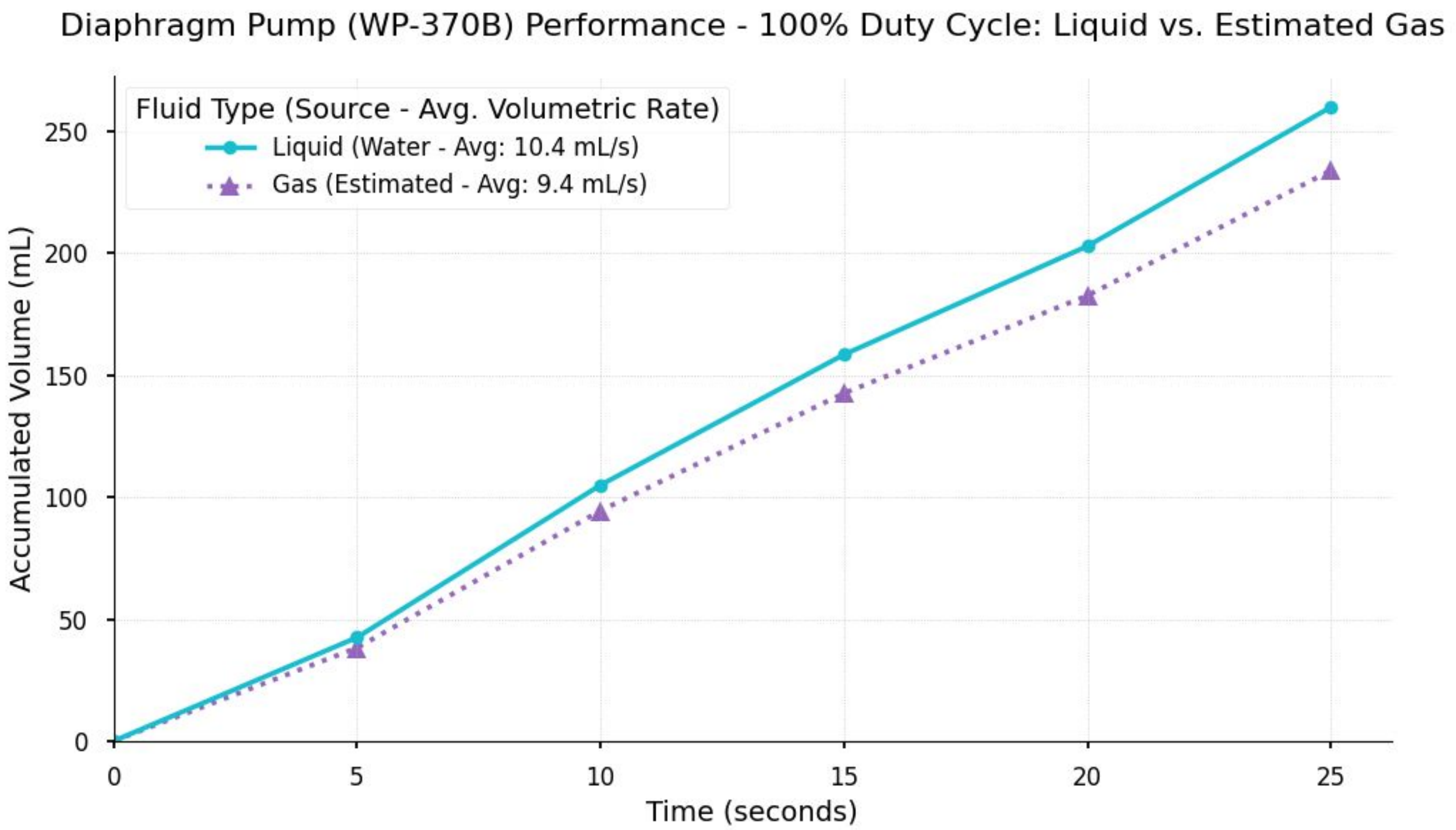
Sea Water Filling Pump

Sea Water Filling Pump
WP-370B Diaphragm Pump

Pumps sea water into the hull, overcoming a 13.8KPa differential, forcing out the gas



Pump Testing



Challenge 3: Enclosure & Waterproofing

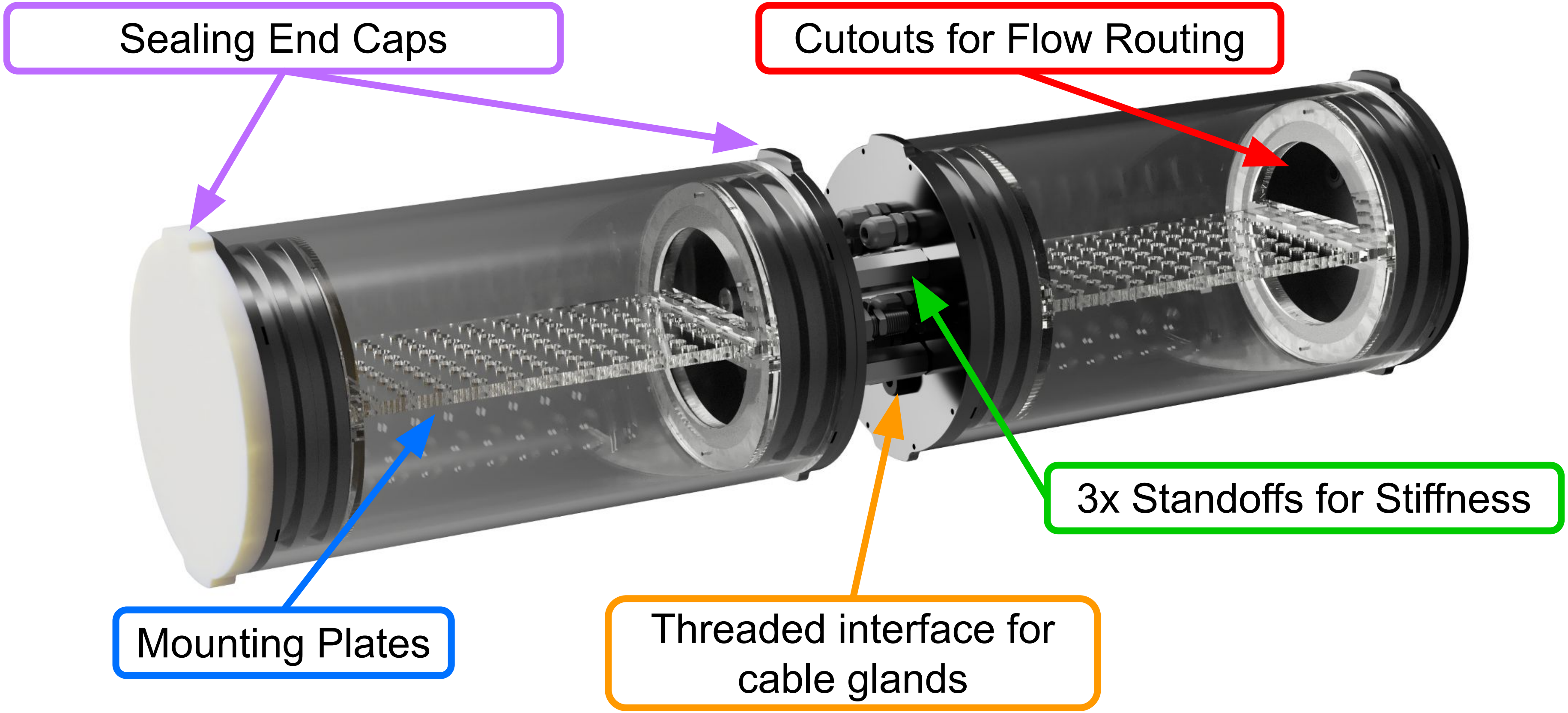
Sealed Environment

- Prevent shorting of electronics
- Maintain reaction produced gases
- Maintain instrumentation functionality

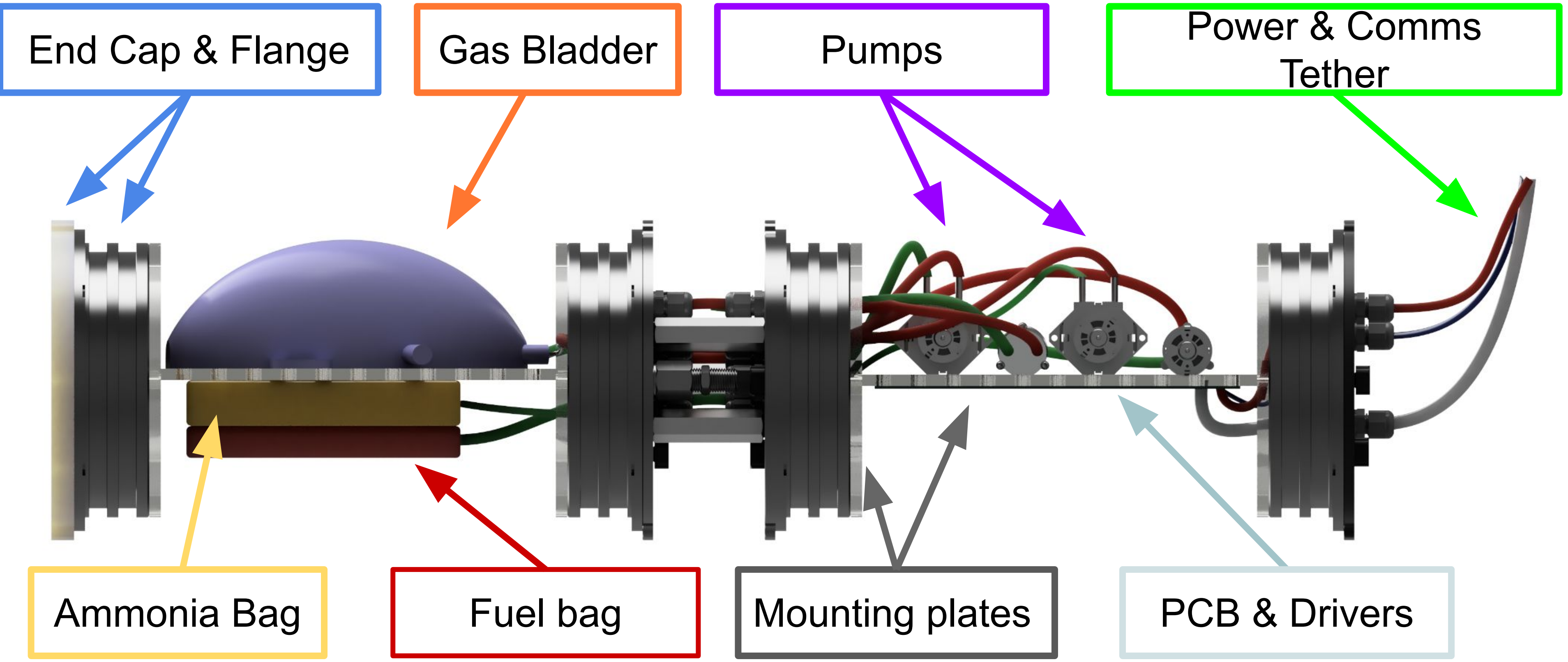
Testing Solutions

- Designed around commercial enclosure
- Sealed interfaces
- Easily removable enclosure for rapid test iteration

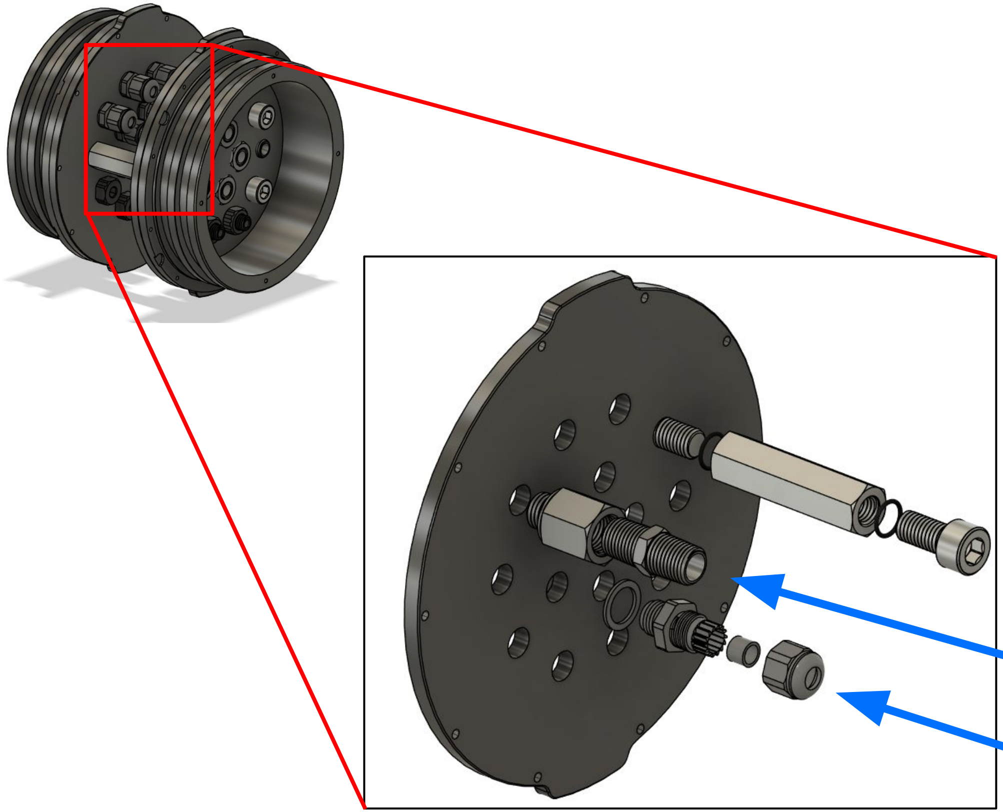
Enclosure Design

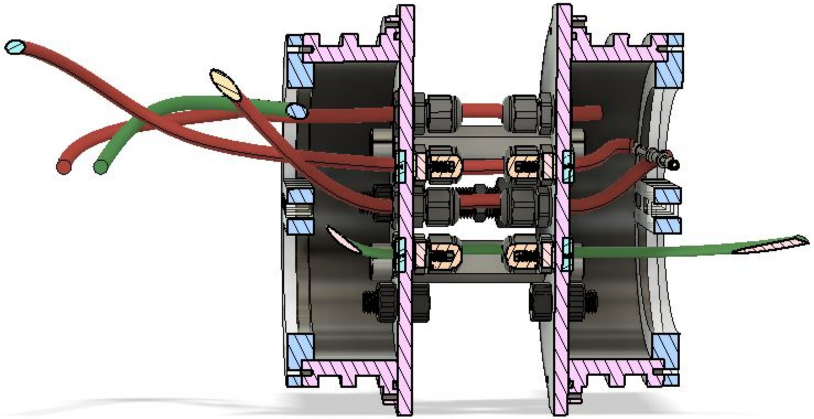


Current Housing Layout



Flow Routing





Standoff

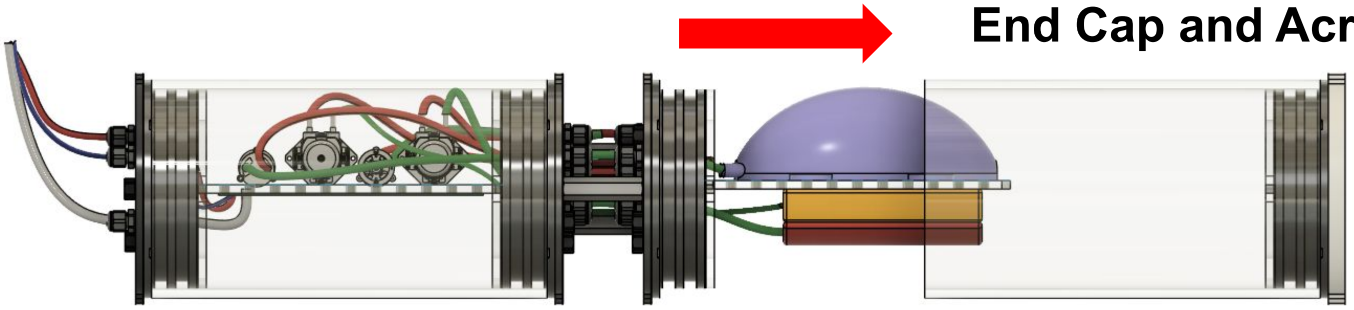
Check Valve

Cable Glands

Mounting & Refueling

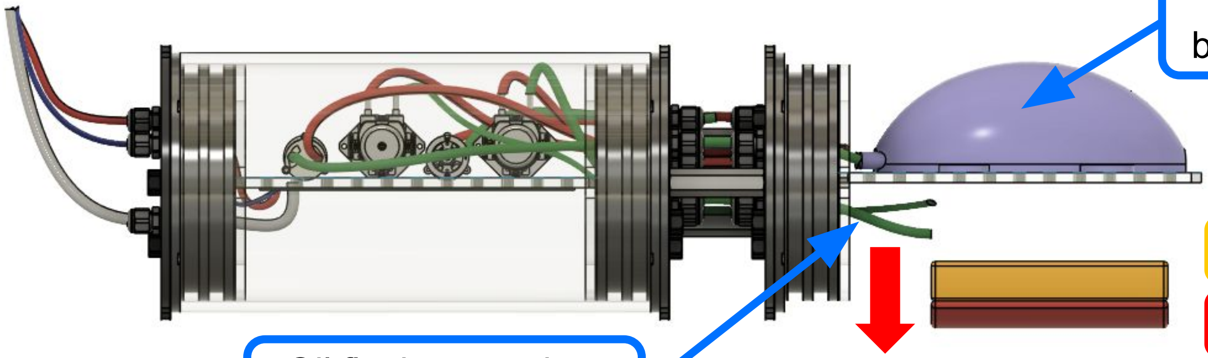
Step 1

Slide out
End Cap and Acrylic Tube



Step 2

Remove and flush bladder of byproducts



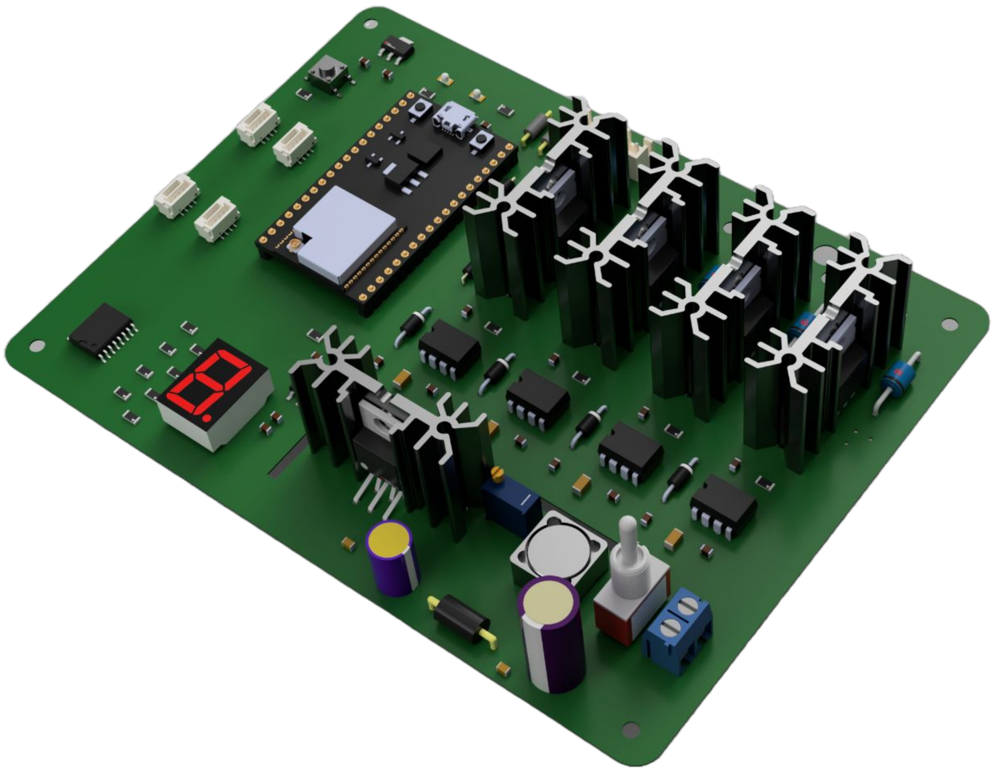
Water Bag

Al Fuel Bag

Oil flush every time when refueling

Disconnect tubes to refuel

Challenge 4: Sensing and Control



Testing Solutions

EE Requirements:

Motor control

- ❑ DI water pump
- ❑ Fuel pump
- ❑ Gas venting pump
- ❑ Seawater pump

Controller

- ❑ ESP32

Power

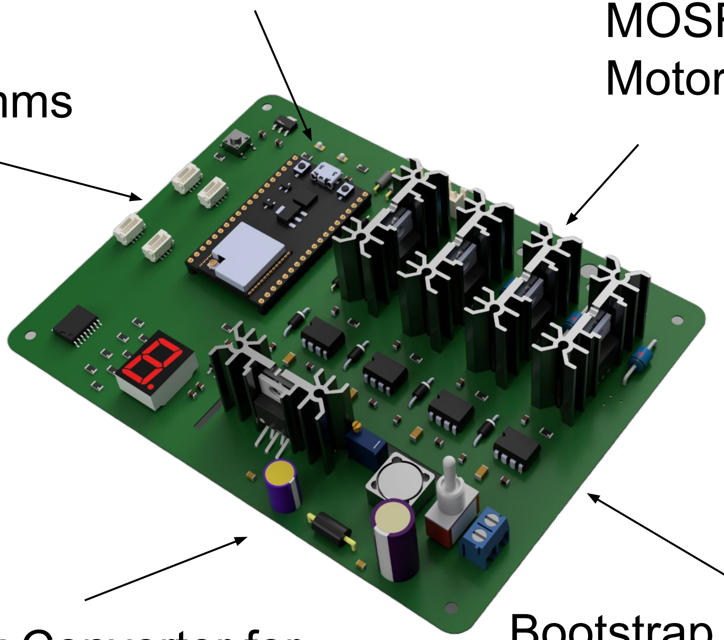
- ❑ 12V and 6V supply

Interface

- ❑ Serial Monitor

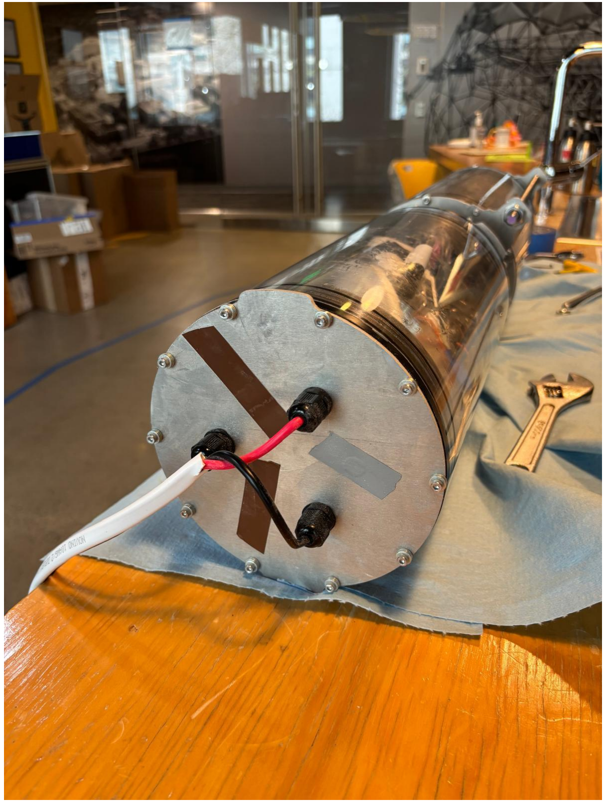
PCB

- Actuate pumps
- Real time measuring and instrumentation
- Rapid iteration and replaceable hardware



Power and Routing

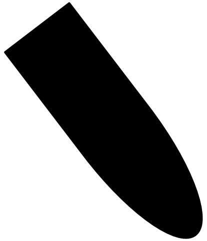
- ❑ Powered and communications sent out by a bundled tether
- ❑ Routes through water-sealing cable glands
- ❑ 12V power supply and laptop control



Control System

Ascend

Dispense fuel & DI water at proper ratio
- Fuel pump
- DI Water pump



Descend

Vent gas out of the bladder
- Gas vent pump

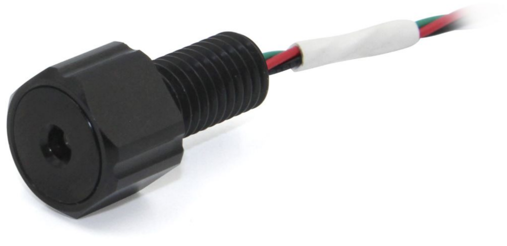
Intake water into chamber
- Sea Water pump

Sensing

- **Hull Pressure** for starting/stopping venting
- **External / Reaction Bag temperature** for reaction timing & abort commands
- In testing, a **tension spring** communicates buoyancy change *(testing visual to follow)*

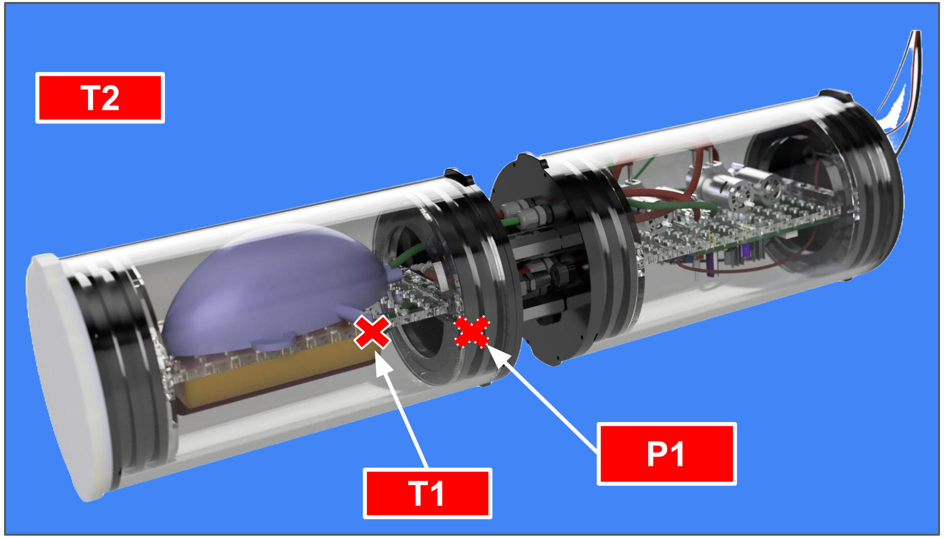
A pressure increase means the chamber is being filled or bladder inflating.

pressure sensor (flooded hull)



A temperature increase means the reaction is happening.

Thermistor (1) bladder and (2) external tank

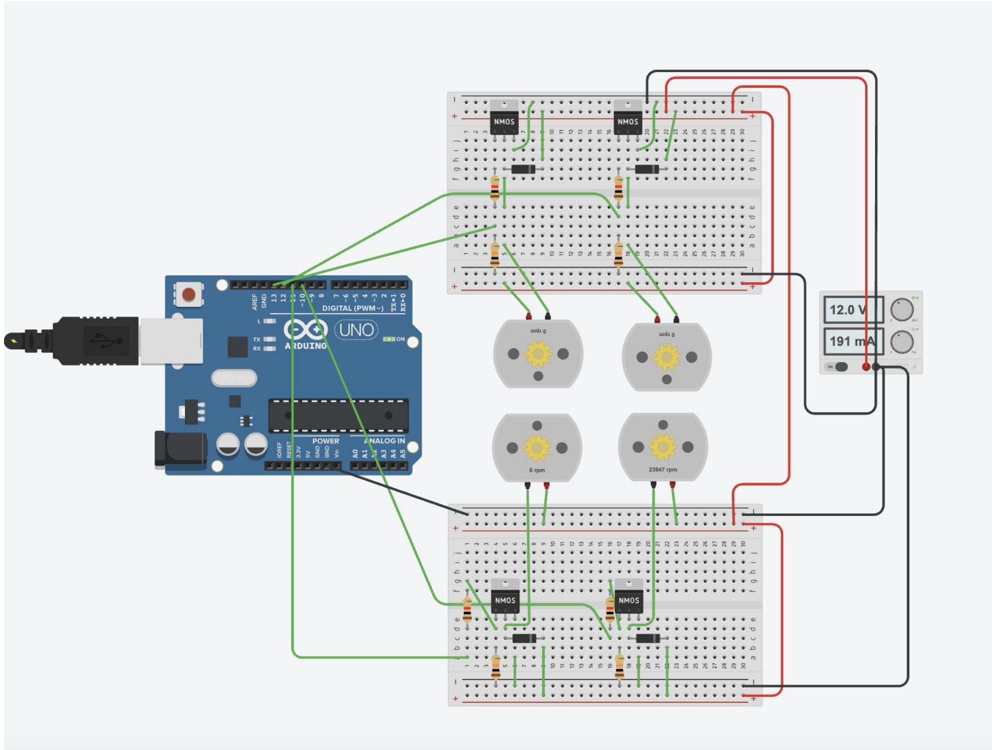


Breadboard Layout

Control Circuit

Straightforward control of 4 12V DC motors

Testing in TinkerCAD allowed for easy experimentation and component selection



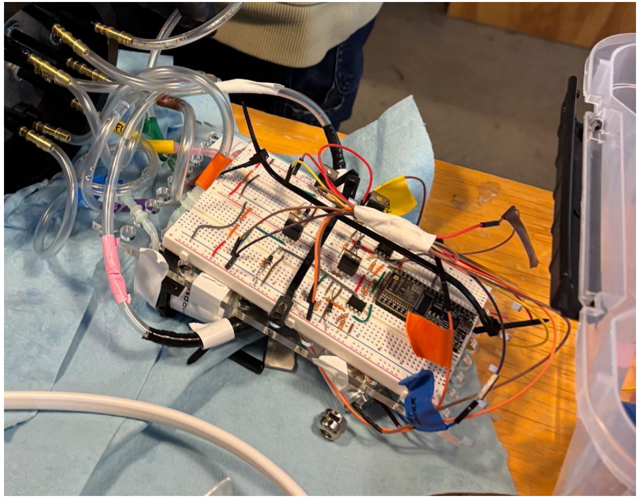
Breadboard

Control Circuit

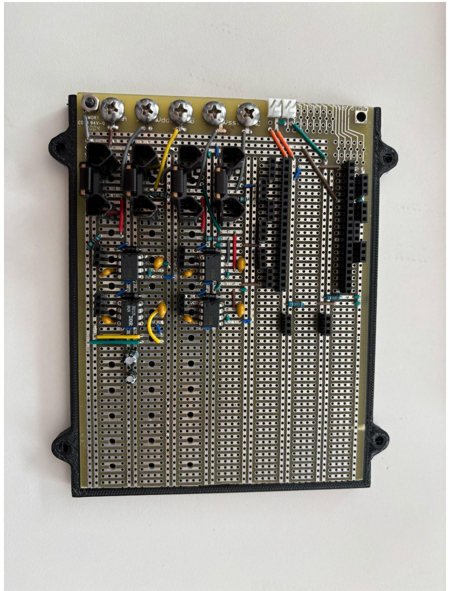
Built on a breadboard

Op-amps used as signal amplifiers to get sufficient gate-source voltage

Breadboard

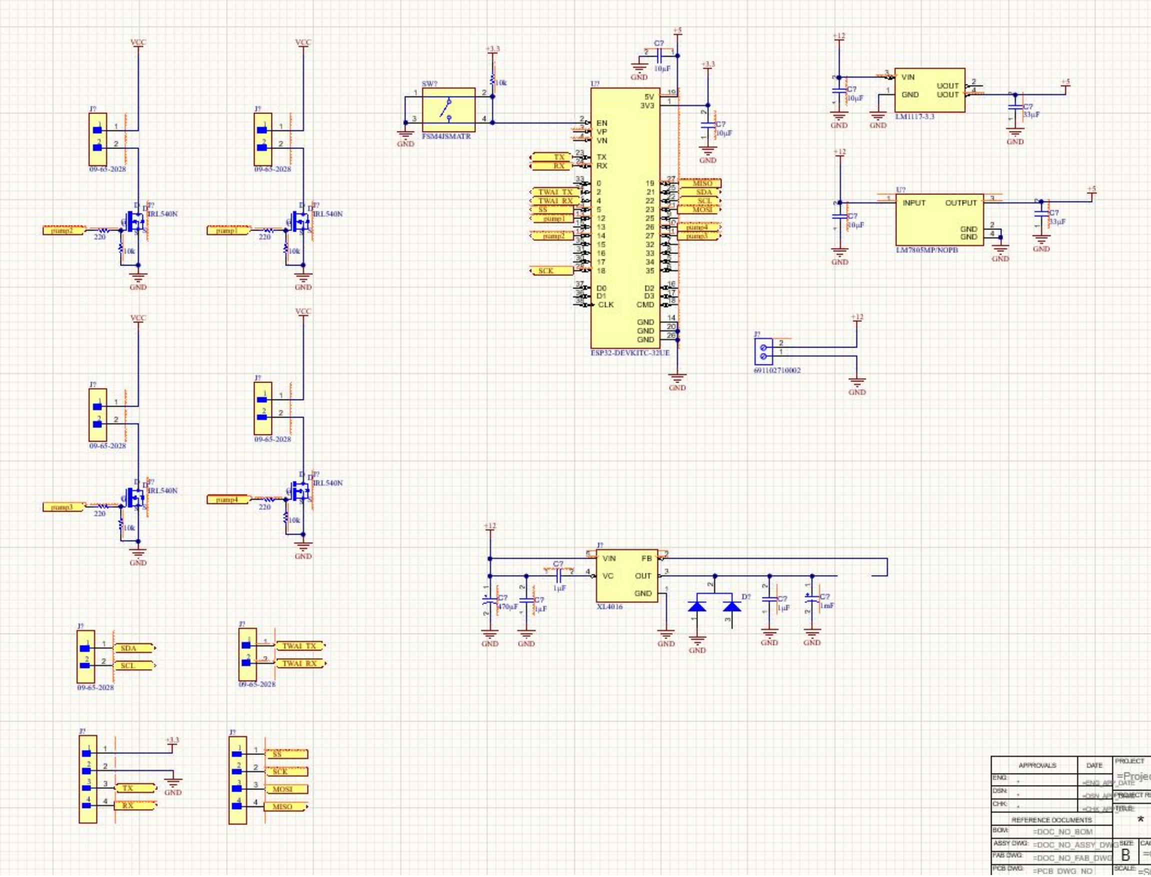


Perfboard

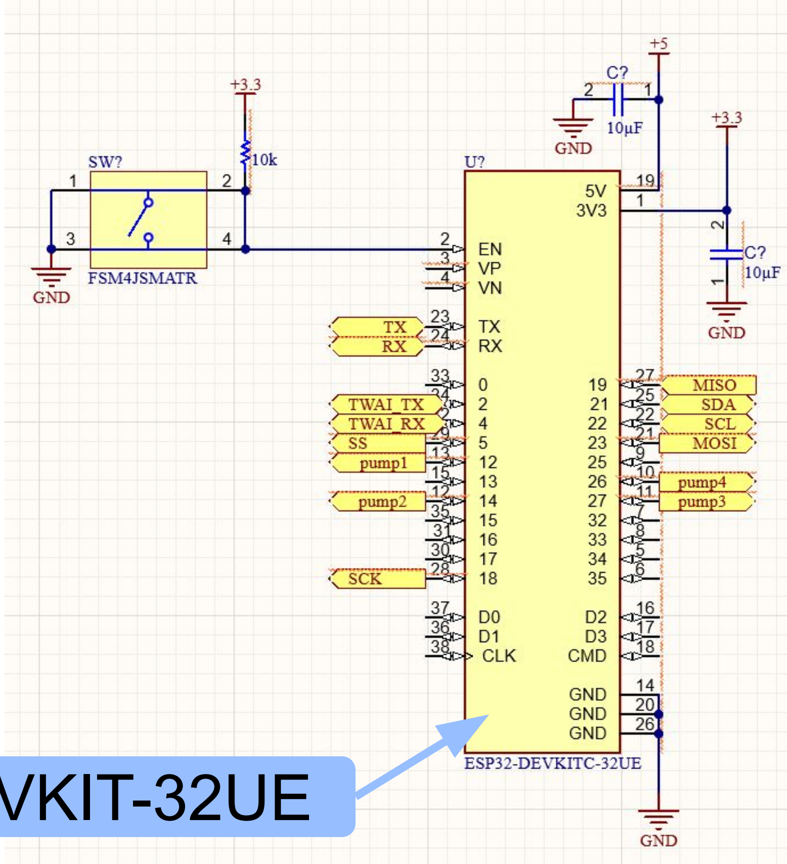


New PCB Schematic

- ❑ Detailed views on next slide
- ❑ Simplified due to Scope reduction



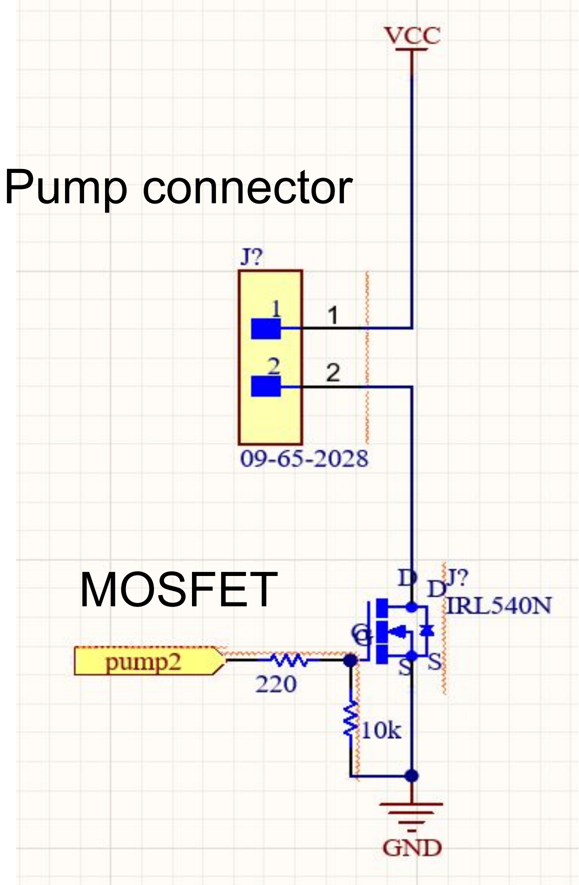
Microcontroller Wiring



ESP32-DEVKIT-32UE

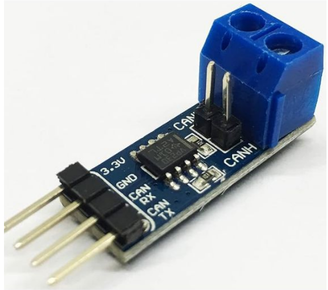
Pump Wiring

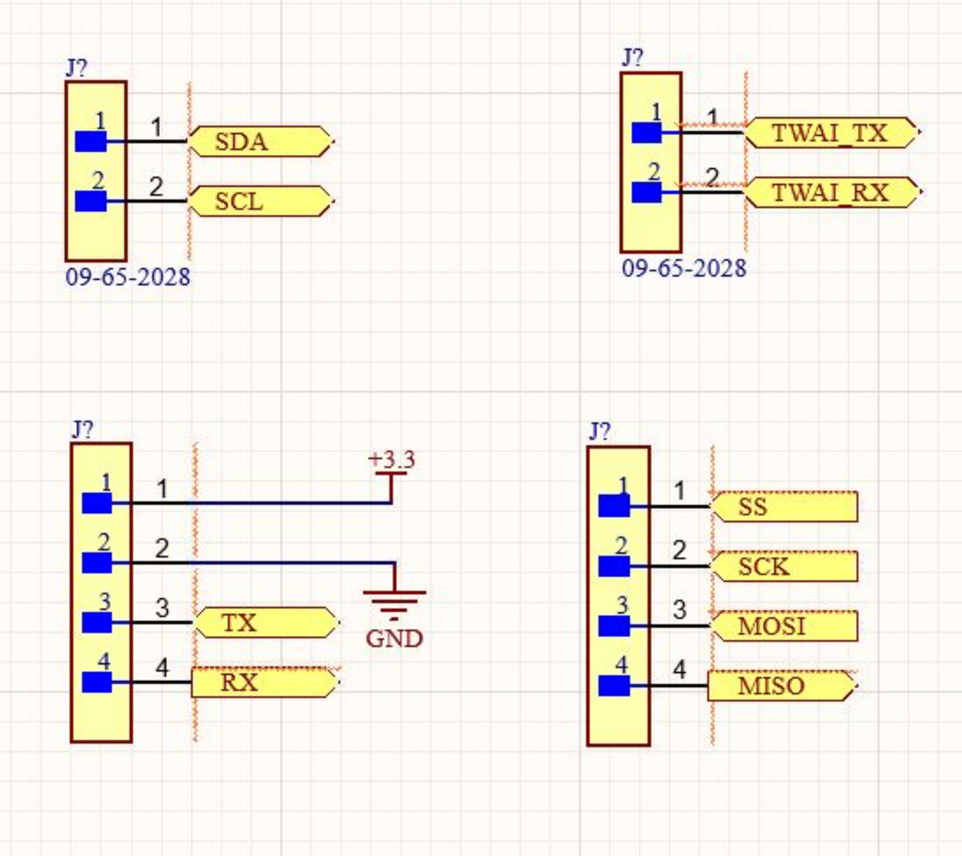
- ❑ Eliminated bladder
  - Makes all pumps unidirectional
- ❑ Control with only MOSFETS



Communication Wiring

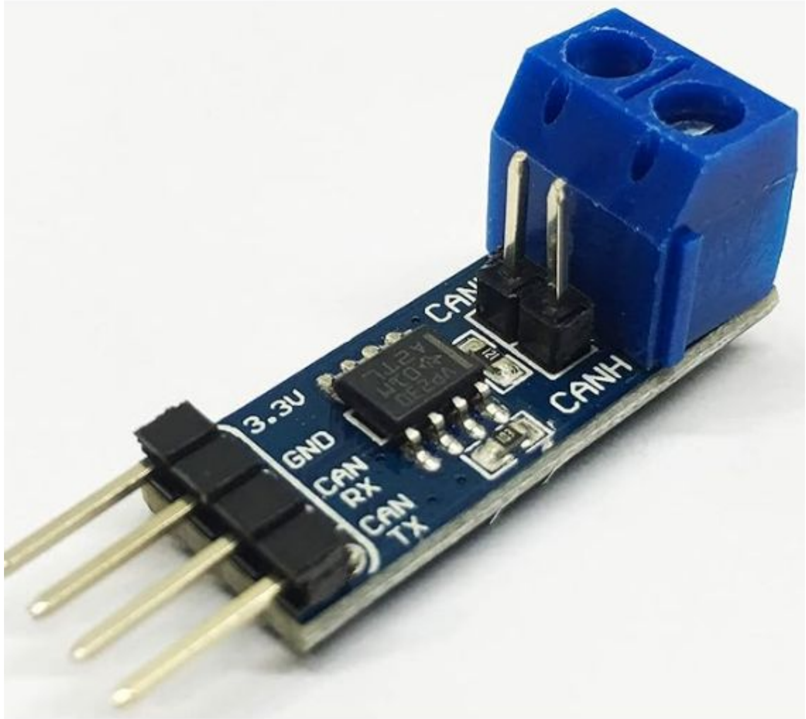
- SPI, UART, CAN, and I2C enabled for versatility
- CAN bus requires conversion from TWAI protocol





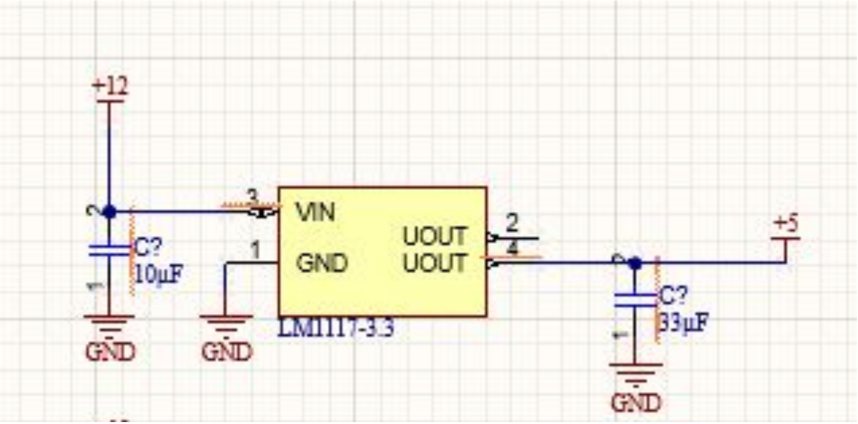
Electronics

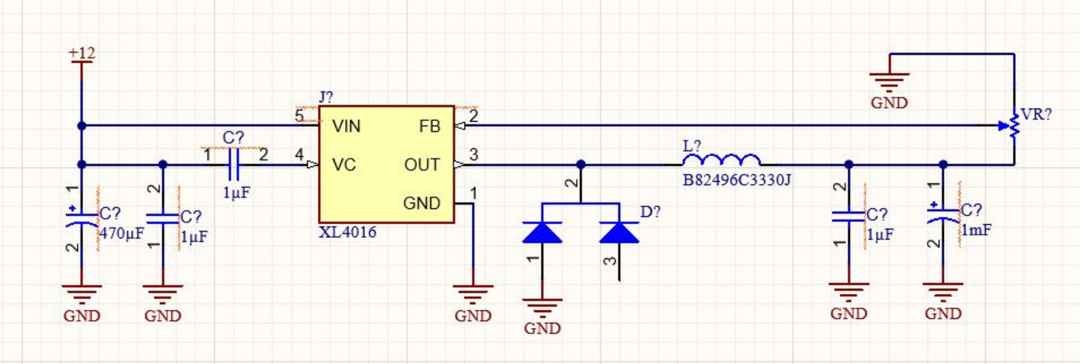
- CAN bus requires conversion from TWAI protocol



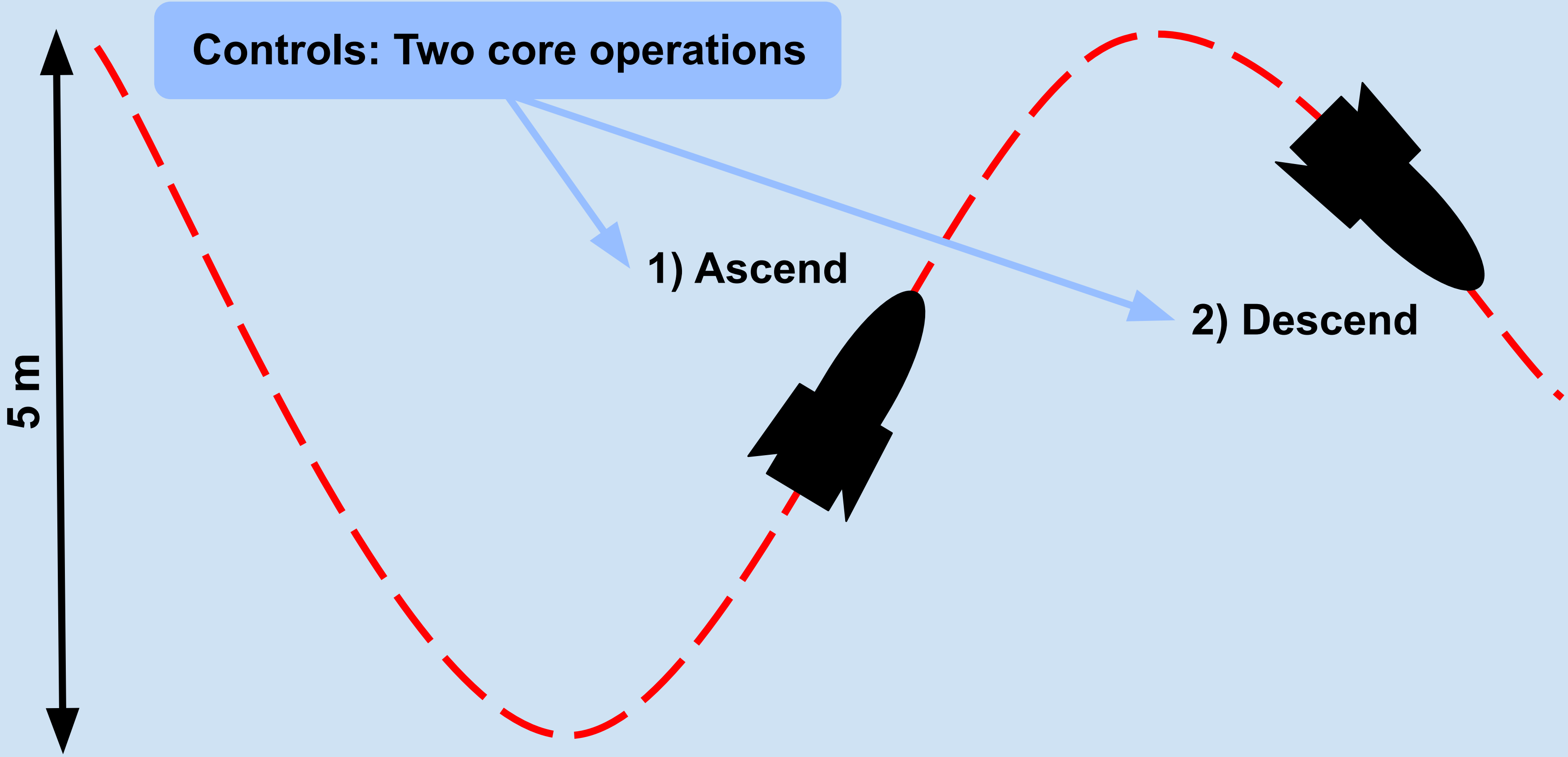
Power Schematic

- ❑ 5V logic level
  - Linear regulator
- ❑ 12V/6V Power
  - Buck converter

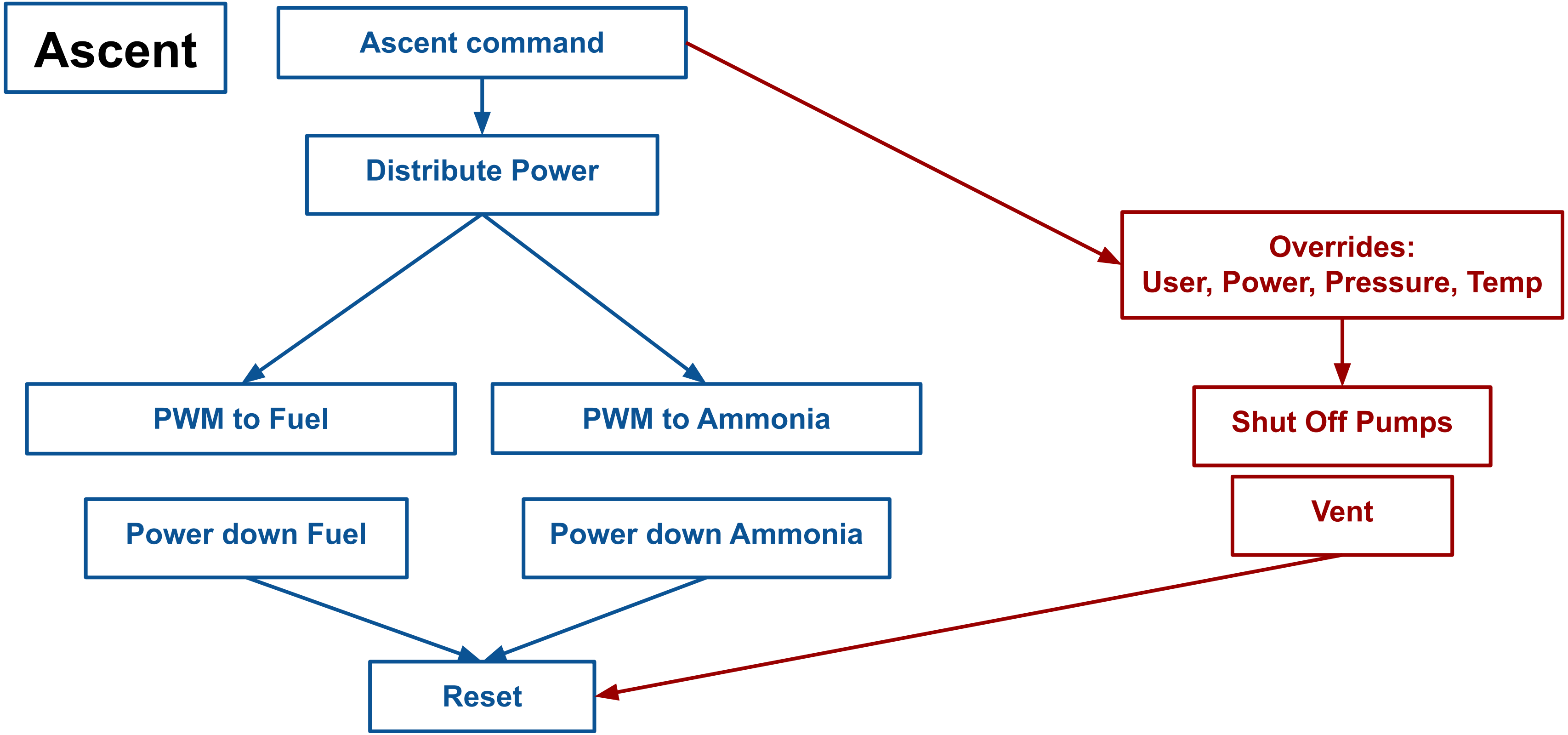




Basic Operations

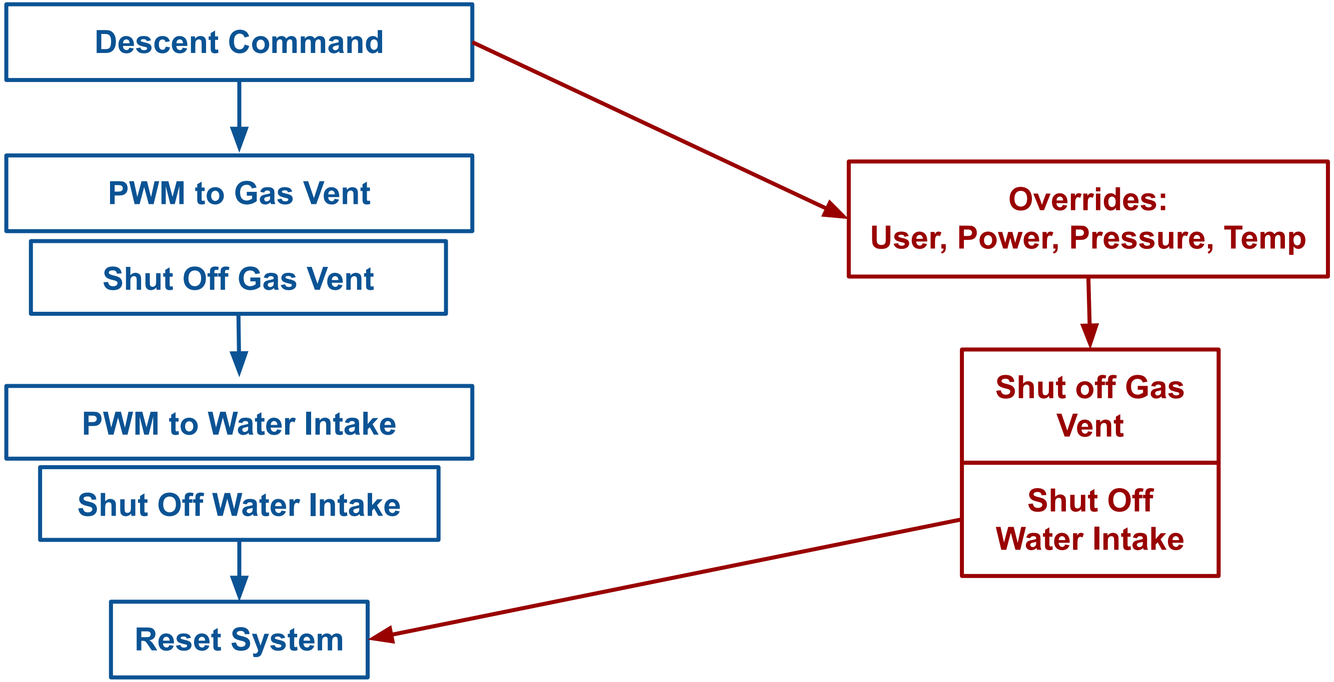


Ascent Flow Chart



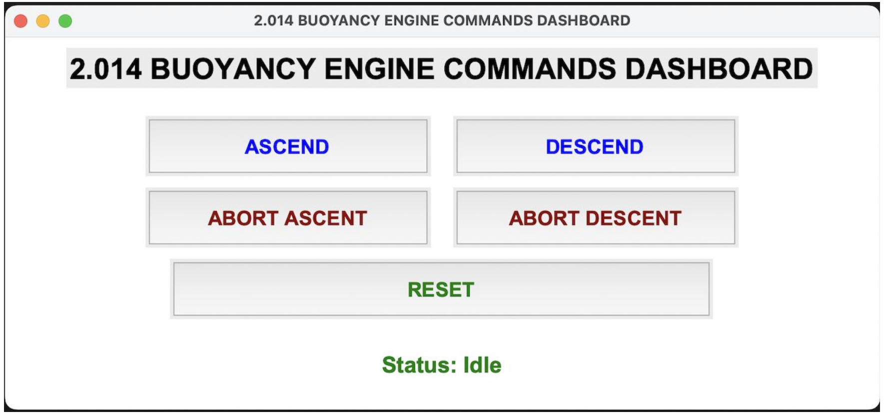
Descent Flow Chart

Descent

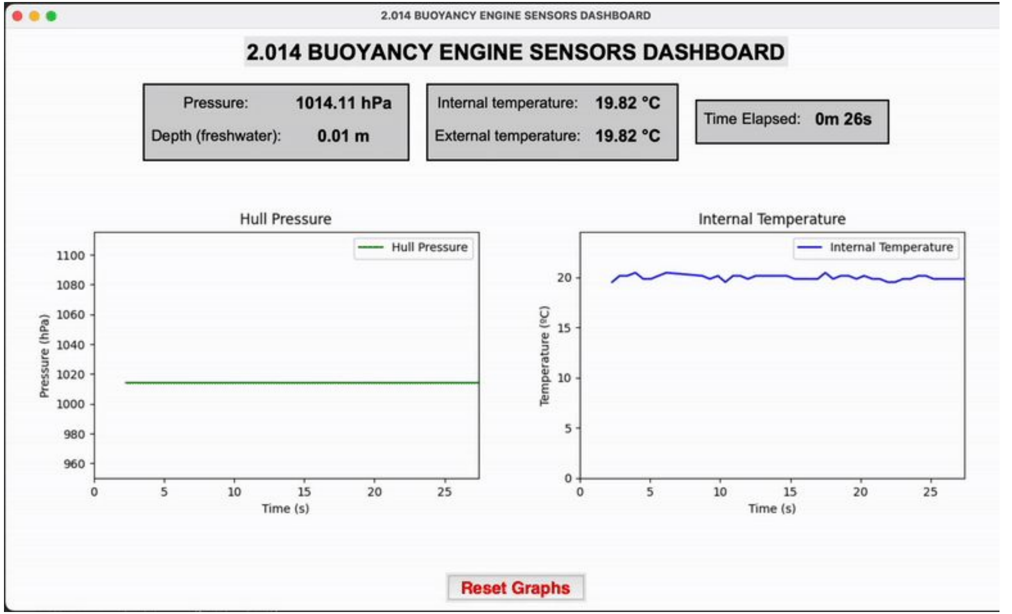


CALIPSO GUI

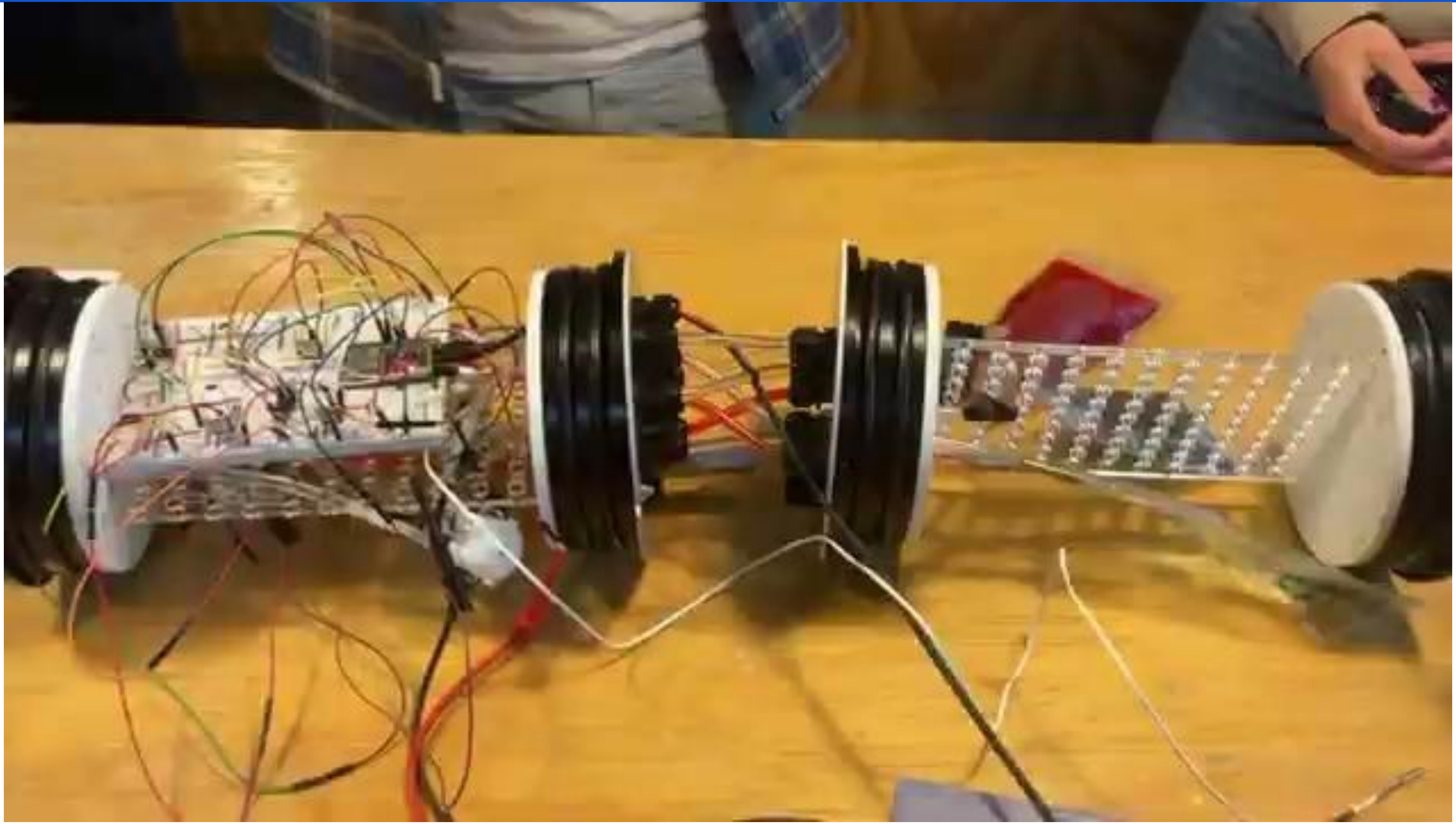
CALIPSO Commands Dashboard



CALIPSO Sensors Dashboard



Mix Testing

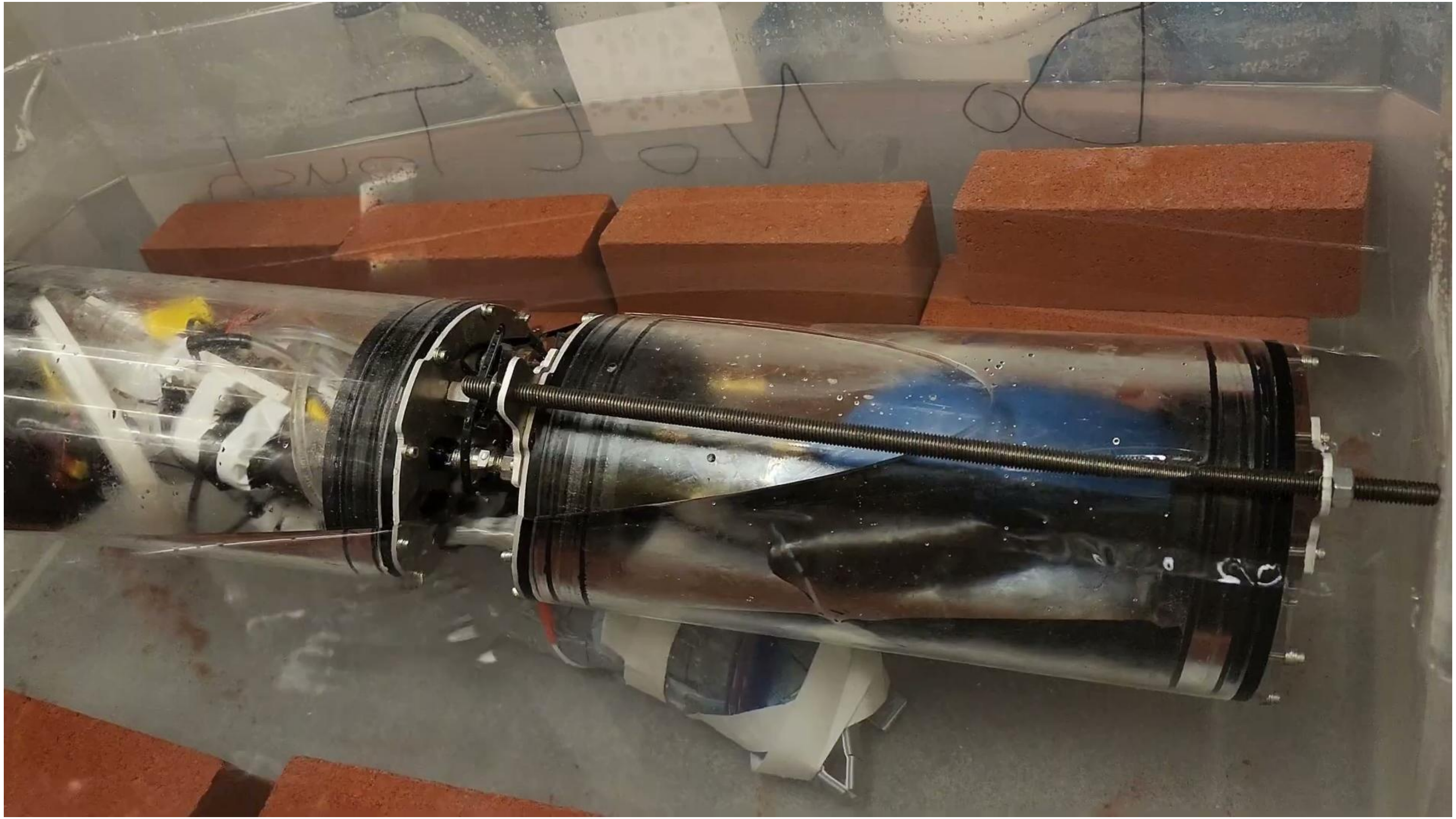


Bladder Testing

Deflated

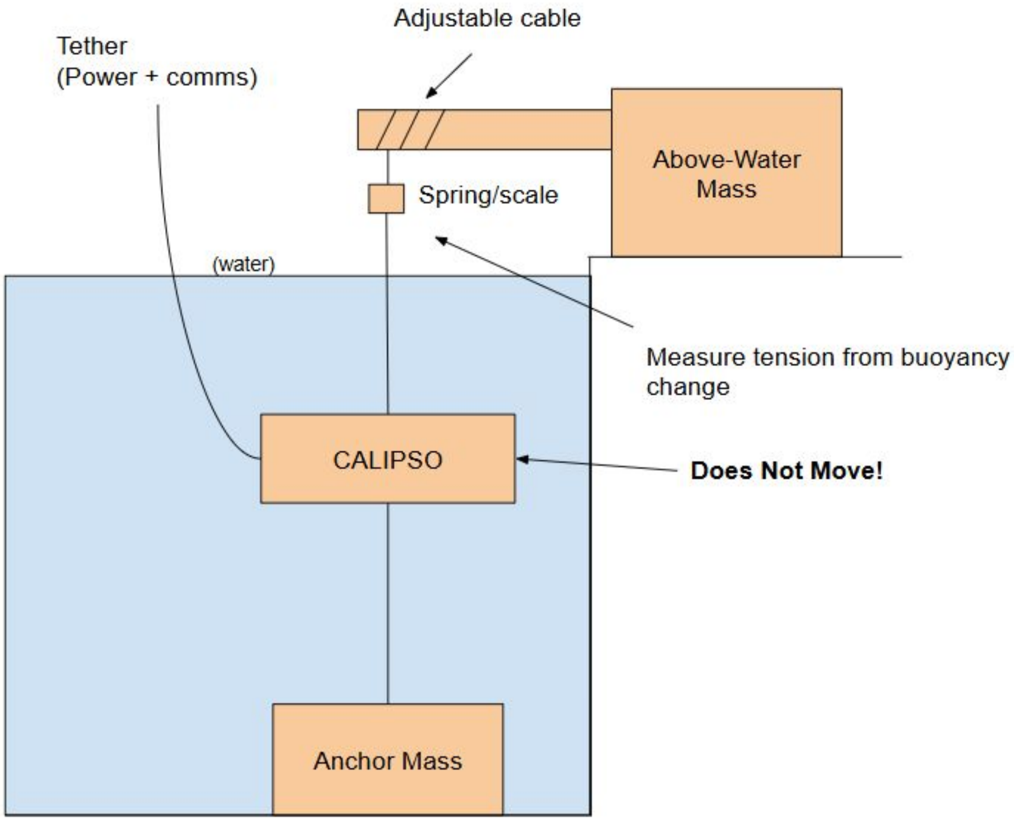
Inflated

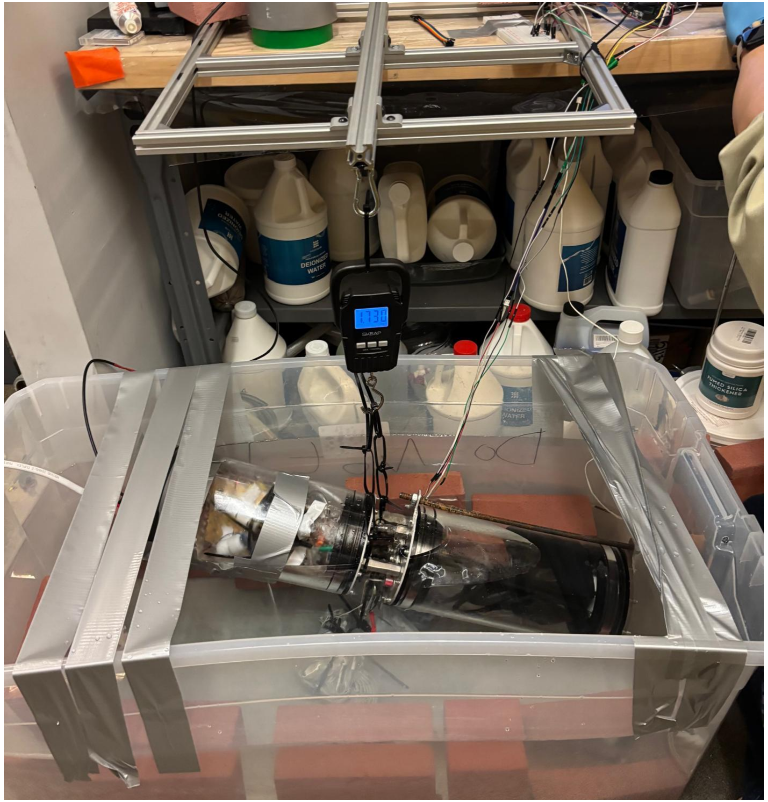
Hull Filling Testing



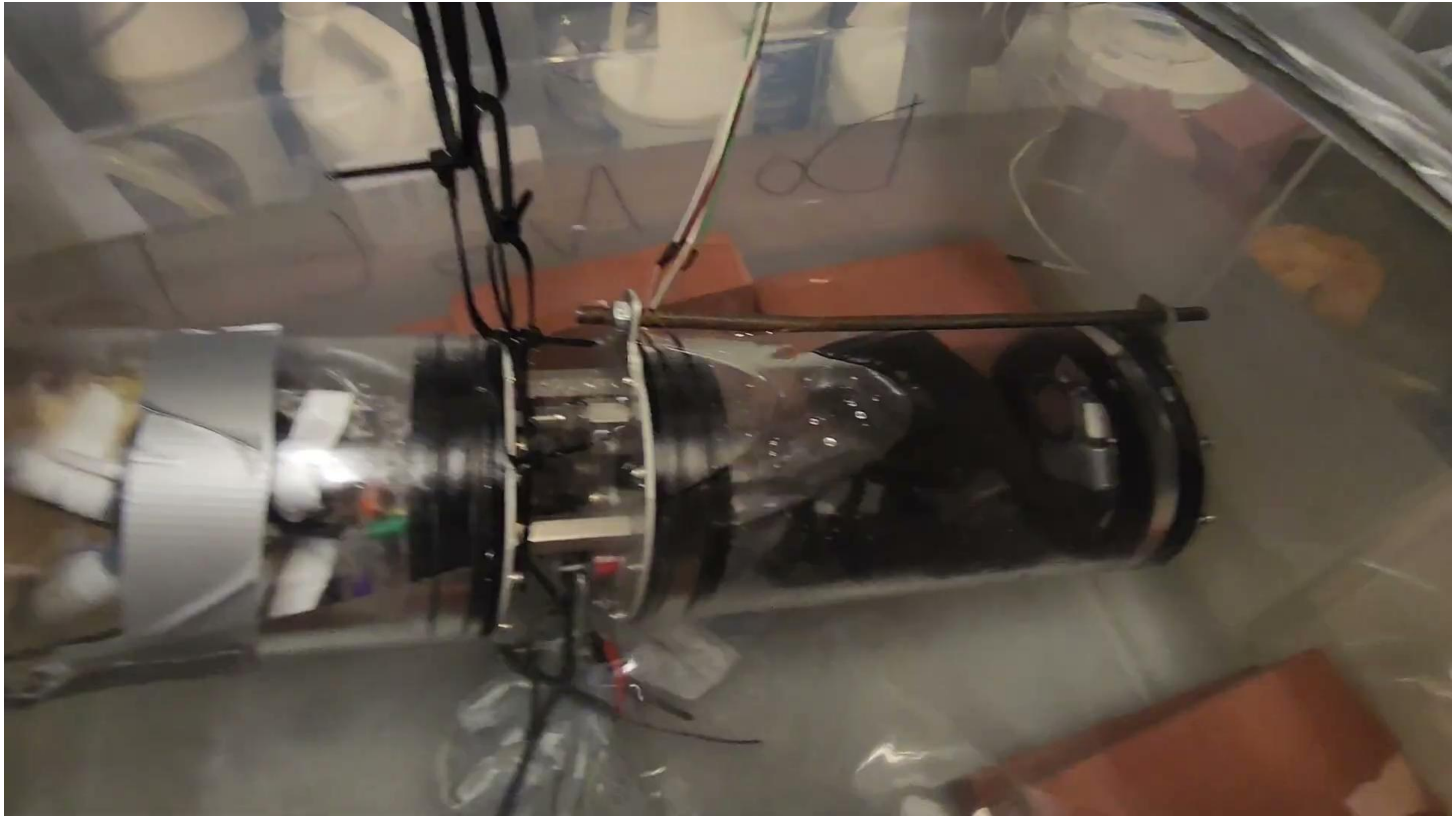
Gas Venting Testing

Testing Setup

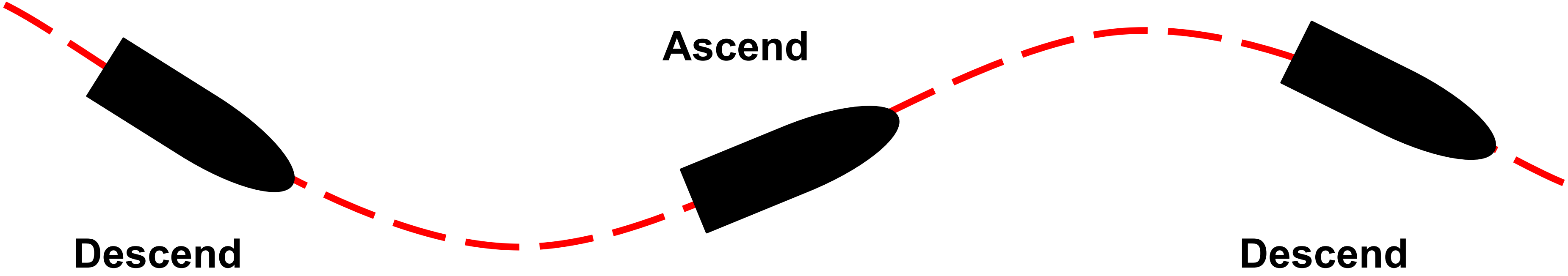




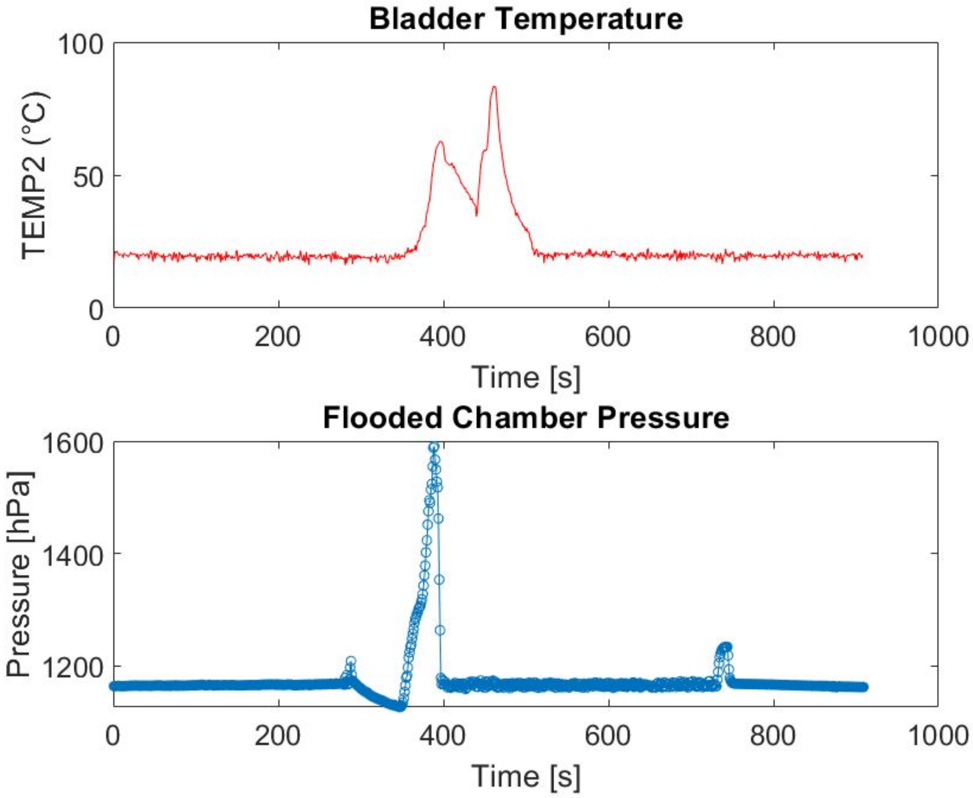
Testing



Testing



Pellet Reaction Test



Results

~50 ∘C increase

~400 hPa increase

~ 1 kg buoyancy

*22 ∘C tank

Next Steps: Scaling

CALIPSO

System Changes for Scaling

- ❑ Al + Water -> Al + Ammonia (more H_2 generation, more inflation)
  - ❑ This will require more careful material handling
- ❑ Improved waterproofing for higher pressures
- ❑ Completely free-standing system, drawing from UUV power instead of power supply
- ❑ Autonomous control or depth-targeted control instead of manual

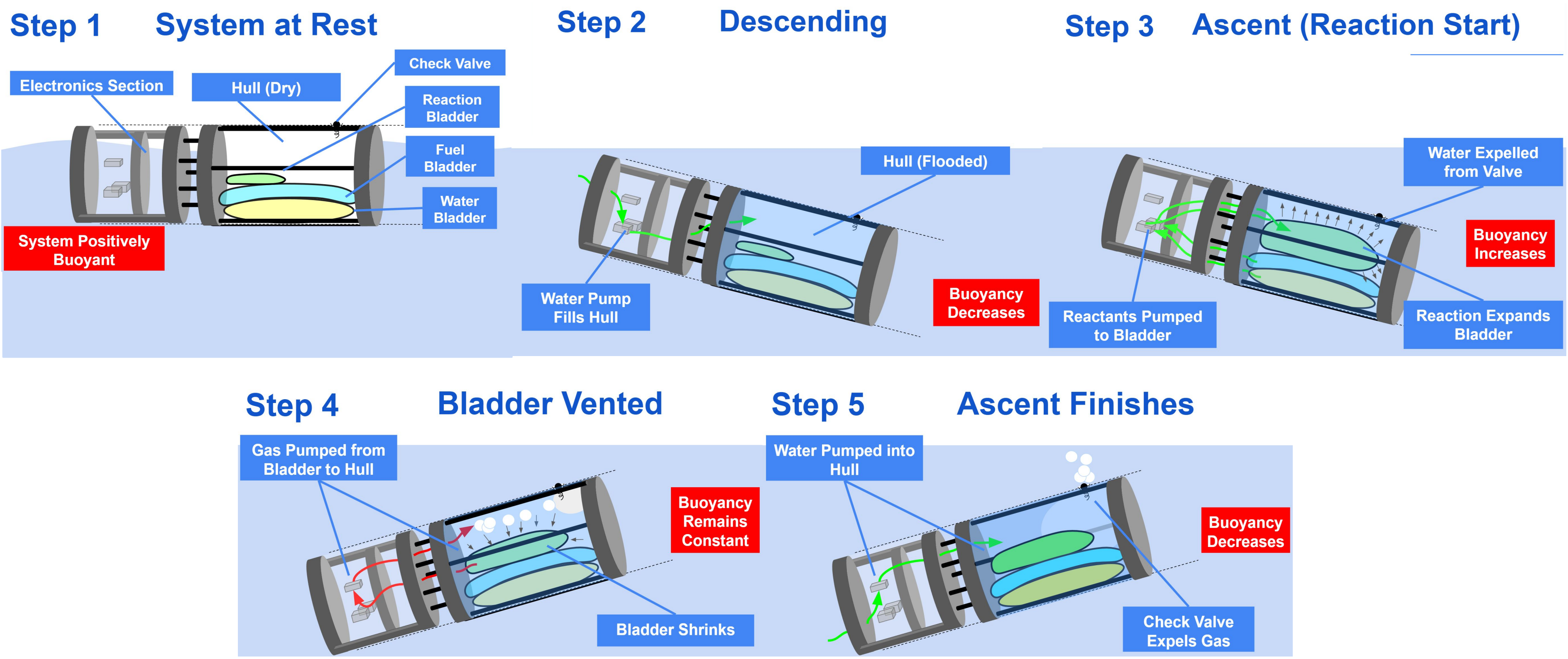
Achievements

- ❑ Developed outline for building a functional Al-fueled buoyancy engine
- ❑ Proved the concept of a aluminum fuel as a gas inflation source
- ❑ Created and tested a custom inflatable bladder that can handle the reaction
- ❑ Characterized key mechanical & chemical properties of aluminum paste with real testing data
- ❑ Successful electronic control, routing, and waterproofing of all systems necessary for an engine

Summary

CALIPSO

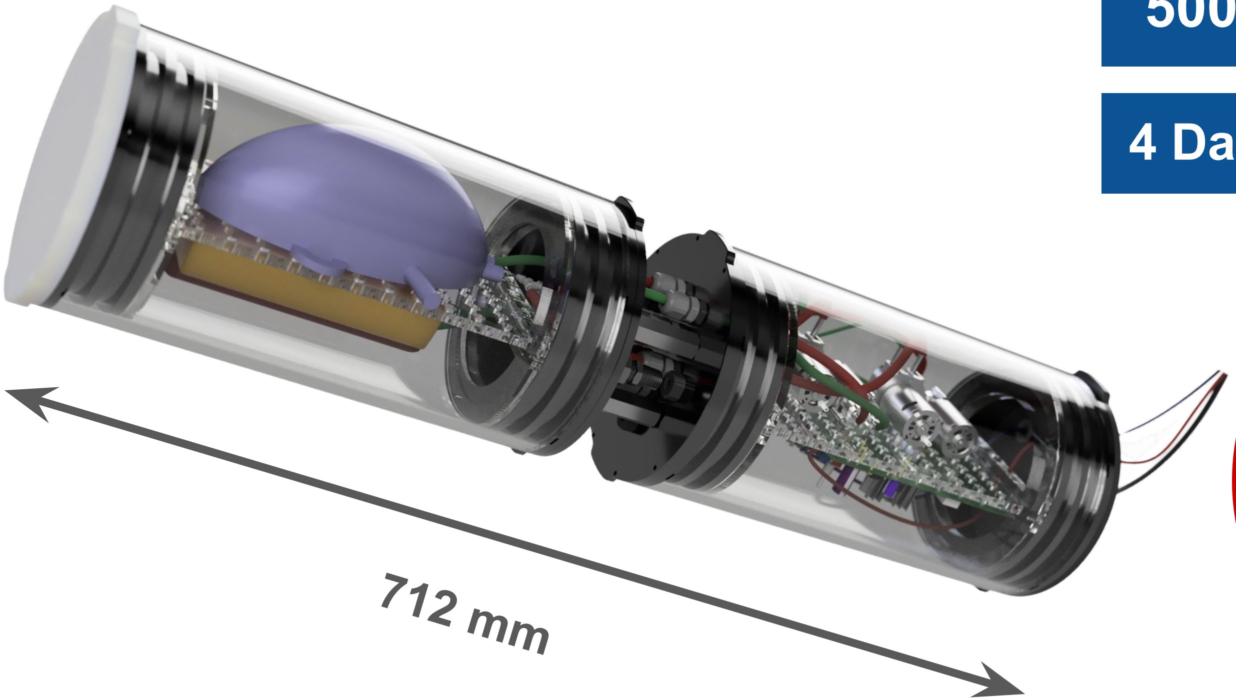
CALIPSO Dive Cycle Actions

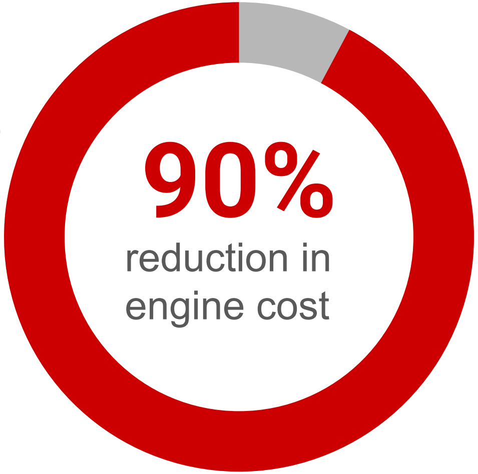


The Future: CALIPSO Lowers Cost for UUVs

500m Operational Depth

4 Days/100 Dives Per Refuel





Appendix: System Requirements

Bladder Volume - By eliminating one of the bladders, the bladder volume halves.

Volume Fuel - start with n = moles of gas in bladder @ 18.5°C and 2 atm
18.5°C is the boiling point of NH3 at 2 atm)

$$n = n_{H_2} + n_{NH_3}$$

$$n_{H_2}\left(\frac{6\ mol\ H_2O}{3\ mol\ H_2}\right)\left(\frac{18.015\ g\ H_2O}{1\ mol\ H_2O}\right)\left(\frac{20\ g\ NH_3}{40\ g\ H_2O}\right)\left(\frac{1\ mol\ NH_3}{17.03\ g\ NH_3}\right) = n_{NH_3}$$

$$n_{H_2}(0.5289) = n_{NH_3}$$

$$n = n_{H_2} + n_{H_2}(0.5289)$$

$$n_{H_2} = (9.58)10^{-3}\ mol$$

$$n_{H_2}\left(\frac{2\ mol\ AL}{3\ mol\ H_2}\right)\left(\frac{27\ g\ AL}{1\ mol\ AL}\right) = m_{AL} = 0.172\ g$$

$$V_{fuel} = \frac{m_{AL}}{(0.60)\rho_{paste}} = 0.175\ \text{mL}$$

Appendix: Viscosity

Einstein's Viscosity Equation for suspensions: $\mu = \mu_0(1 + \alpha_f \phi)$

- μ_0: liquid viscosity
- α_f: constant coefficient used for nanofluid lubricants
- ϕ : fraction of suspended particles by volume

Therefore matching volume percentage allows us to create Stainless steel substitute that mimics the viscosity of aluminum fuel.

Appendix: Viscosity

Poiseuille's Law:

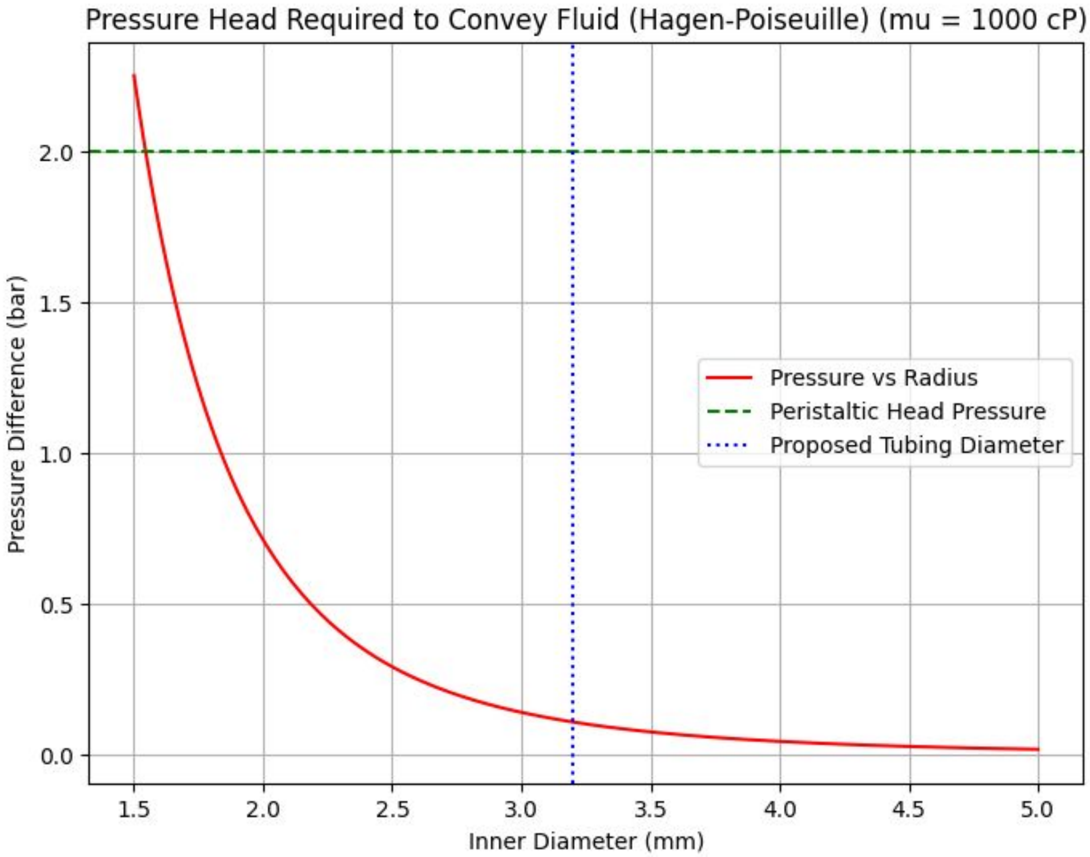
$$Q = \frac{\Delta P \pi r^4}{8 \eta l} = \frac{\rho \pi r^4 g h}{8 \eta l} = \frac{\rho \pi r^4 g Q t}{8 \eta l A}$$

$$\therefore \frac{\eta}{\rho} = \left(\frac{\pi r^4 g}{8 l A}\right) t \sim kt \rightarrow \frac{\eta_1}{\rho_1 t_1} = \frac{\eta_2}{\rho_2 t_2}$$

To mimic the Aluminum Fuel viscosity, for a given weight of oil, we calculated the volume of aluminum from the weight ratio. We then input that same volume of stainless steel since we found that viscosity of a suspension is proportional to the volume of particles in the fluid – from Einstein's equation. (resulted in a denser liquid that in theory has the same viscosity).

$$E = E_0(1 + 2.5V) \rightarrow E \sim V$$

Appendix: Fuel Pump Specification



$$\Delta P = \frac{8\mu L Q}{\pi r^4}$$

ΔP = Pressure difference (Pa)

μ = Dynamic viscosity (Pa·s)

L = Pipe Length (m)

Q = Volumetric flow rate (m^3/s)

R = Radius (m)